电力系统分析(1)

Power System Analysis （1）

（八）

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第八章

电力系统简单不对称故障的分析计算

1、什么是对称分量法?
2、为什么要引入对称分量法?

- 对称分量法

分析过程是什么?

1、各元件的序参数是怎样的?
制电力系统的序网图?

- 对称分量法在

如何利用对称分量法对简单不对称故障进行分析与计算?

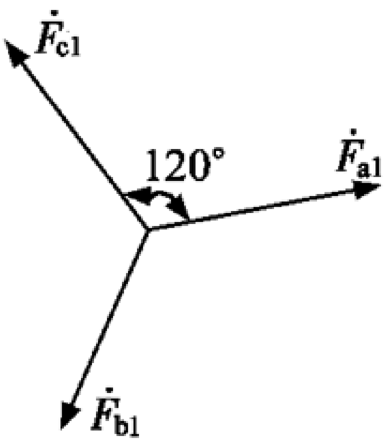
- 电力系统元件 序网图
- 简单不对称故障的分析计算

8.1 对称分量法在不对称短路计算中的应用

一、对称分量法

在三相系统中，任意不对称的三相量可分为对称的三序分量之和

$$\left.\begin{aligned}\dot{F}_a &= \dot{F}_{a1} + \dot{F}_{a2} + \dot{F}_{a0} \\ \dot{F}_b &= \dot{F}_{b1} + \dot{F}_{b2} + \dot{F}_{b0} \\ \dot{F}_c &= \dot{F}_{c1} + \dot{F}_{c2} + \dot{F}_{c0}\end{aligned}\right\}$$



正序分量

一、对称分量法

- **正序分量**：三相量大小相等，互差120^0，且与系统正常运行相序相同。
- **负序分量**：三相量大小相等，互差120^0，且与系统正常运行相序相反。
- **零序分量**：三相量大小相等，相位一致。

逆时针旋转120^0

$$\left.\begin{aligned}
\dot{F}_{b1} &= a^2\dot{F}_{a1}, \dot{F}_{c1} = a\dot{F}_{a1} \\
\dot{F}_{b2} &= a\dot{F}_{a2}, \dot{F}_{c2} = a^2\dot{F}_{a2} \\
\dot{F}_{b0} &= \dot{F}_{c0} = \dot{F}_{a0}
\end{aligned}\right\}$$

$$a = e^{j120^\circ}$$

$$\boldsymbol{a^2 = e^{j240^\circ}}$$

$$\boldsymbol{a^2 + a + 1 = 0}$$

一、对称分量法

- 三相量用三序量表示

$$\mathbf{T}=\begin{bmatrix}1 & 1 & 1\\ a^2 & a & 1\\ a & a^2 & 1\end{bmatrix}$$

$$\left.\begin{aligned}\dot{F}_a &= \dot{F}_{a1}+\dot{F}_{a2}+\dot{F}_{a0}\\ \dot{F}_b &= \dot{F}_{b1}+\dot{F}_{b2}+\dot{F}_{b0}=a^2\dot{F}_{a1}+a\dot{F}_{a2}+\dot{F}_{a0}\\ \dot{F}_c &= \dot{F}_{c1}+\dot{F}_{c2}+\dot{F}_{c0}=a\dot{F}_{a1}+a^2\dot{F}_{a2}+\dot{F}_{a0}\end{aligned}\right\}\Rightarrow \dot{\boldsymbol{F}}_{abc}=\mathbf{T}\dot{\boldsymbol{F}}_{120}$$

- 三序量用三相量表示

$$\mathbf{T}^{-1}=\frac{1}{3}\begin{bmatrix}1 & a & a^2\\ 1 & a^2 & a\\ 1 & 1 & 1\end{bmatrix}$$

$$\begin{bmatrix}\dot{F}_{a1}\\ \dot{F}_{a2}\\ \dot{F}_{a0}\end{bmatrix}=\frac{1}{3}\begin{bmatrix}1 & a & a^2\\ 1 & a^2 & a\\ 1 & 1 & 1\end{bmatrix}\begin{bmatrix}\dot{F}_a\\ \dot{F}_b\\ \dot{F}_c\end{bmatrix}\Rightarrow \dot{\boldsymbol{F}}_{120}=\mathbf{T}^{-1}\dot{\boldsymbol{F}}_{abc}$$

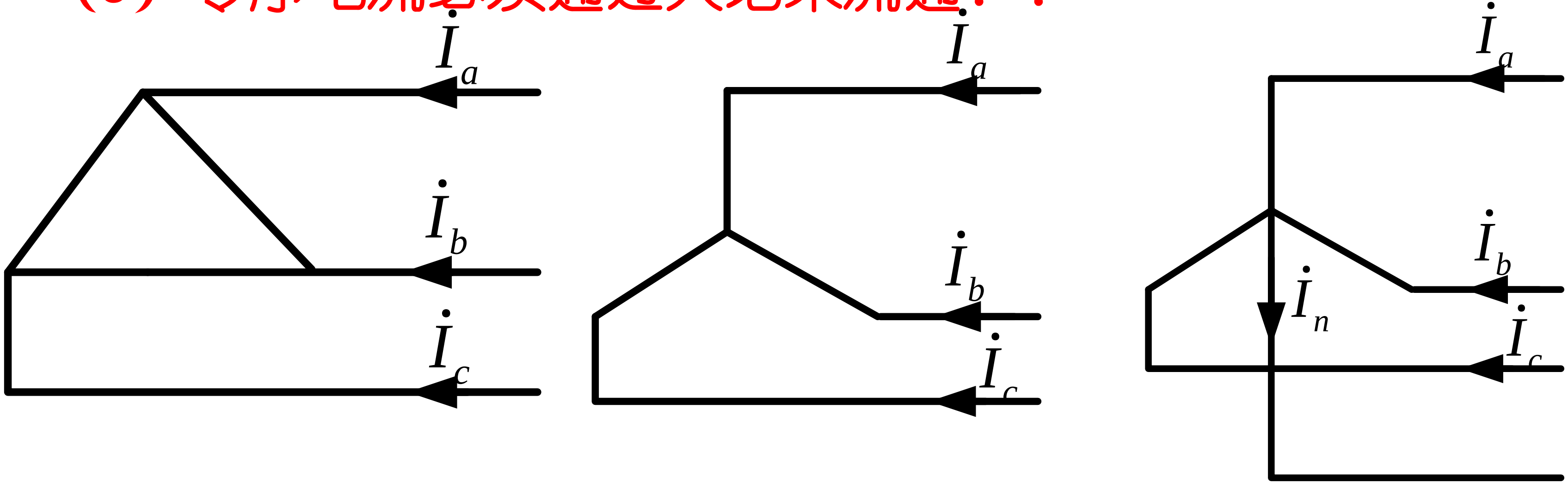
注意：

（1）只有当三相电流之和不等于零时才有零序分量。如果三相系统是三角形接法，或者没有中性线（包括以地代中线）的星形接法，三相线电流之和总为零，不可能有零序分量电流。

（2）在有中性线的星形接法中，则中性线中的电流为

$$\dot{I}_n = \dot{I}_a + \dot{I}_b + \dot{I}_c = 3\dot{I}_{a(0)}$$

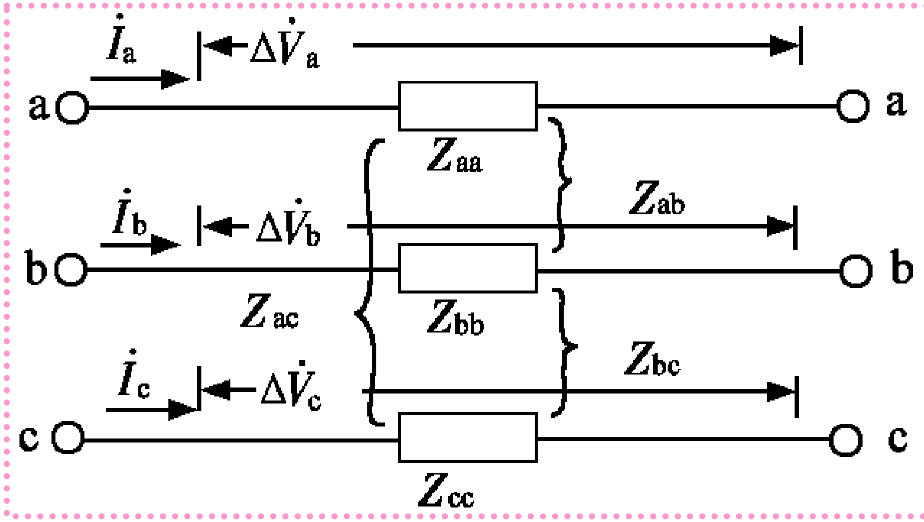
（3）零序电流必须通过大地来流通！！



二、各序分量的独立性

- 静止的三相电路元件

$$\begin{bmatrix} \Delta\dot{V}_a \\ \Delta\dot{V}_b \\ \Delta\dot{V}_c \end{bmatrix} = \begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ab} & Z_{bb} & Z_{bc} \\ Z_{ac} & Z_{bc} & Z_{cc} \end{bmatrix} \begin{bmatrix} \dot{I}_a \\ \dot{I}_b \\ \dot{I}_c \end{bmatrix}$$



$$\Downarrow$$

$$\Delta\mathbf{V}_{abc} = \boldsymbol{ZI}_{abc}$$

$$\Downarrow$$

$$\Delta\mathbf{V}_{120} = \mathbf{T}^{-1}\mathbf{ZTI}_{120} = \mathbf{Z}_{sc}\mathbf{I}_{120}$$

$$\mathbf{Z}_{sc} = \mathbf{T}^{-1}\mathbf{ZT}$$

称为序阻抗矩阵

二、各序分量的独立性

- 当元件参数完全对称时 $z_{aa}=z_{bb}=z_{cc}=z_s \qquad z_{ab}=z_{bc}=z_{ca}=z_m$

$$\mathbf{Z}_{sc}=\begin{bmatrix} \mathbf{Z}_s-\mathbf{Z}_m & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{Z}_s-\mathbf{Z}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{Z}_s+2\mathbf{Z}_m \end{bmatrix}=\begin{bmatrix} \mathbf{Z}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{Z}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{Z}_0 \end{bmatrix}$$

$$\boldsymbol{\Delta V}_{120}=\mathbf{Z}_{sc}\boldsymbol{I}_{120} \Rightarrow \left.\begin{aligned} \Delta\dot{V}_{a1}&=Z_1\dot{I}_{a1} \\ \Delta\dot{V}_{a2}&=Z_2\dot{I}_{a2} \\ \Delta\dot{V}_{a0}&=Z_0\dot{I}_{a0} \end{aligned}\right\}$$

结论：在三相参数对称的线性电路中，各序对称分量具有独立性，因此，可以对正序、负序、零序分量分别进行计算。

注意：各序分量是对称的，只分析一相。

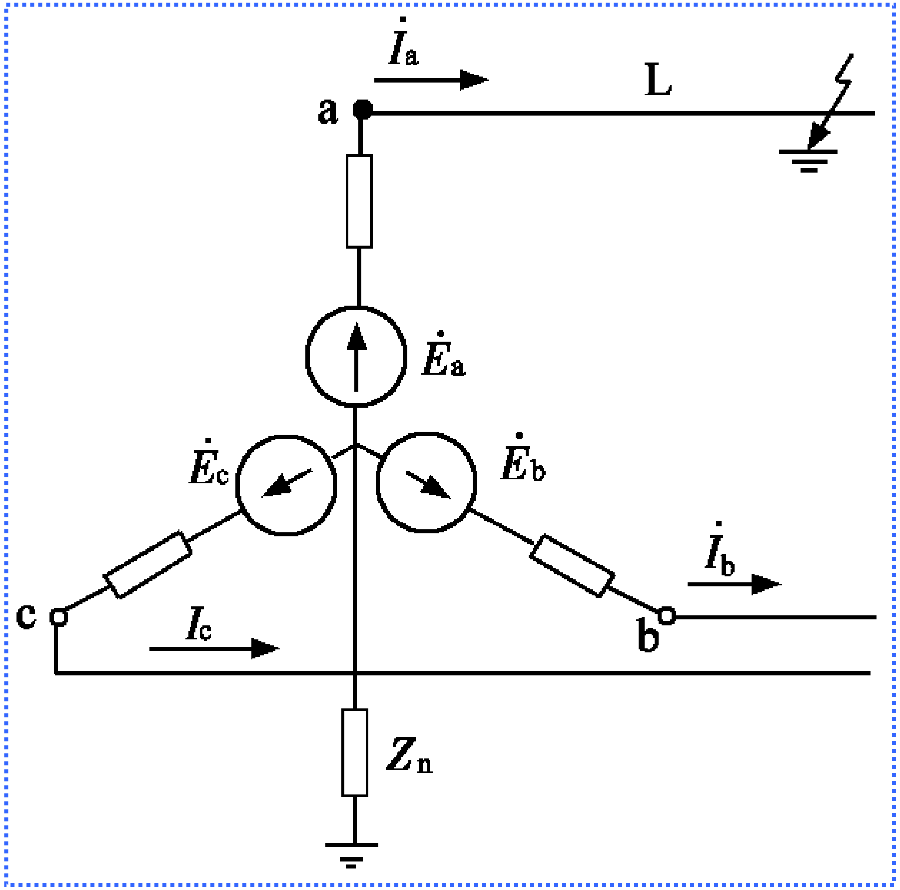
二、各序分量的独立性

- 序阻抗：元件三相参数对称时，元件两端某一序的电压降与通过该元件的同一序电流的比值。

$$\left.\begin{array}{ll}\text{正序阻抗} \longrightarrow & Z_1 = \Delta \dot{V}_{a1} / \dot{I}_{a1} \\ \text{负序阻抗} \longrightarrow & Z_2 = \Delta \dot{V}_{a2} / \dot{I}_{a2} \\ \text{零序阻抗} \longrightarrow & Z_0 = \Delta \dot{V}_{a0} / \dot{I}_{a0}\end{array}\right\}$$

三、对称分量法在不对称短路计算中的应用

- 一台发电机接于空载线路，发电机中性点经阻抗Z_n接地。
- a相发生单相接地



$$\dot{V}_a = 0 \qquad \dot{I}_a \neq 0$$

$$\dot{V}_b \neq 0 \qquad \dot{I}_b = 0$$

$$\dot{V}_c \neq 0 \qquad \dot{I}_c = 0$$

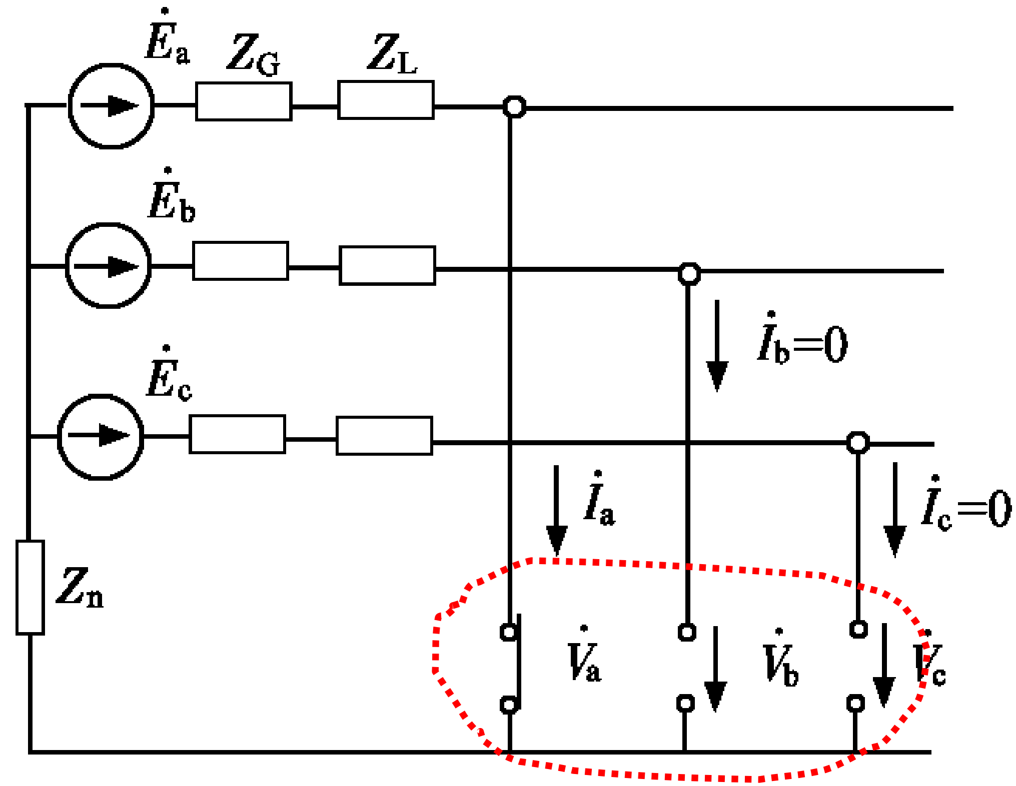
三、对称分量法在不对称短路计算中的应用

- a相接地的模拟

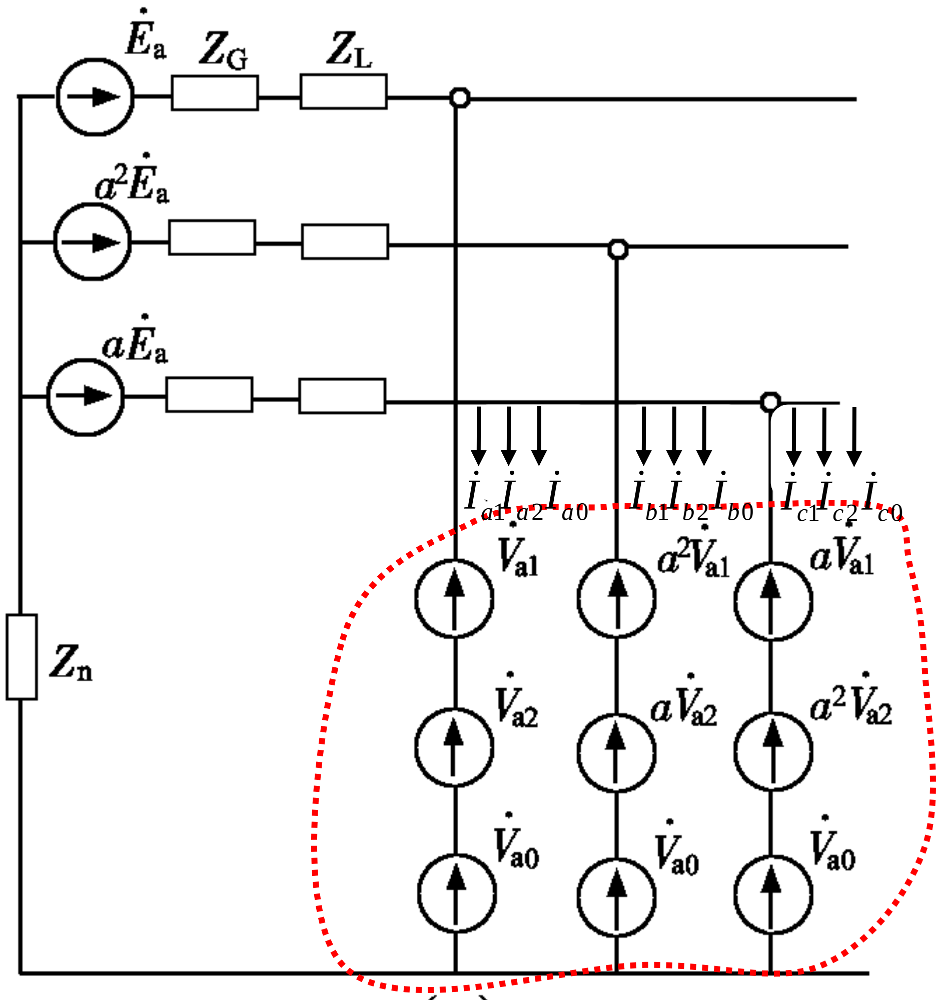
$$\dot{V}_a = 0 \qquad \dot{I}_a \neq 0$$

$$\dot{V}_b \neq 0 \qquad \dot{I}_b = 0$$

$$\dot{V}_c \neq 0 \qquad \dot{I}_c = 0$$



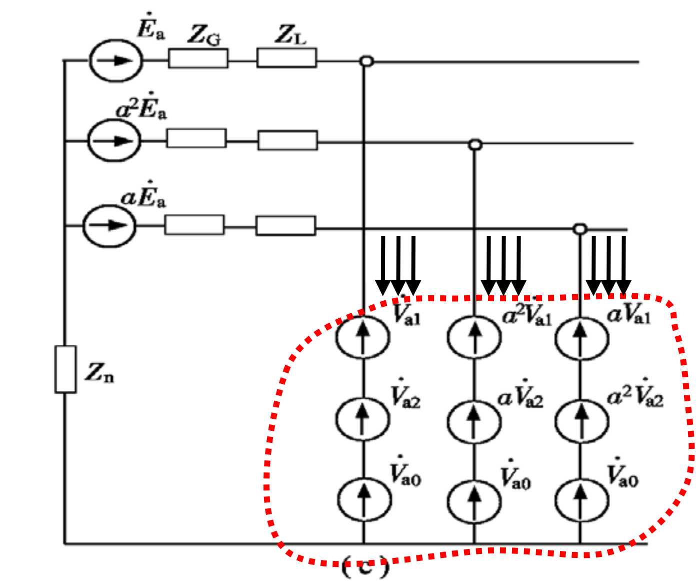
三、对称分量法在不对称短路计算中的应用



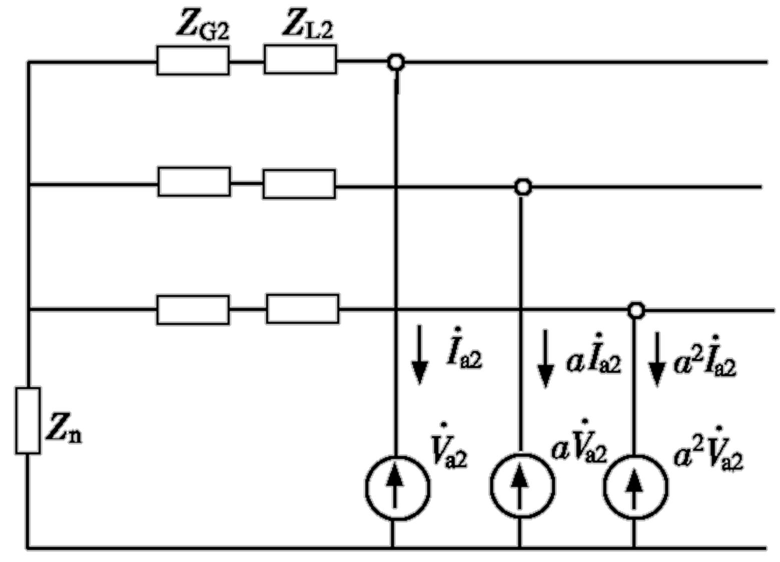
(c)

将不对称部分用三序分量表示

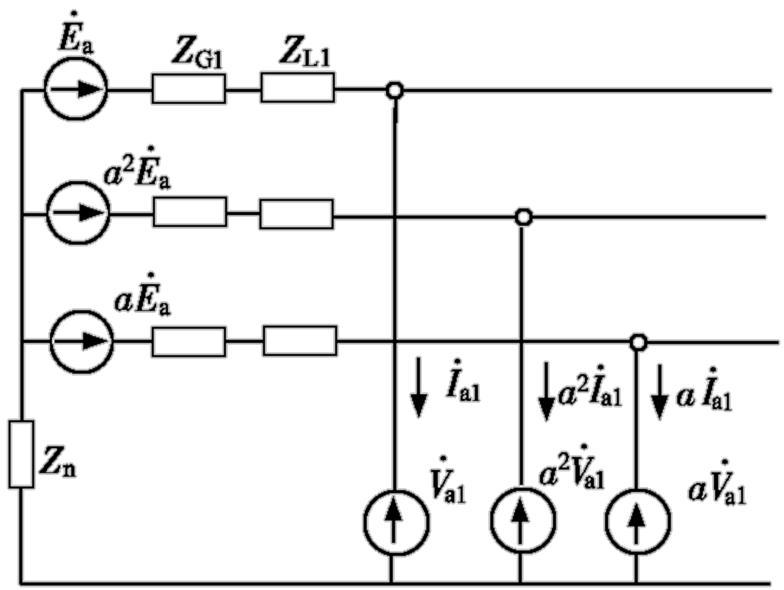
应用叠加原理进行分解



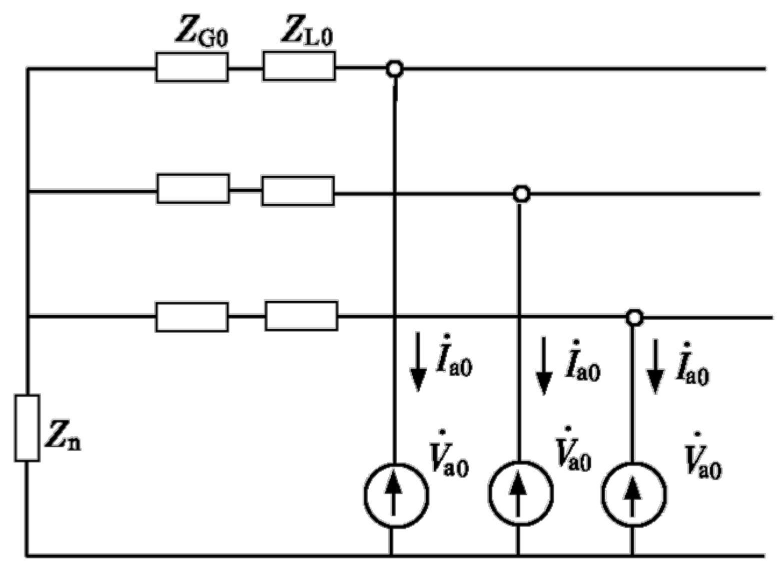
Ėa
ZG
ZL
a²Ėa
aĖa
Zn
V̇a1
a²V̇a1
aV̇a1
V̇a2
aV̇a2
a²V̇a2
V̇a0
V̇a0
V̇a0
(c)



ZG2
ZL2
Zn
İa2
aİa2
a²İa2
V̇a2
aV̇a2
a²V̇a2



Ėa
ZG1
ZL1
a²Ėa
aĖa
Zn
İa1
a²İa1
aİa1
V̇a1
a²V̇a1
aV̇a1



ZG0
ZL0
Zn
İa0
İa0
İa0
V̇a0
V̇a0
V̇a0

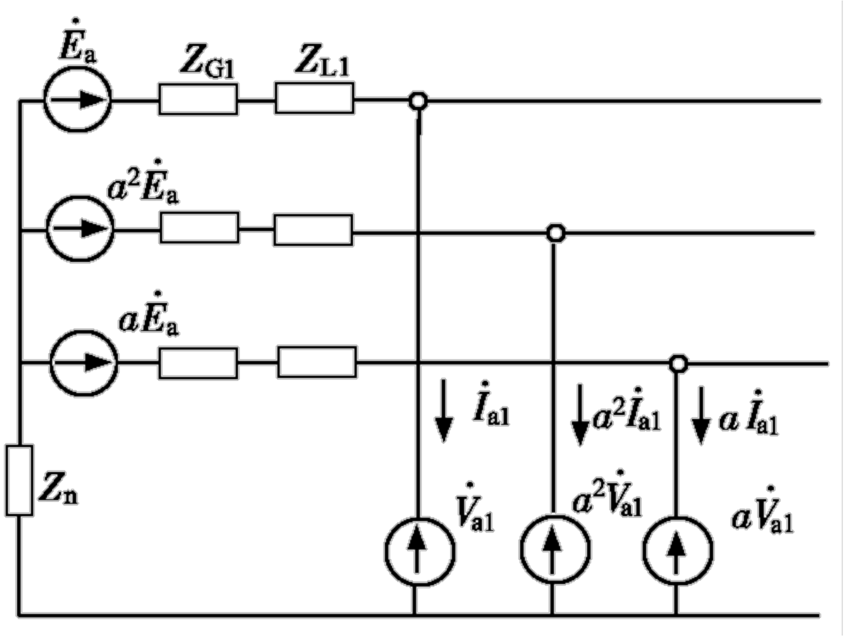
三、对称分量法在不对称短路计算中的应用

➢正序网

$$\dot{E}_a - \dot{I}_{a1}(Z_{G1} + Z_{L1}) - (\dot{I}_{a1} + a^2\dot{I}_{a1} + a\dot{I}_{a1})Z_n = \dot{V}_{a1}$$

$$\begin{aligned}&\dot{I}_{a1} + \dot{I}_{b1} + \dot{I}_{c1}\\&= \dot{I}_{a1} + \alpha^2\dot{I}_{a1} + \alpha\dot{I}_{a1}\\&= 0\end{aligned}$$

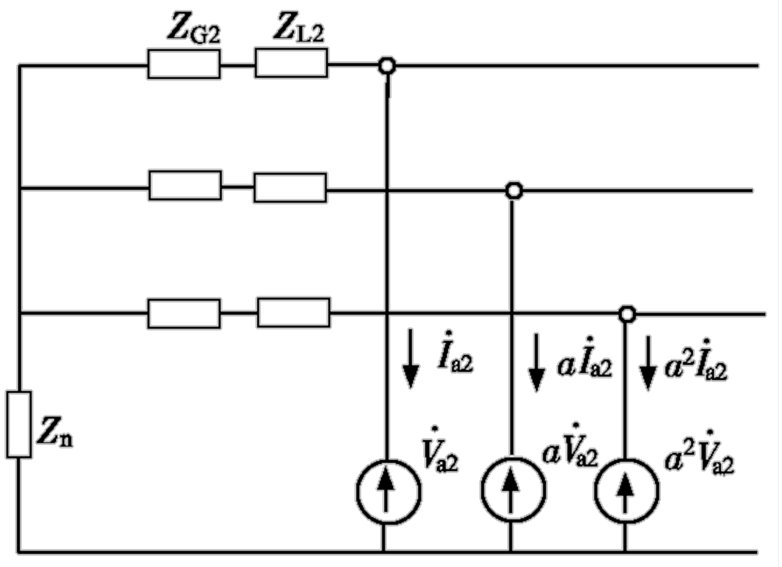
$$\dot{E}_a - \dot{I}_{a1}(Z_{G1} + Z_{L1}) = \dot{V}_{a1}$$



三、对称分量法在不对称短路计算中的应用

➢ **负序网**

$$0 - \dot{I}_{a2}(Z_{G2} + Z_{12}) = \dot{V}_{a2}$$



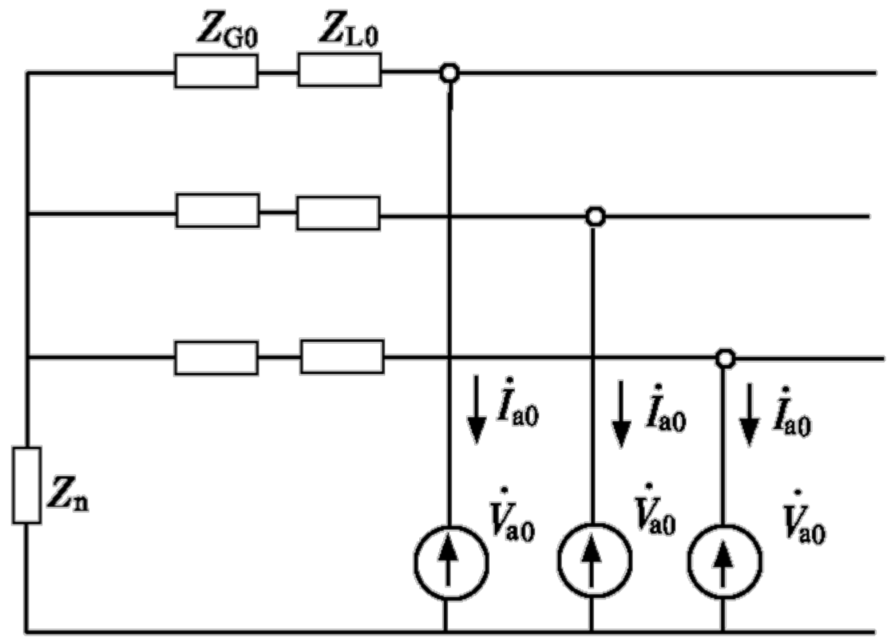
三、对称分量法在不对称短路计算中的应用

➢ 零序网

$$\dot{I}_{a0}+\dot{I}_{b0}+\dot{I}_{c0}=3\dot{I}_{a0}$$

$$0-\dot{I}_{a0}(Z_{G0}+Z_{L0})-3\dot{I}_{a0}Z_n=\dot{V}_{a0}$$

$$0-\dot{I}_{a0}(Z_{G0}+Z_{L0}+3Z_n)=\dot{V}_{a0}$$

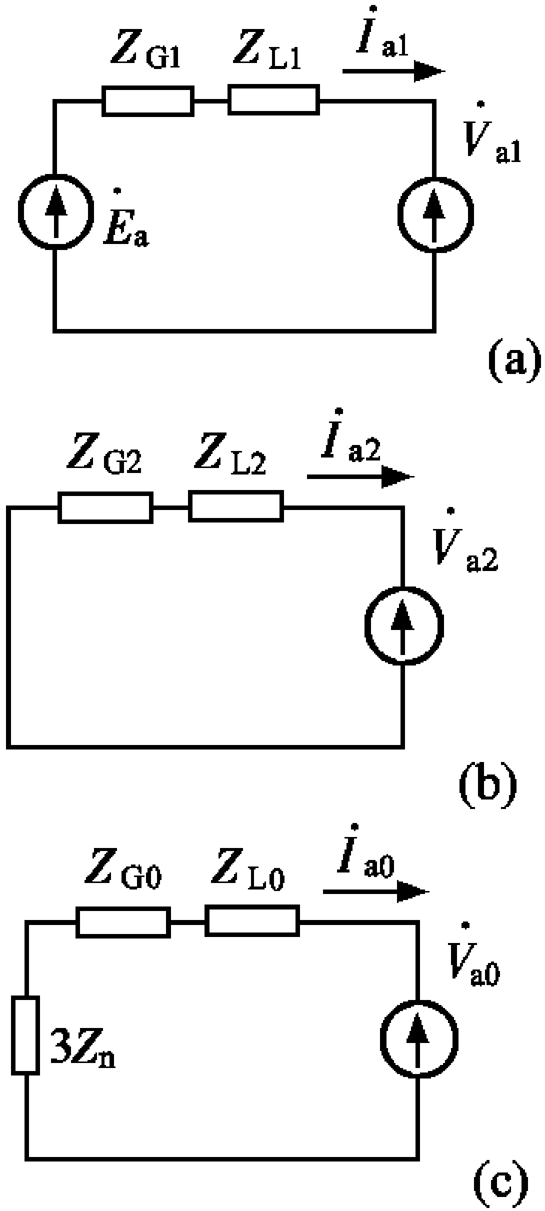


$$\dot{E}_a - \dot{I}_{a1}(Z_{G1} + Z_{L1}) = \dot{V}_{a1}$$

$$0 - \dot{I}_{a2}(Z_{G2} + Z_{12}) = \dot{V}_{a2}$$

$$0 - \dot{I}_{a0}(Z_{G0} + Z_{L0} + 3Z_n) = \dot{V}_{a0}$$

$$\left.\begin{aligned} \dot{E}_{\Sigma} - \dot{I}_{a1} Z_{1\Sigma} &= \dot{V}_{a1} \\ 0 - \dot{I}_{a2} Z_{2\Sigma} &= \dot{V}_{a2} \\ 0 - \dot{I}_{a0} Z_{0\Sigma} &= \dot{V}_{a0} \end{aligned}\right\}$$

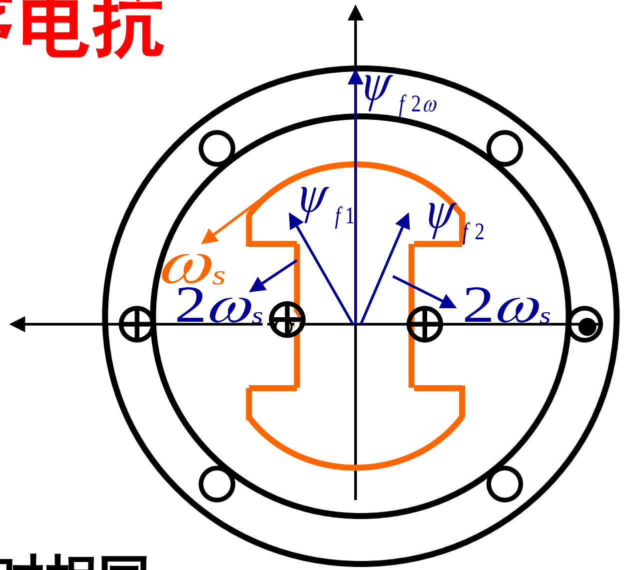


8.2 电力系统各序网络

- **静止元件**：正序阻抗等于负序阻抗，不等于零序阻抗。如：变压器、输电线路等。

- **旋转元件**：各序阻抗均不相同。如：发电机、电动机等元件。

一、同步发电机的负序和零序电抗



1、不对称短路时的高次谐波

定子电流
- 基波
  - 正序
  - 负序
  - 零序
- 直流分量

➢正序：与转子相对静止，与三相短路时相同

➢负序：与转子方向相反，与转子绕组相互作用，在定子中产生奇次谐波，在转子中产生偶次谐波。

➢零序：空间对称，不形成合成磁场，不影响转子绕组，只有漏磁场

➢直流分量：产生静止的磁场，与转子绕组相互作用，在定子中产生偶次谐波，在转子中产生奇次谐波，衰减到零。

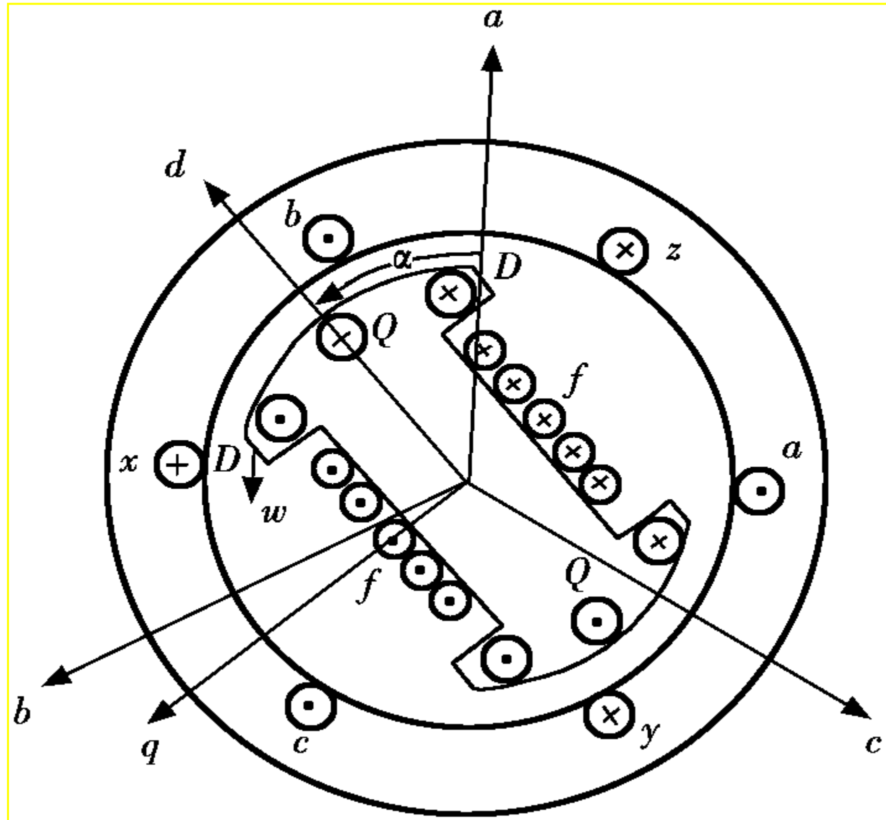
➢注意：如交、直轴方向有相同的绕组，则定子、转子中不会产生高次谐波。

一、同步发电机的负序和零序电抗

2 同步发电机的负序电抗：发电机端点的负序电压基频分量和流入定子绕组的负序电流基频分量的比值。

•负序旋转磁场与转子旋转方向相反，因而在不同的位置会遇到不同的磁阻（因转子不是任意对称的），负序电抗会发生周期性变化。

- 有阻尼绕组发电机 $X''_d \sim X''_q$
- 无阻尼绕组发电机 $X'_d \sim X_q$



2 同步发电机的负序电抗

- 实用计算中发电机负序电抗计算

有阻尼绕组 $X_2=\frac{1}{2}(X_d''+X_q'')$ 无阻尼绕组 $X_2=\sqrt{X_d'X_q}$

- 发电机负序电抗近似估算值

有阻尼绕组 $X_2=1.22X_d''$ 无阻尼绕组 $X_2=1.45X_d'$

- 无确切数值，可取典型值

<table>
<tr><td rowspan="2">电机类型
电抗</td><td colspan="2">水轮发电机</td><td rowspan="2">汽轮发电机</td><td rowspan="2">调相机和
大型同步电动机</td></tr>
<tr><td>有阻尼绕组</td><td>无阻尼绕组</td></tr>
<tr><td>X_2</td><td>0.15~0.35</td><td>0.32~0.55</td><td>0.134~0.18</td><td>0.24</td></tr>
<tr><td>X_0</td><td>0.04~0.125</td><td>0.04~0.125</td><td>0.036~0.08</td><td>0.08</td></tr>
</table>

3. 同步发电机的零序电抗

- 定义：发电机端点的零序电压基频分量和流入定子绕组的零序电流基频分量的比值。

- 三相零序电流在气隙中产生的合成磁势为零，因此其零序电抗仅由定子线圈的漏磁通确定。

- 同步发电机零序电抗在数值上相差很大（绕组结构形式不同）： $X_0=(0.15\sim0.6)X_d''$

- 零序电抗典型值

- 发电机中性点通常不接地 $X_0=\infty$

二、异步电动机和综合负荷的序阻抗

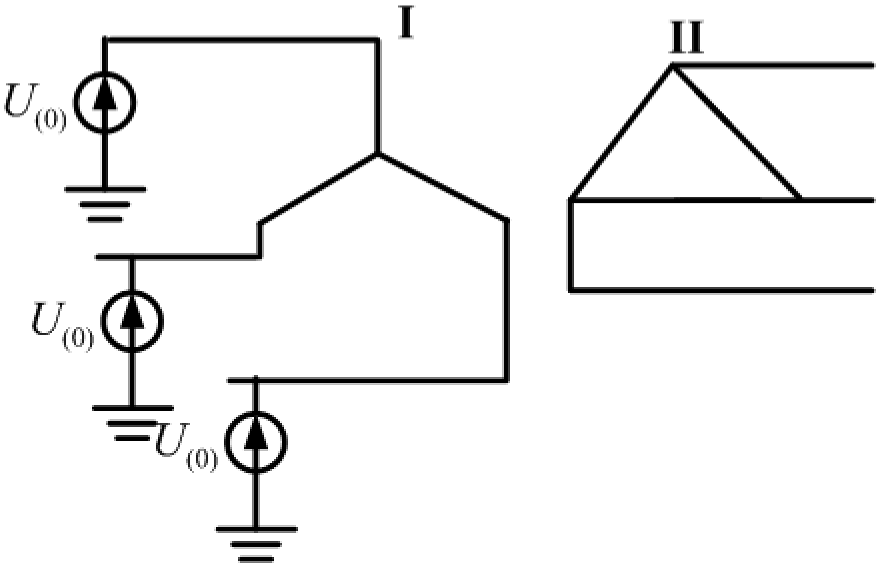
- 异步电机和综合负荷的正序阻抗：

  $Z_1=0.8+j0.6$或$X_1=1.2$；
- 异步电机负序阻抗：$X_2=X''=0.2$；
- 综合负荷负序阻抗：$X_2=0.35$；
- 异步电机和综合负荷的零序电抗：$X_0=\infty$。（绕组通常接成三角形和不接地星形）

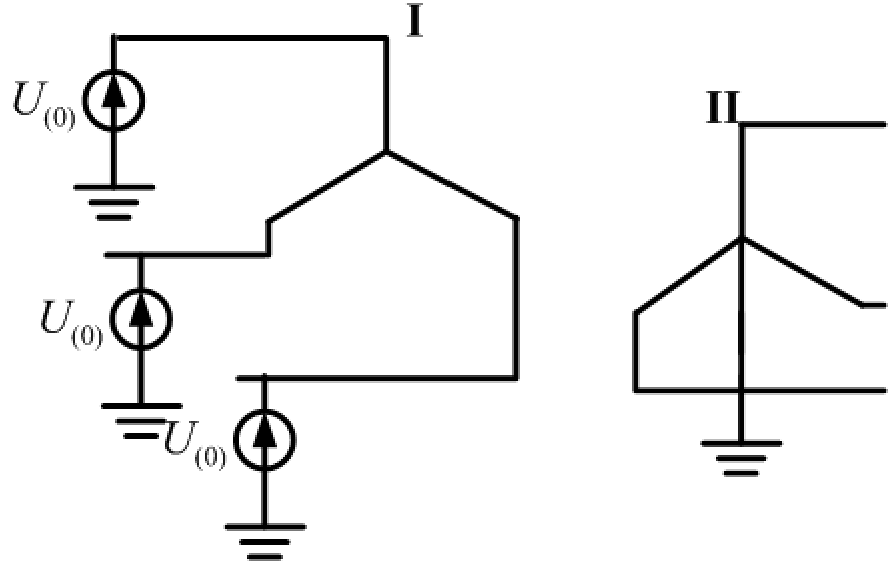
三、变压器的零序电抗及其等效电路

（1）稳态运行时变压器的等值电抗（双绕组变压器即为两个绕组漏抗之和）就是它的正序或负序电抗。

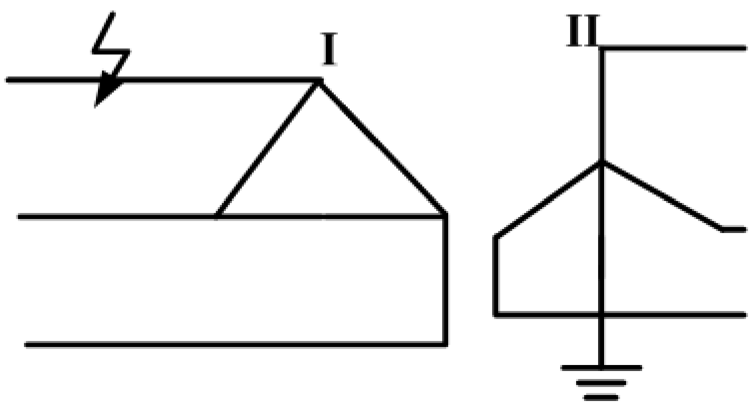
（2）变压器的零序电抗和正序、负序电抗不同。当在变压器端点施加零序电压时，其绕组中有无零序电流，以及零序电流的大小与变压器三绕组的接线方式和变压器的结构密切相关。



零序电抗为：$x_{(0)}=\infty$



零序电抗为：$x_{(0)}=\infty$



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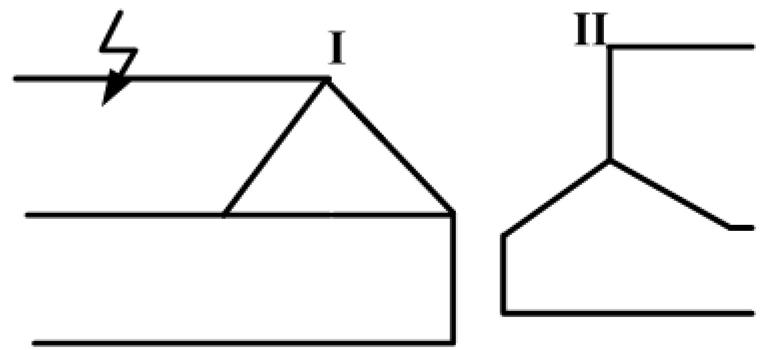
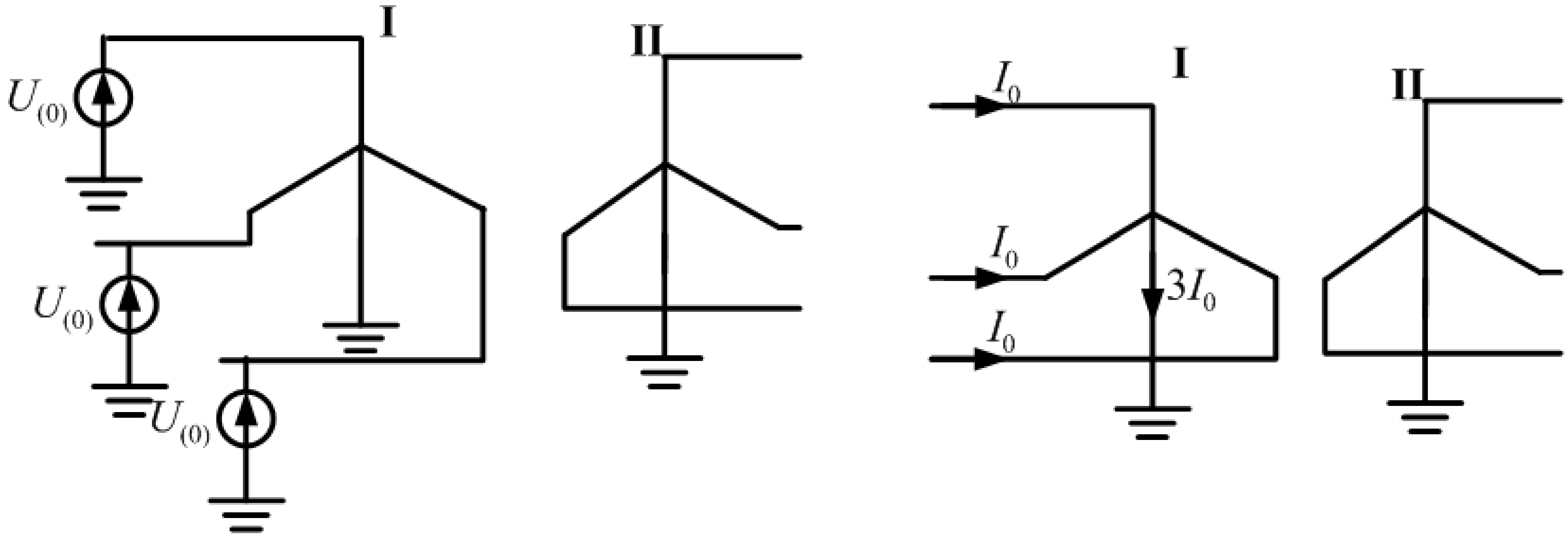
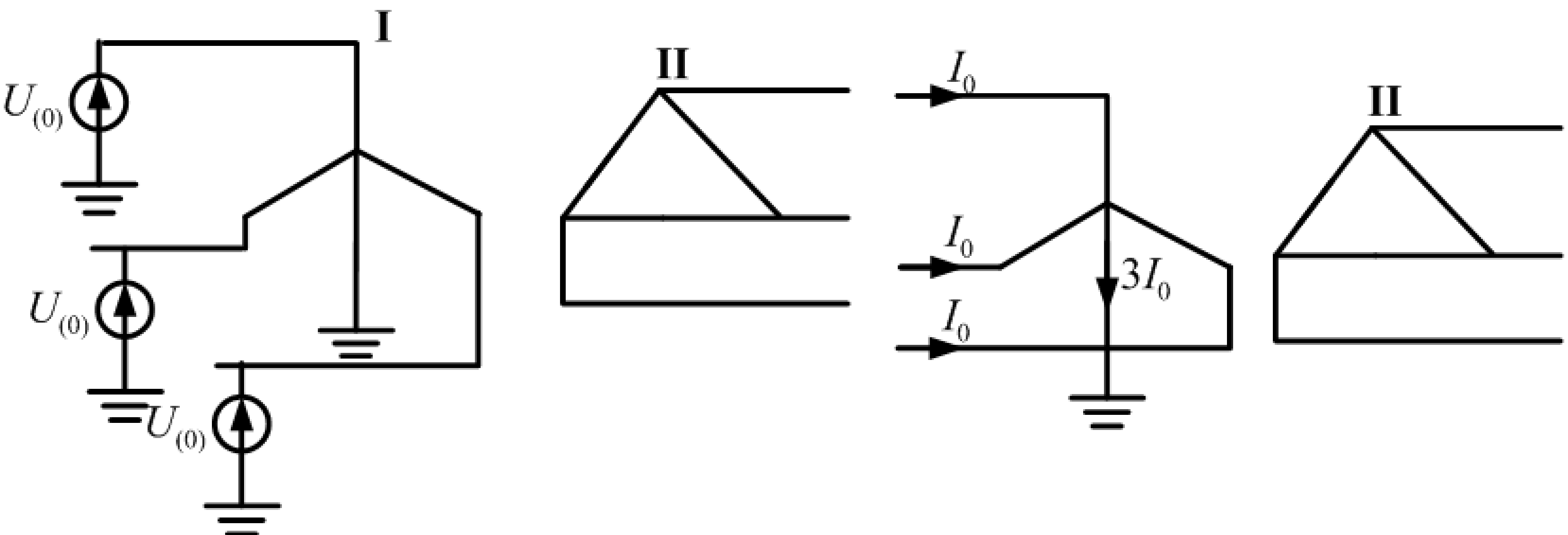


零序电抗为：$x_{(0)}=\infty$





有零序电流流入与短路点相连的绕组！

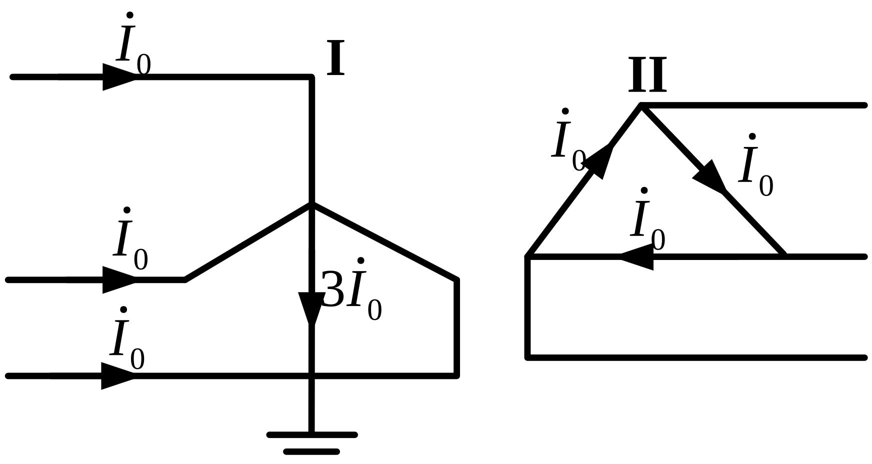
（3）零序电压施加在变压器绕组的三角形侧或不接地星形侧时，无论另一侧绕组的接线方式如何，变压器中都没有零序电流流通。这种情况下，变压器的零序电抗为无穷大。

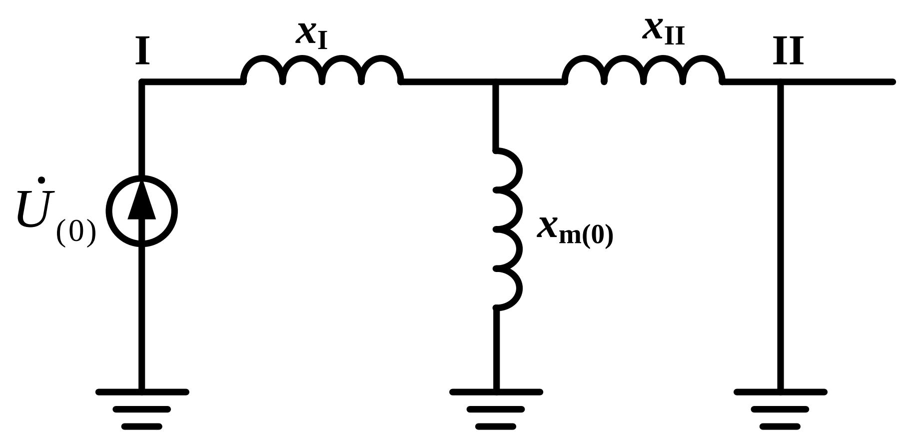
（4）零序电压施加在绕组连接成接地星形一侧时，大小相等、相位相同的零序电流将通过三相绕组经中性点流入大地，构成回路。但在另一侧，零序电流流通的情况则随该侧的接线方式而异。

（一）双绕组变压器

1. Y_0/Δ接线变压器

变压器星形侧流过零序电流时，在三角形侧各绕组中将感应零序电动势。但因零序电流三相大小相等、相位相同，它只在三角形绕组中形成环流，而流不到绕组以外的线路上去 。

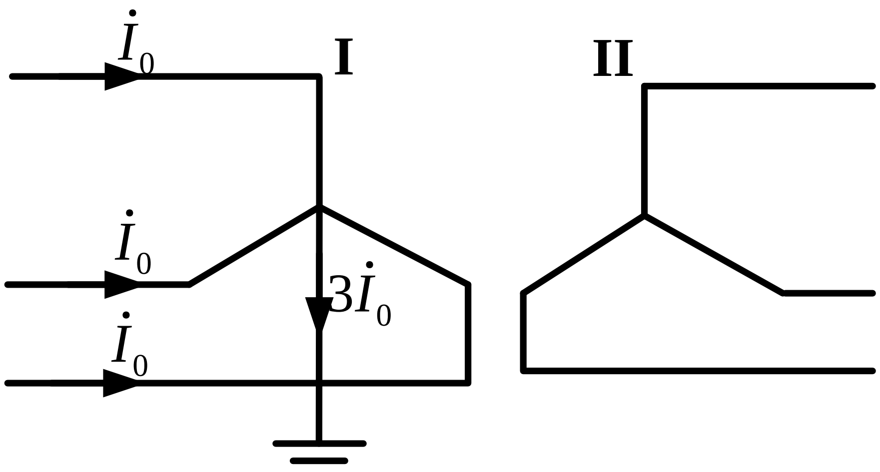


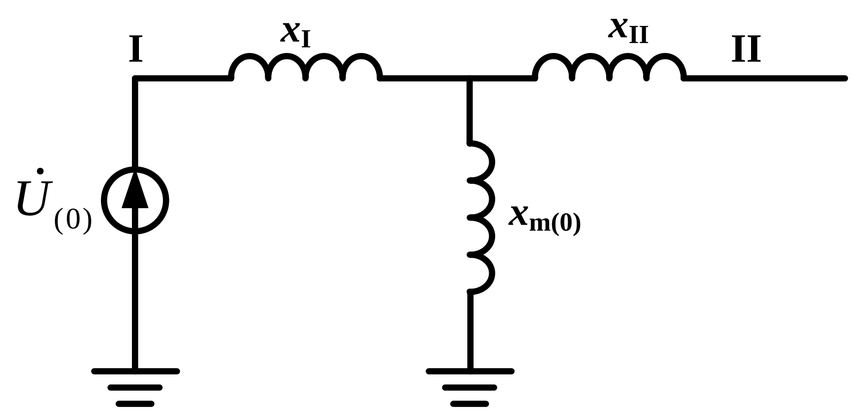


零序电抗为： $x_{(0)} = x_{\mathrm{I}} + \dfrac{x_{\Pi} x_{m(0)}}{x_{\Pi} + x_{m(0)}}$

2. Y_0/Y 接线变压器

变压器接地星形侧流过零序电流，各相绕组中将感应零序电动势。但另一侧中性点不接地，零序电流没有通路，这种情况下变压器相当于空载。

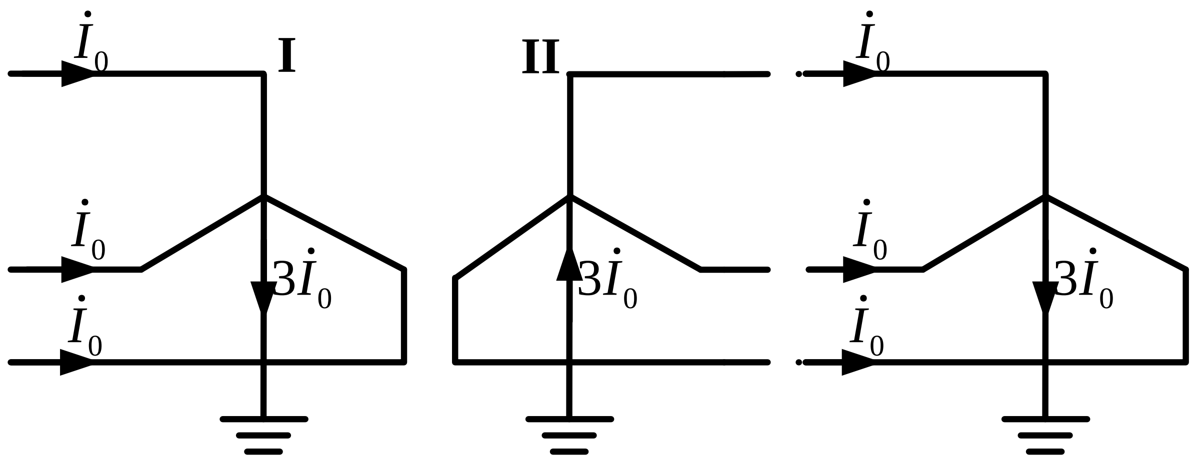


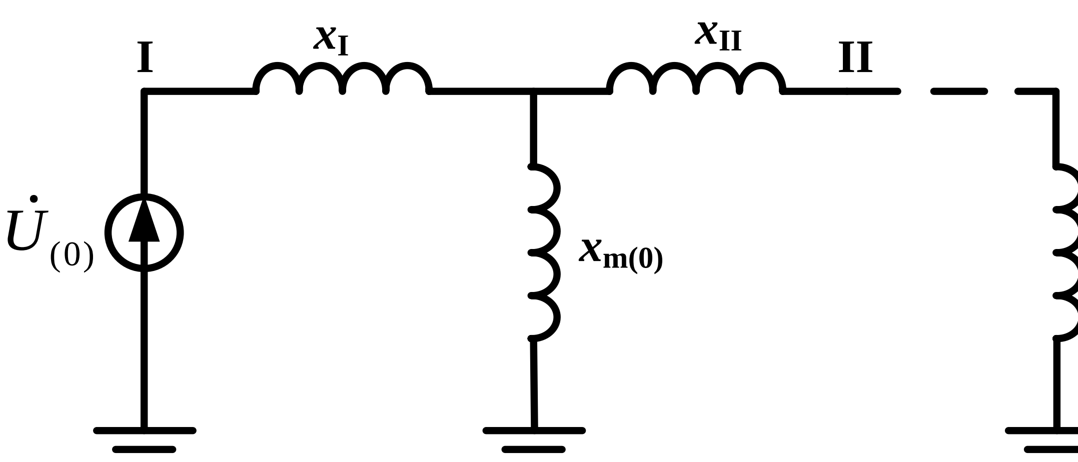


零序电抗为： $X_{(0)} = X_{\mathrm{I}} + X_{m(0)}$

3. Y_0/Y_0 接线变压器

变压器一次星形侧流过零序电流，二次星形侧各绕组中将感应零序电势。如与二次星形侧相连的电路中还有另一个接地中性点，则二次绕组中将有零序电流流通．





$$x_{(0)} = x_{\mathrm{I}} + x_{m(0)} // (x_{II} + x)$$

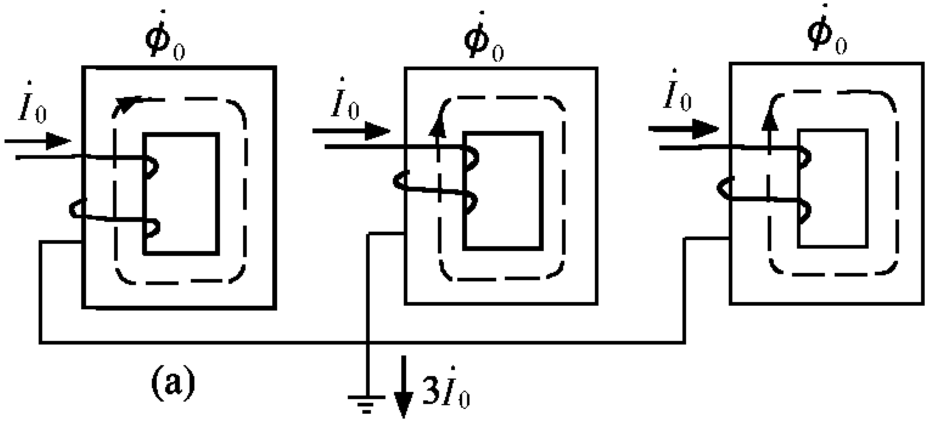
4. 关于激磁电抗 $x_{m(0)}$

正序的激磁电抗都很大。这是由于正序激磁磁通均在铁芯内部，磁阻较小。零序的激磁电抗与变压器的结构有很大关系。

(1) 由三个单相变压器组成的三相变压器，各相磁路独立，正序和零序磁通都按相在本身的铁芯中形成回路，因而各序激磁电抗相等，而且数值很大，近似认为激磁电抗为无限大。

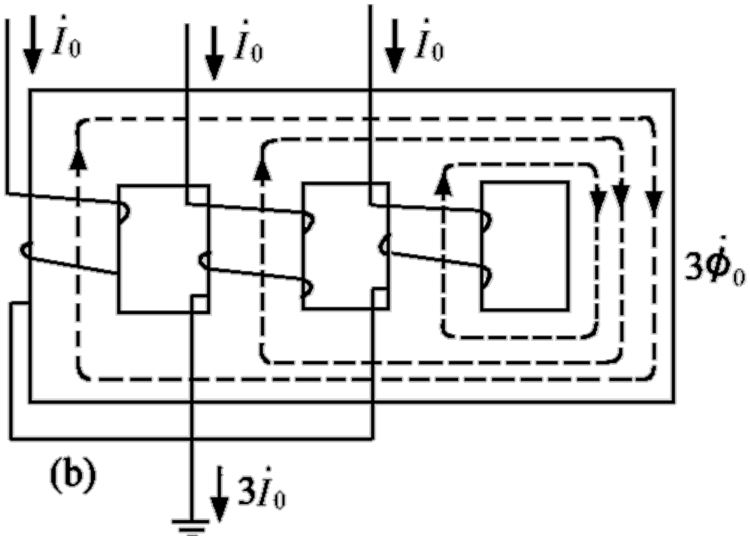
(2) 对于三相五柱式和壳式变压器，零序磁通可以通过没有绕组的铁芯部分形成回路，零序激磁电抗也相当大，也可近似认为无限大。

(3) 三相三柱式变压器的零序激磁电抗较小，其值可用试验方法求得，它的标幺值一般很少超过1.0。



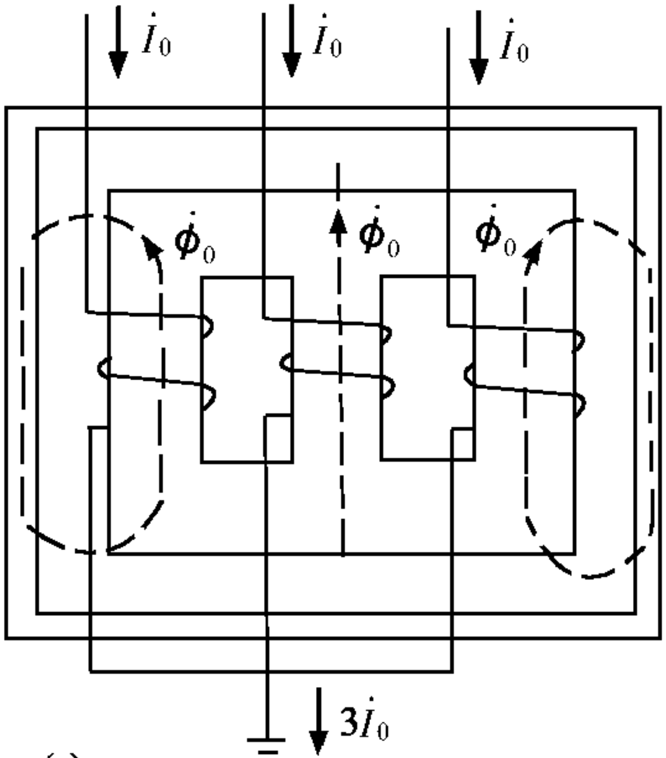
(a)

零序励磁电抗等于正序励磁电抗



(b)

零序励磁电抗等于正序励磁电抗



(c)

零序励磁电抗比正序励磁电抗小得多：

$X_{m(0)}$=0.3~1.0

注意：

(1) 非三相三柱式变压器，$x_{m(0)}=\infty$ 。

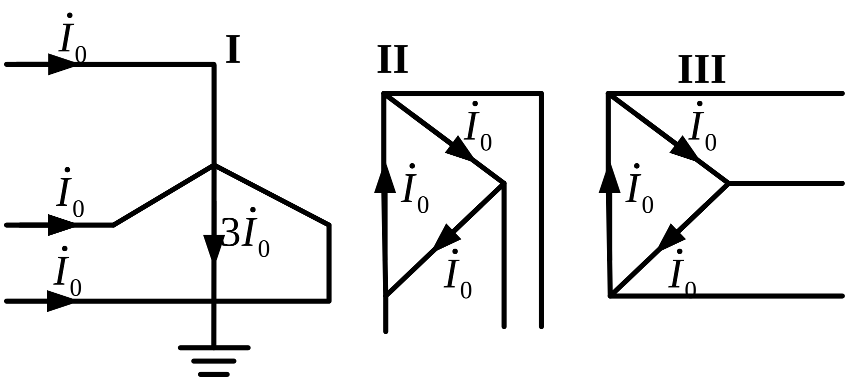
- 当接线为$Y_0/\triangle$和Y_0/Y_0时，$x_{(0)}=x_{\mathrm{I}}+x_{\mathrm{II}}=x_{(1)}$。
- 当接线为Y_0/Y时，$x_{(0)}=\infty$。

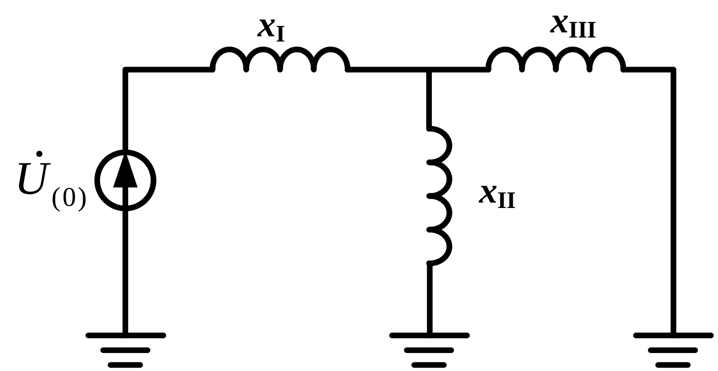
(2) 对于三相三柱式变压器，由于$x_{m(0)}\neq\infty$，需计入 $x_{m(0)}$的具体数值。

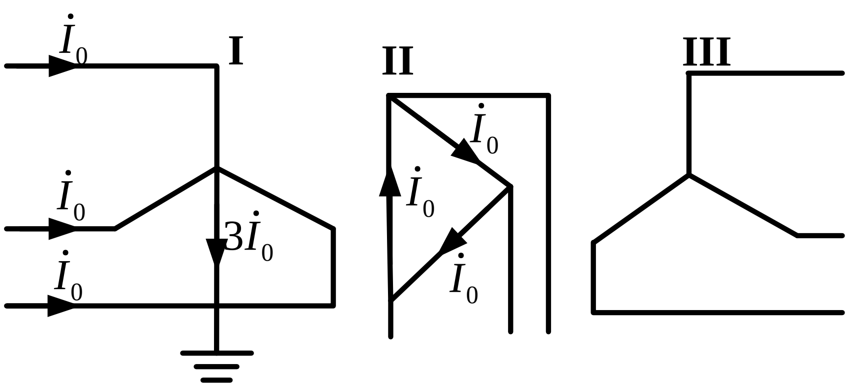
- 当接线为$Y_0/\triangle$时，由于$x_{m(0)}$比x_{II}大很多，实用计算中可取$x_{(0)}\approx x_{(1)}$。

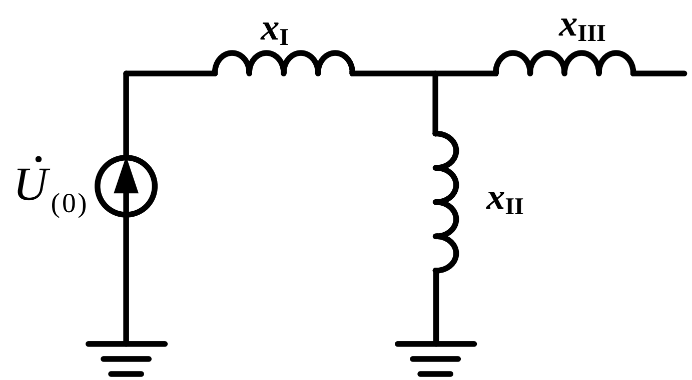
（二）三绕组变压器

1. 在三绕组变压器中，为了消除三次谐波磁通的影响，总有一个绕组是连成三角形的，以提供三次谐波电流的通路。可以不计入$x_{m(0)}$。

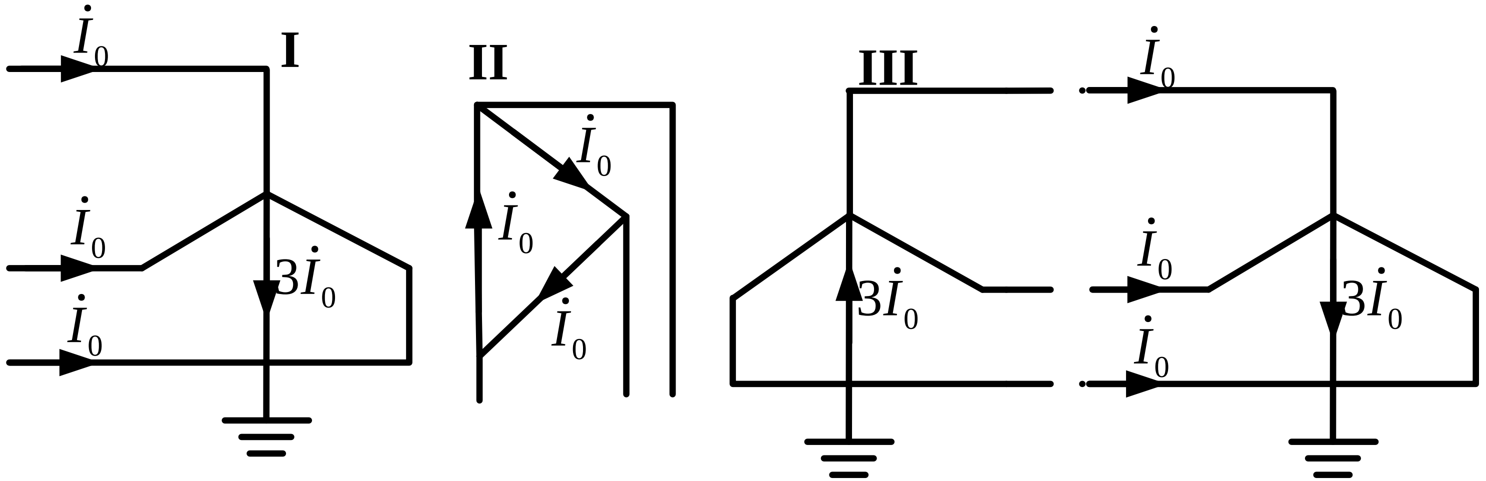




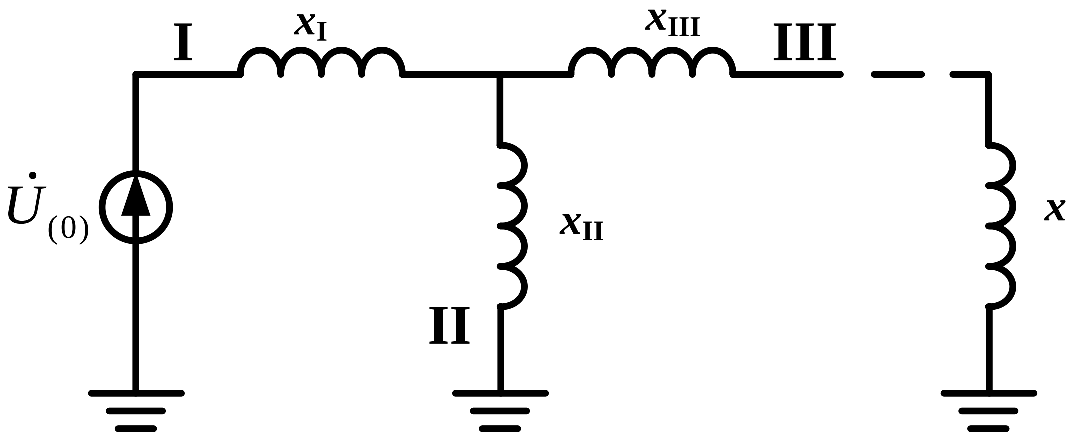




（二）三绕组变压器

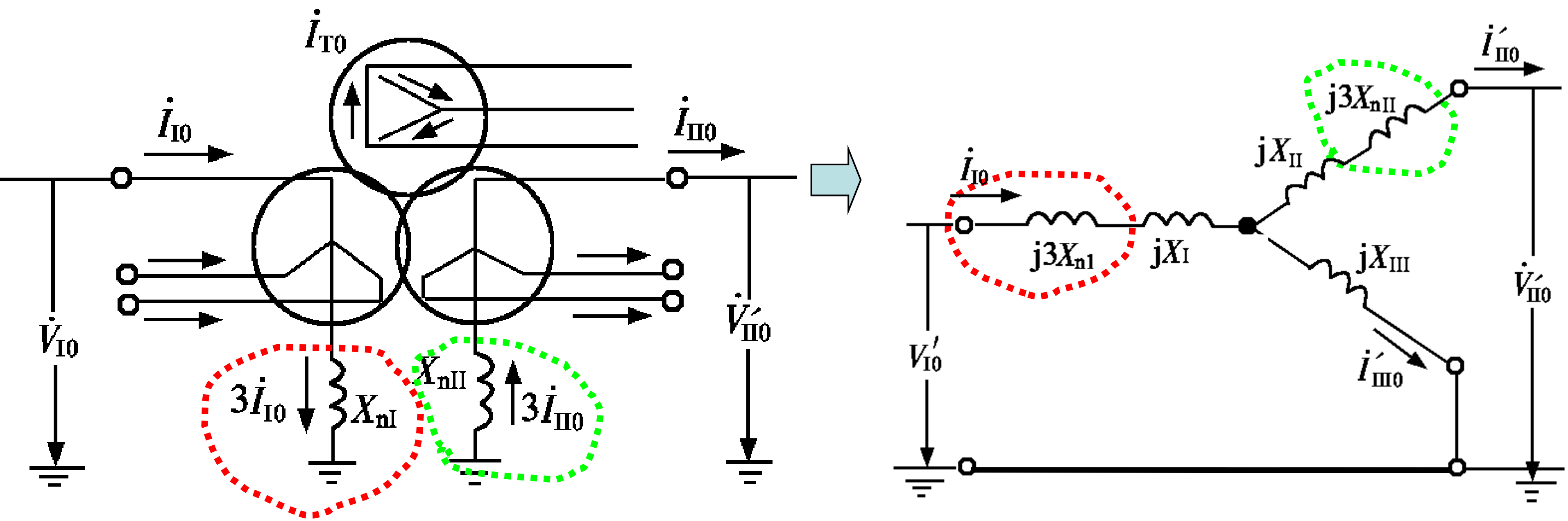


$\dot{I}_0$
I
$\dot{I}_0$
$3\dot{I}_0$
$\dot{I}_0$
II
$\dot{I}_0$
$\dot{I}_0$
$\dot{I}_0$
III
$\dot{I}_0$
$3\dot{I}_0$
$\dot{I}_0$
$3\dot{I}_0$
$\dot{I}_0$



I
x_{I}
x_{III}
III
$\dot{U}_{(0)}$
x_{II}
x
II

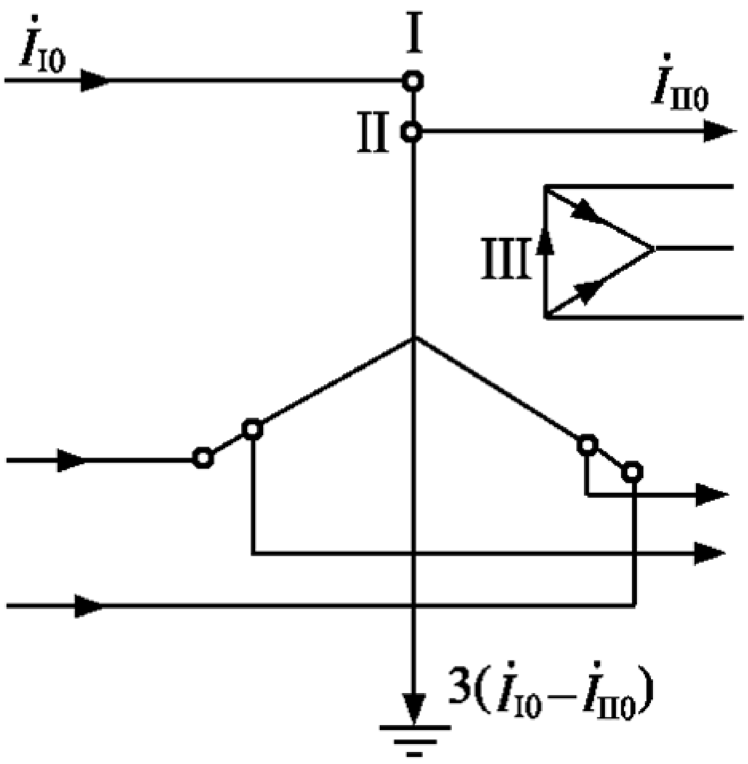
2. 中性点有接地电阻时变压器的零序等值电路

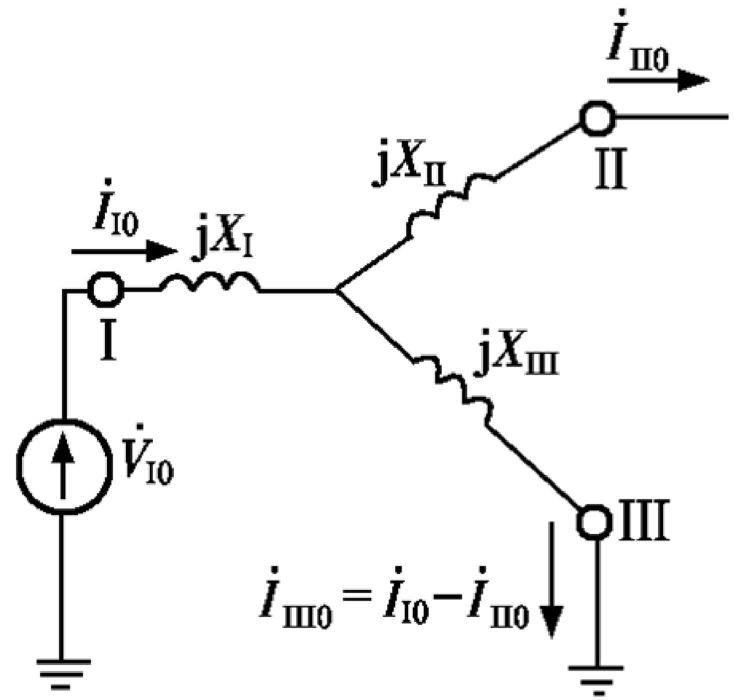


变压器中性点经电抗接地时的零序等值电路

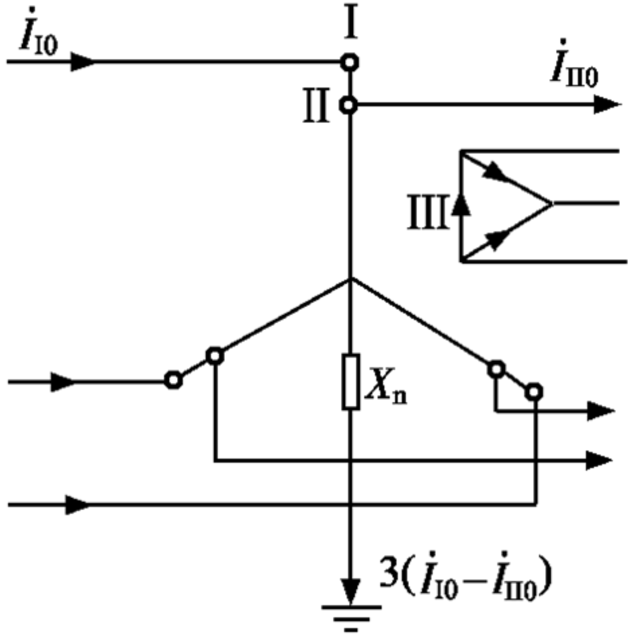
3. 自耦变压器的零序阻抗及其等值电路

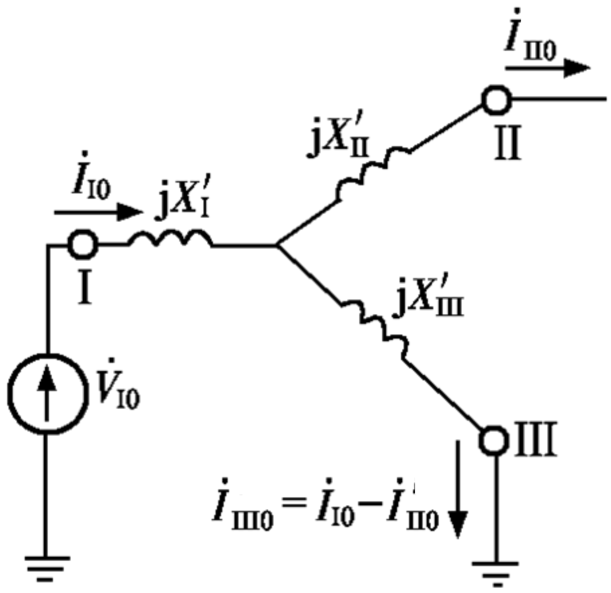
- 中性点直接接地的自耦变压器





中性点经电抗接地的自耦变压器

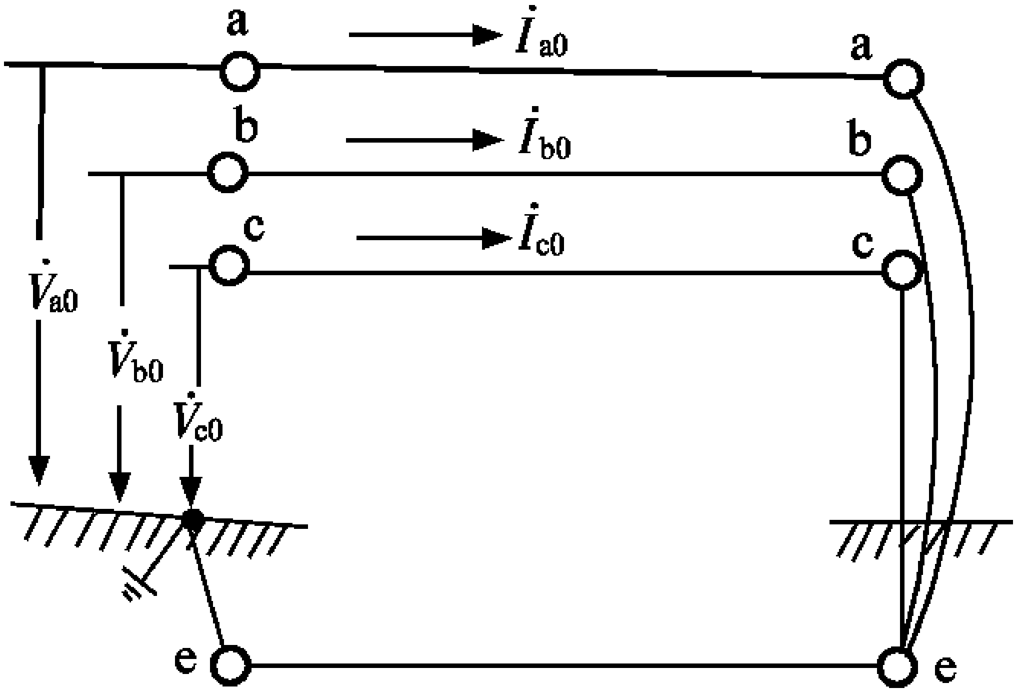




$$\left.\begin{aligned}
X'_{\mathrm{I}} &= X_{\mathrm{I}} + 3X_n(1-k_{12}) \\
X'_{\mathrm{II}} &= X_{\mathrm{II}} + 3X_n k_{12}(k_{12}-1) \\
X'_{\mathrm{III}} &= X_{\mathrm{III}} + 3X_n k_{12}
\end{aligned}\right\}$$

四、架空线路的零序阻抗及其等值电路

- 零序电流必须借助大地或架空地线构成通路

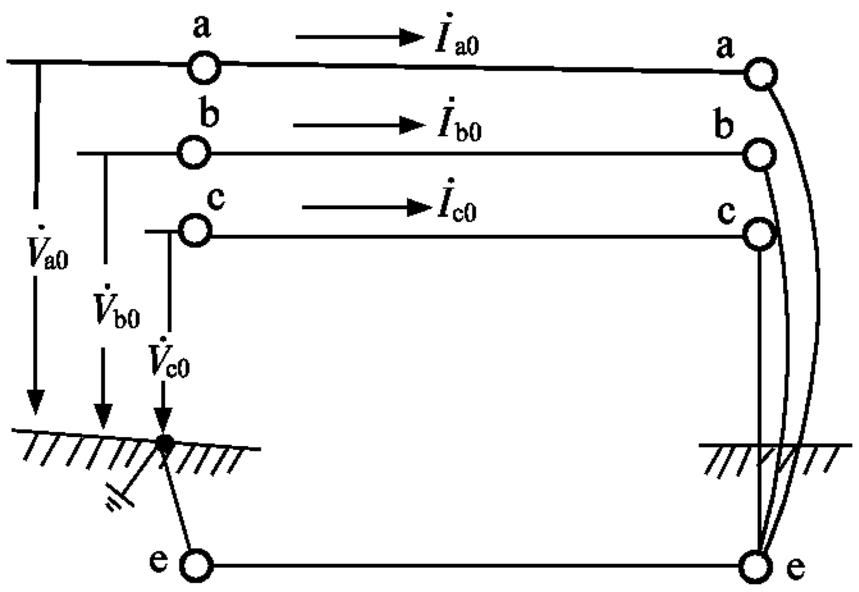


四、架空线路的零序阻抗及其等值电路

- 零序阻抗比正序阻抗大

（1）回路中包含了大地电阻

（2）自感磁通和互感磁通是助增的



(1)、单根导线——大地回路的自阻抗

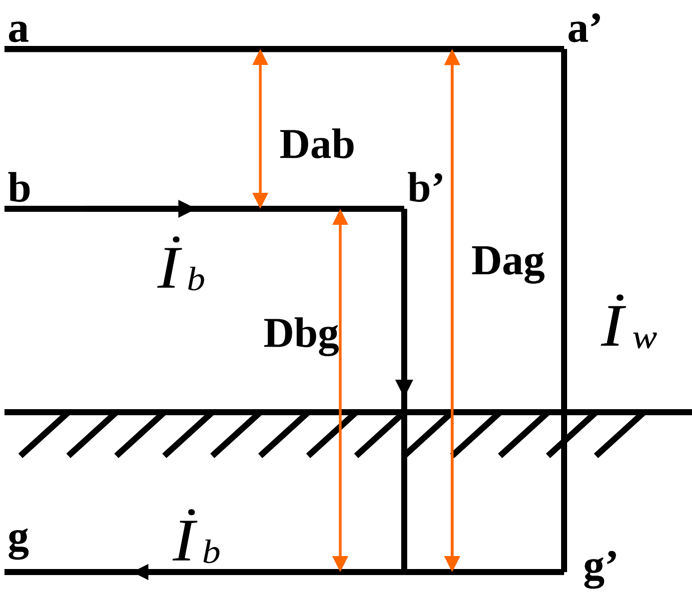
$$Z_s = R_a + R_g + j0.1445\lg\frac{D_g}{r'}(\Omega/km) \qquad D_g = \frac{D_{ag}^2}{r_g}$$

Ra为导线电阻，Rg为大地等值电阻约为0.05Ω/km

r’为导线的等值半径，Dg为等值深度，一般取Dg＝1000m

(2)、两个“导线——大地”回路间的互阻抗

$$Z_m = R_g + j0.1445\lg\frac{D_g}{D_{ab}}(\Omega/km)$$



(3)、单回路架空线的零序阻抗

➢三相不对称排列，互感为：

$$\begin{cases} x_{ab}=0.1445\lg\dfrac{D_g}{D_{ab}} \\ x_{ac}=0.1445\lg\dfrac{D_g}{D_{ac}} \\ x_{bc}=0.1445\lg\dfrac{D_g}{D_{bc}} \end{cases}$$

➢经完全换位后，互感接近相等为：

$$x_m=\frac{1}{3}(x_{ab}+x_{ac}+x_{bc})=0.1445\lg\frac{D_g}{D_m}$$

Dm为几何均距： $D_m=\sqrt[3]{D_{ab}D_{ac}D_{bc}}$

➢每一相的零序阻抗为：

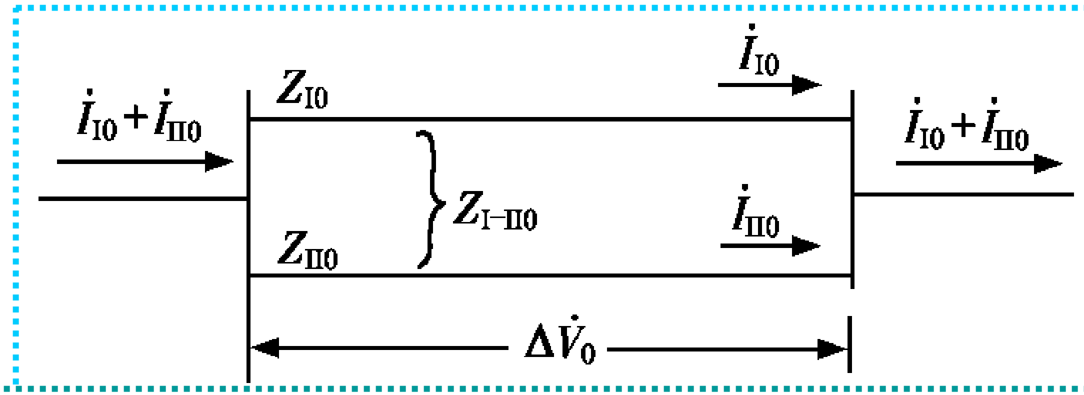
$$Z_0=Z_s+2Z_m=R_a+R_g+j0.1445\lg\frac{D_g}{r'}+2R_g+j2\times0.1445\lg\frac{D_g}{D_m}$$

$$\Rightarrow Z_0=R_a+3R_g+j0.4335\lg\frac{D_g}{D_s}$$

Ds为组合导线的等值半径： $D_s=\sqrt[3]{r' D_m^2}$

四、架空线路的零序阻抗及其等值电路

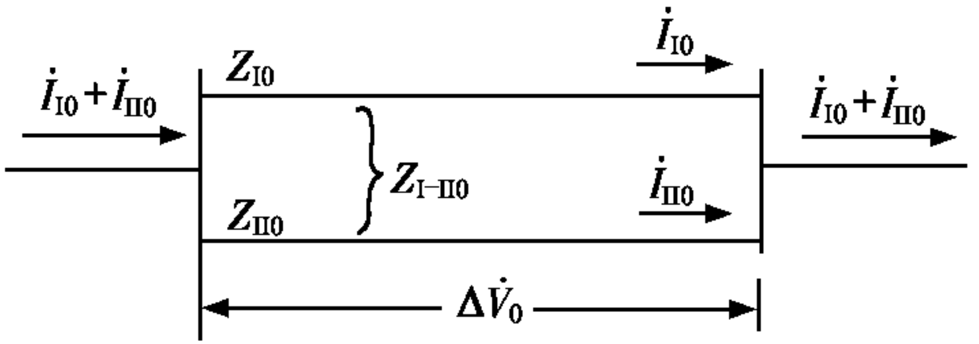
(4) 平行架设双回线零序等值电路：两个回路互感产生助磁作用，零序阻抗进一步增大



➢双回路等值电路：

$$\Delta\dot{U}_0 = Z_{I0}\dot{I}_{I0} + Z_{I-II0}\dot{I}_{II0} = \left(Z_{I0} - Z_{I-II0}\right)\dot{I}_{I0} + Z_{I-II0}\left(\dot{I}_{I0} + \dot{I}_{II0}\right)$$

$$\Delta\dot{U}_0 = Z_{II0}\dot{I}_{II0} + Z_{I-II0}\dot{I}_{I0} = \left(Z_{II0} - Z_{I-II0}\right)\dot{I}_{II0} + Z_{I-II0}\left(\dot{I}_{I0} + \dot{I}_{II0}\right)$$



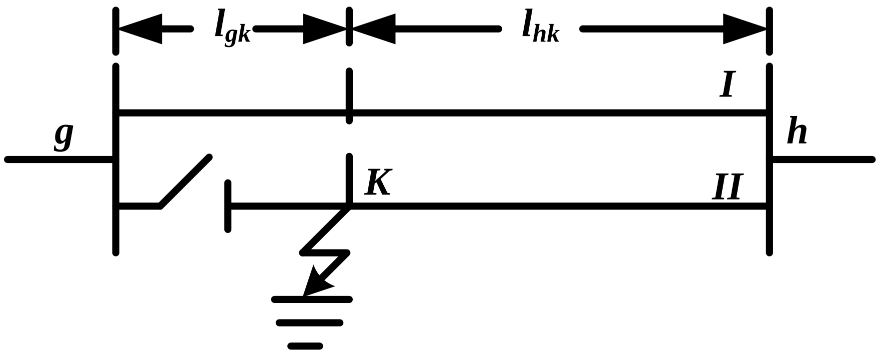
如果两个回路完全相同，

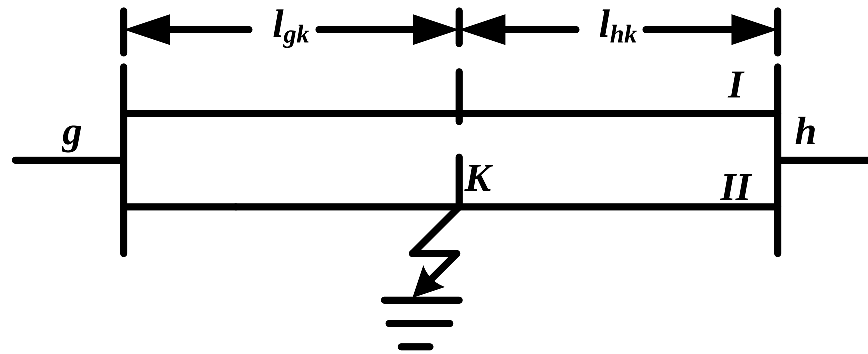
$$z_{I(0)} = z_{II(0)} = z_{(0)}$$

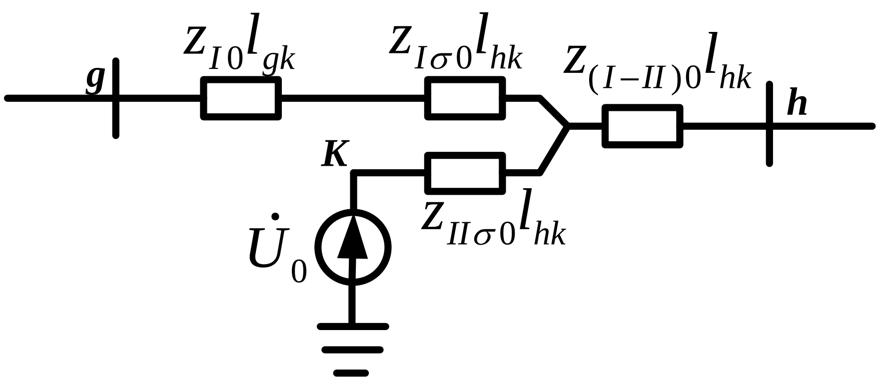
则每一回路的零序阻抗为：

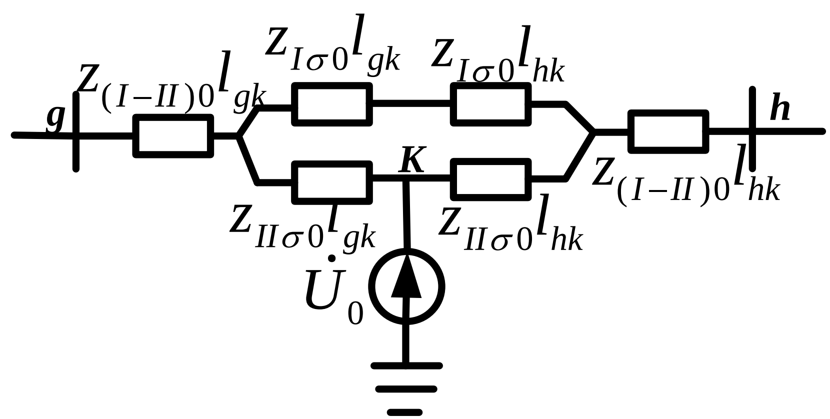
$$z^{(2)}_{(0)} = z_{(0)} + z_{I-II(0)}$$

➢ 如果不对称故障发生在线路中间，则可对于故障点两侧应用等值电路。







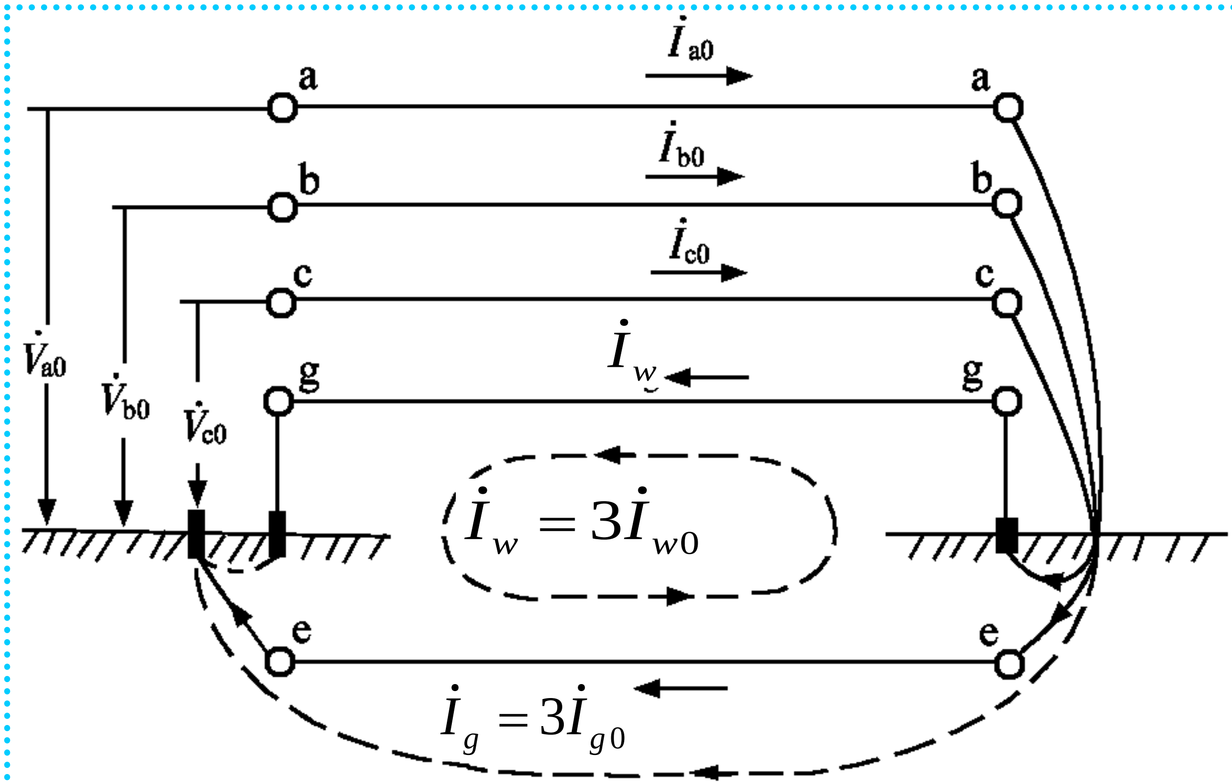


$$Z_{I\sigma 0} = Z_{I0} - Z_{I-II0}$$

$$Z_{II\sigma 0} = Z_{II0} - Z_{I-II0}$$

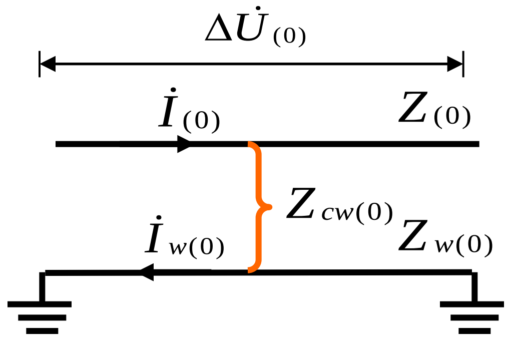
四、架空线路的零序阻抗及其等值电路

- 有架空地线的情况：架空地线对导线起去磁作用，零序阻抗有所减小。



➤架空地线的自阻抗为：

$$Z_{w(0)} = 3R_w + 0.15 + j0.4335\lg\frac{D_g}{r_w'}(\Omega / km)$$



➤三相导线与架空地线的互阻抗为：

$$Z_{cw(0)} = 0.15 + j0.4335\lg\frac{D_g}{D_{c-w}}(\Omega / km)$$

➤等值电路：

$$\Delta\dot{U}_{(0)} = Z_{(0)}\dot{I}_{(0)} - Z_{cw(0)}\dot{I}_{w(0)}$$

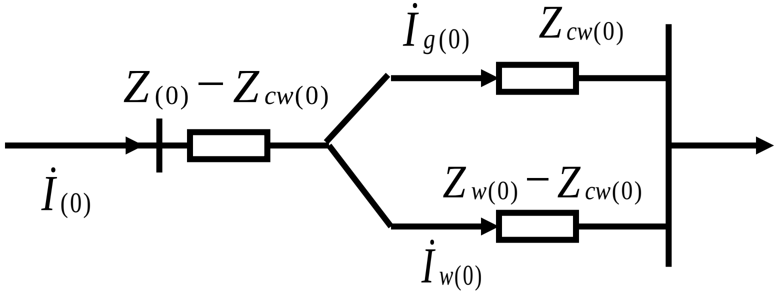
$$0 = Z_{w(0)}\dot{I}_{w(0)} - Z_{cw(0)}\dot{I}_{(0)}$$

$$\Rightarrow \Delta\dot{U}_{(0)} = \left(Z_{(0)} - \frac{Z_{cw(0)}^2}{Z_{w(0)}}\right)\dot{I}_{(0)}$$

➤地线去磁，零序阻抗减少

$$Z_{(0)}^{(w)} = Z_{(0)} - \frac{Z_{cw(0)}^2}{Z_{w(0)}}$$

$$= Z_{(0)} - Z_{cw(0)} + \frac{Z_{cw(0)}(Z_{w(0)} - Z_{cw(0)})}{Z_{cw(0)} + Z_{w(0)} - Z_{cw(0)}}$$



四、架空线路的零序阻抗及其等值电路

➢ 实用计算中一相等值零序电抗

无架空地线的单回线路 ⟶ $x_0 = 3.5x_1$

有钢质架空地线的单回线路 ⟶ $x_0 = 3x_1$

有良导体架空地线的单回线路 ⟶ $x_0 = 2x_1$

无架空地线的双回线路 ⟶ $x_0 = 5.5x_1$

有钢质架空地线的双回线路 ⟶ $x_0 = 4.7x_1$

有良导体架空地线的双回线路 ⟶ $x_0 = 3x_1$

五、电力系统各序网络

- **等值电路的绘制原则**

根据电力系统的原始资料，在故障点分别施加各序电势，从故障点开始，查明各序电流的流通情况，凡是某序电流能流通的元件，必须包含在该序网络中，并用相应的序参数及等值电路表示。

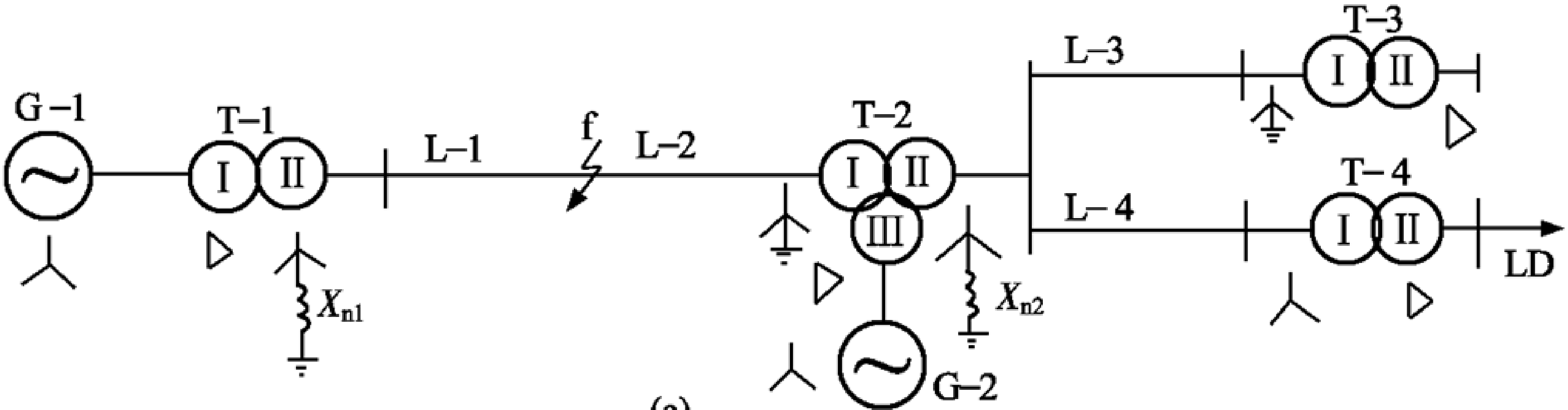
五、电力系统各序网络

1. 正序网
- 正序电势＝发电机电动势
- 正序阻抗＝元件对称参数

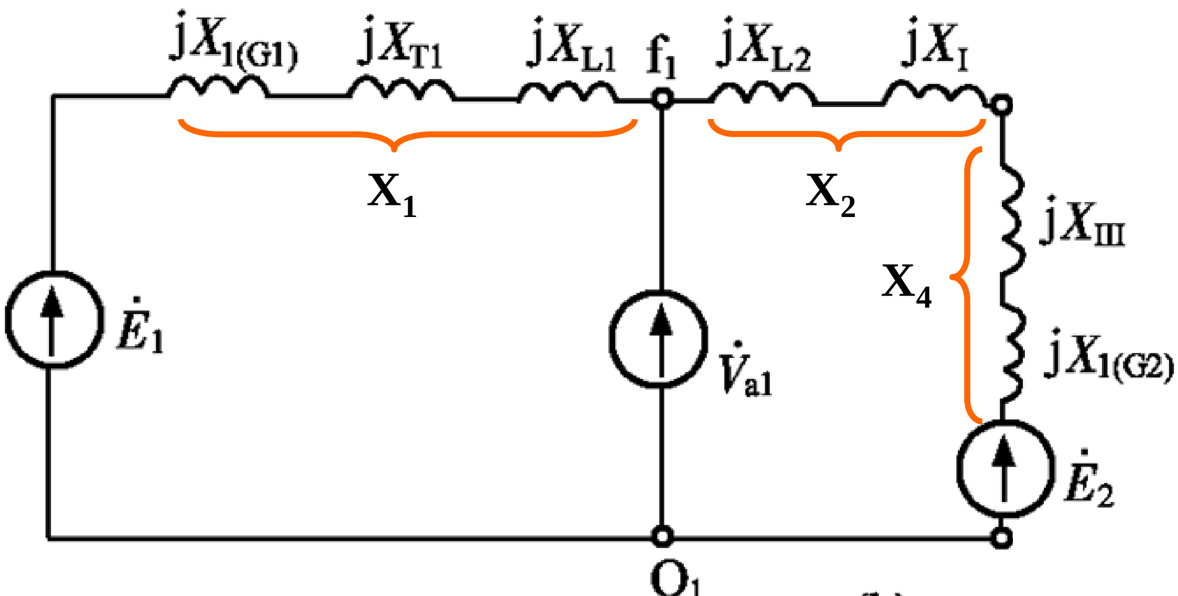
2. 负序网
- 无电源电动势
- 正序回路与负序回路相同，但旋转元件的正、负序参数不同

3. 零序网
- 无电源电动势
- 零序网与正、负序网差异很大
- 某元件零序阻抗的有无，取决与零序电流能否流过它

零序电流如何流通和网络的结构（变压器接线及中性点接地方式）有关



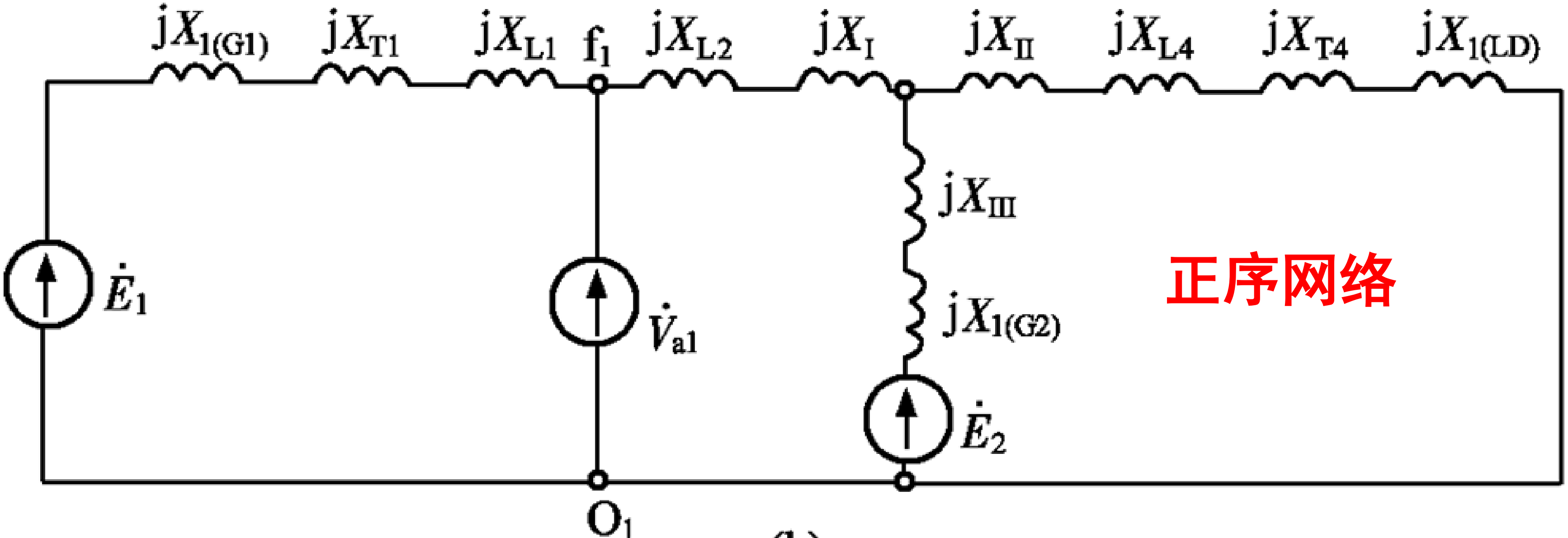
(a)



(b)

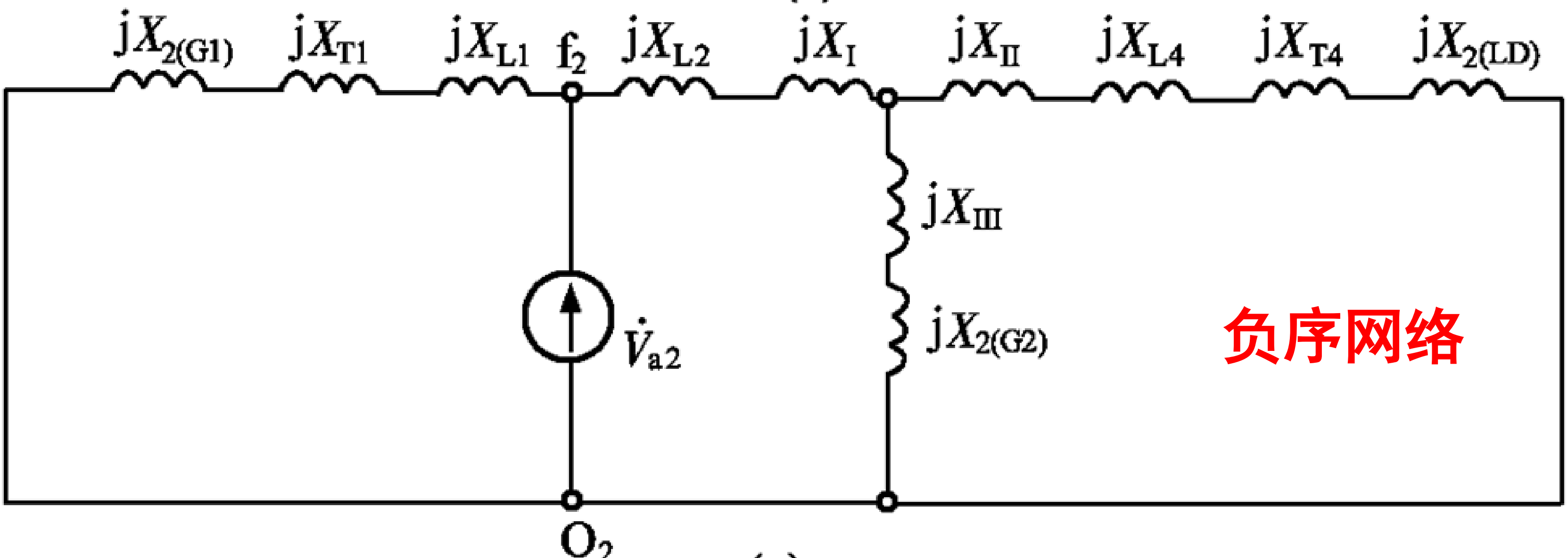
正序网络

$$X_{1\Sigma}=\left(X_3 // X_4+X_2\right) // X_1$$



$jX_{1(G1)}$
jX_{T1}
jX_{L1}
f_1
jX_{L2}
jX_{I}
jX_{II}
jX_{L4}
jX_{T4}
$jX_{1(LD)}$
jX_{III}
$jX_{1(G2)}$
$\dot{E}_1$
$\dot{V}_{a1}$
$\dot{E}_2$
O_1
正序网络

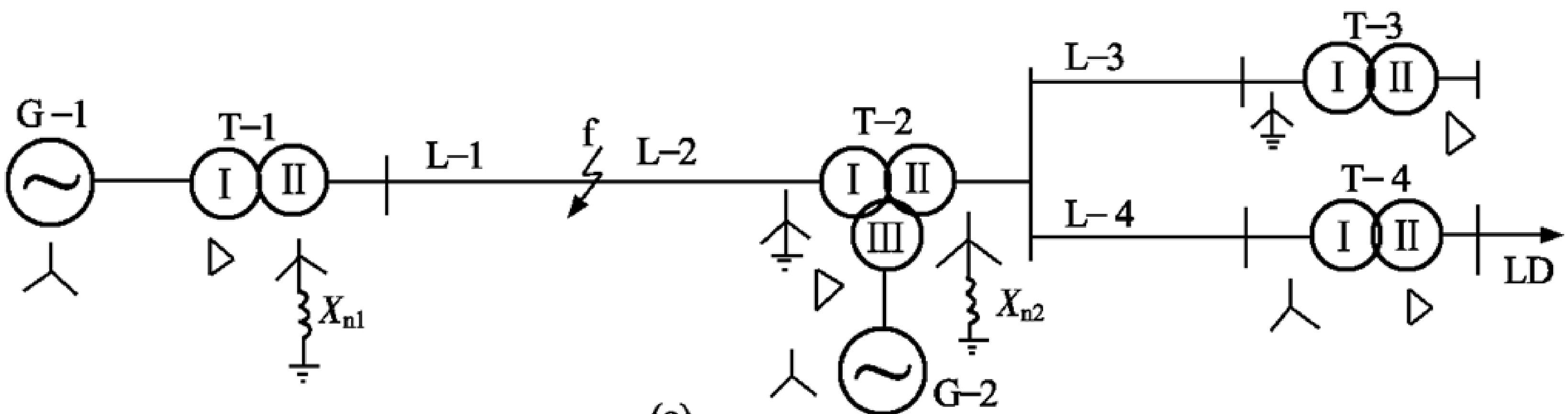
(b)



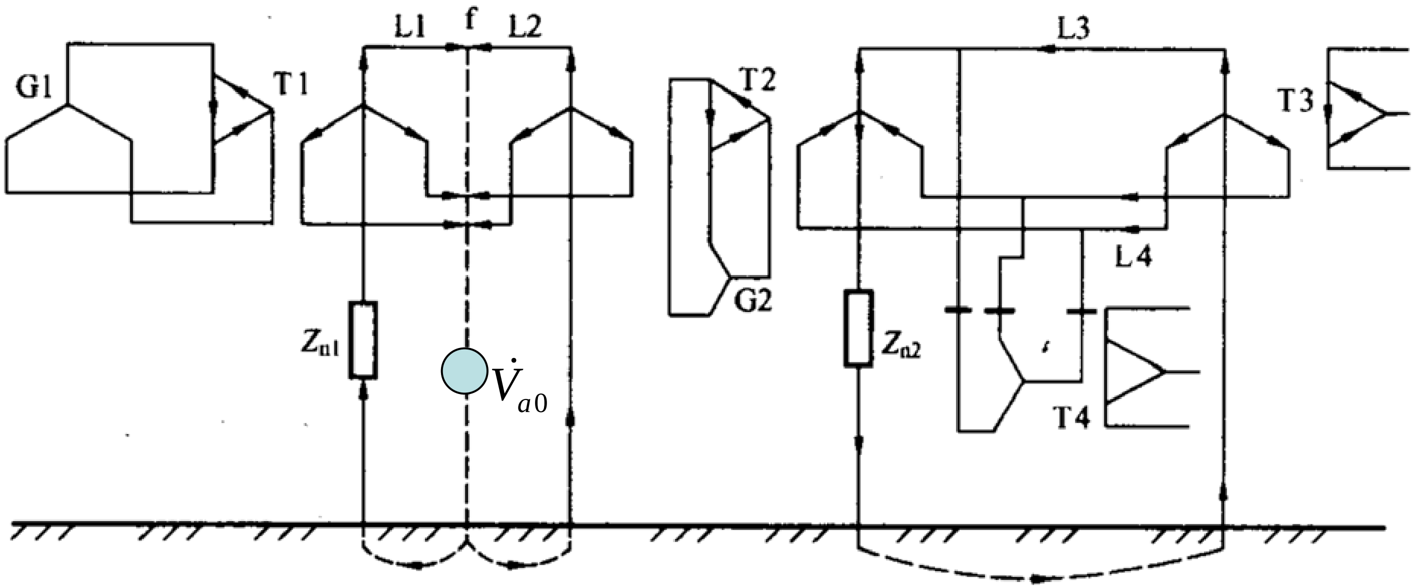
$jX_{2(G1)}$
jX_{T1}
jX_{L1}
f_2
jX_{L2}
jX_{I}
jX_{II}
jX_{L4}
jX_{T4}
$jX_{2(LD)}$
jX_{III}
$jX_{2(G2)}$
$\dot{V}_{a2}$
O_2
负序网络

(c)

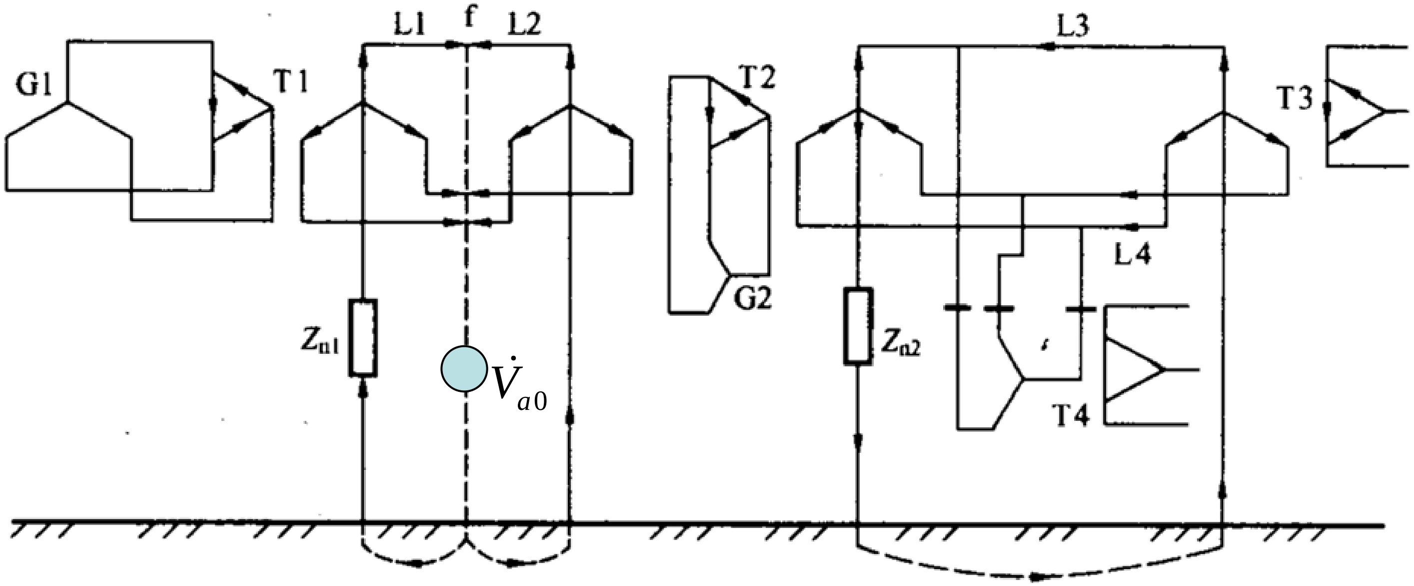
零序网络：必须首先确定零序电流的流通路径。



(a)

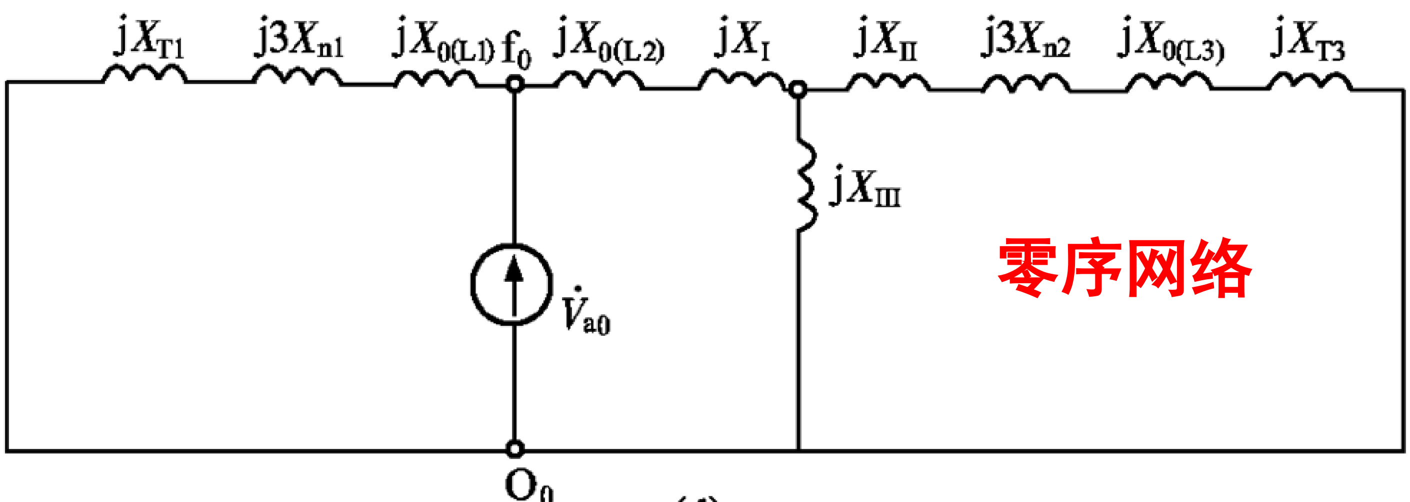


(b)



L1
f
L2
L3
G1
T1
T2
T3
G2
L4
Z_{n1}
Z_{n2}
$\dot{V}_{a0}$
T4

(b)



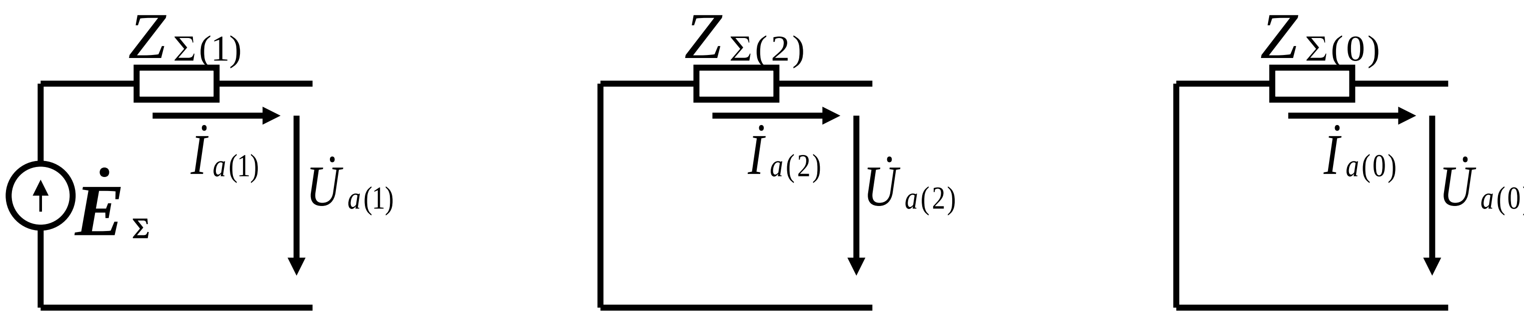
jX_{T1}
$j3X_{n1}$
$jX_{0(L1)}$
f_0
$jX_{0(L2)}$
jX_{I}
jX_{II}
$j3X_{n2}$
$jX_{0(L3)}$
jX_{T3}
jX_{III}
$\dot{V}_{a0}$
O_0

零序网络

(d)

8.3 不对称短路时故障处的短路电流和电压

- 选定参考相——以特殊相为参考相
- 首先制定各序网络



$\dot{E}_{\Sigma}-Z_{\Sigma(1)}\dot{I}_{a(1)}=\dot{U}_{a(1)}$ $-Z_{\Sigma(2)}\dot{I}_{a(2)}=\dot{U}_{a(2)}$ $-Z_{\Sigma(0)}\dot{I}_{a(0)}=\dot{U}_{a(0)}$

- 六个序分量，三个等值电路，还需三个式子。根据故障类型，列出故障点的边界条件（用序分量表示）

8.3 不对称短路时故障处的短路电流和电压

- 求故障点电压和电流的各序分量

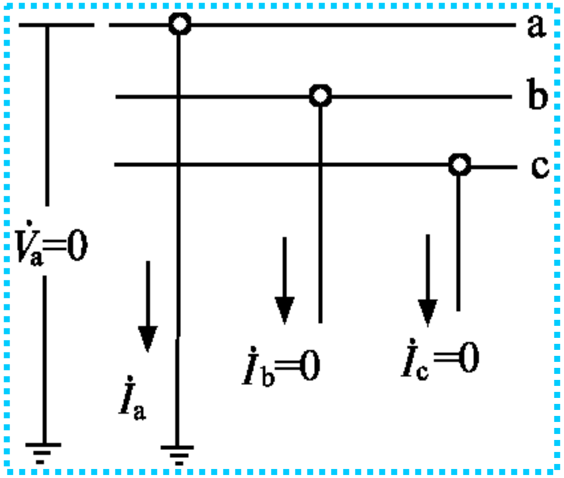
$$\dot{U}_{a(1)}, \dot{U}_{a(2)}, \dot{U}_{a(0)}, \dot{I}_{a(1)}, \dot{I}_{a(2)}, \dot{I}_{a(0)}$$

解析法——联立求解上述六个方程

复合序网法——由边界条件将三序网连成复合序网

- 求故障点电压和电流各相的量。

一、单相接地短路



1、边界条件

$$\left.\begin{aligned}\dot{V}_a &= 0\\ \dot{I}_b &= 0\\ \dot{I}_c &= 0\end{aligned}\right\} \Rightarrow \left.\begin{aligned}\dot{V}_{a1}+\dot{V}_{a2}+\dot{V}_{a0} &= 0\\ a^2\dot{I}_{a1}+a\dot{I}_{a2}+\dot{I}_{a0} &= 0\\ a\dot{I}_{a1}+a^2\dot{I}_{a2}+\dot{I}_{a0} &= 0\end{aligned}\right\}$$

$$\Downarrow$$

$$\left.\begin{aligned}\dot{V}_{a1}+\dot{V}_{a2}+\dot{V}_{a0} &= 0\\ \dot{I}_{a1}=\dot{I}_{a2}=\dot{I}_{a0}\end{aligned}\right\}$$

$$\left.\begin{aligned}\dot{E}_{\Sigma}-jX_{1\Sigma}\dot{I}_{a1} &= \dot{V}_{a1}\\ -jX_{2\Sigma}\dot{I}_{a2} &= \dot{V}_{a2}\\ -jX_{0\Sigma}\dot{I}_{a0} &= \dot{V}_{a0}\end{aligned}\right\}$$

$$\Leftarrow$$

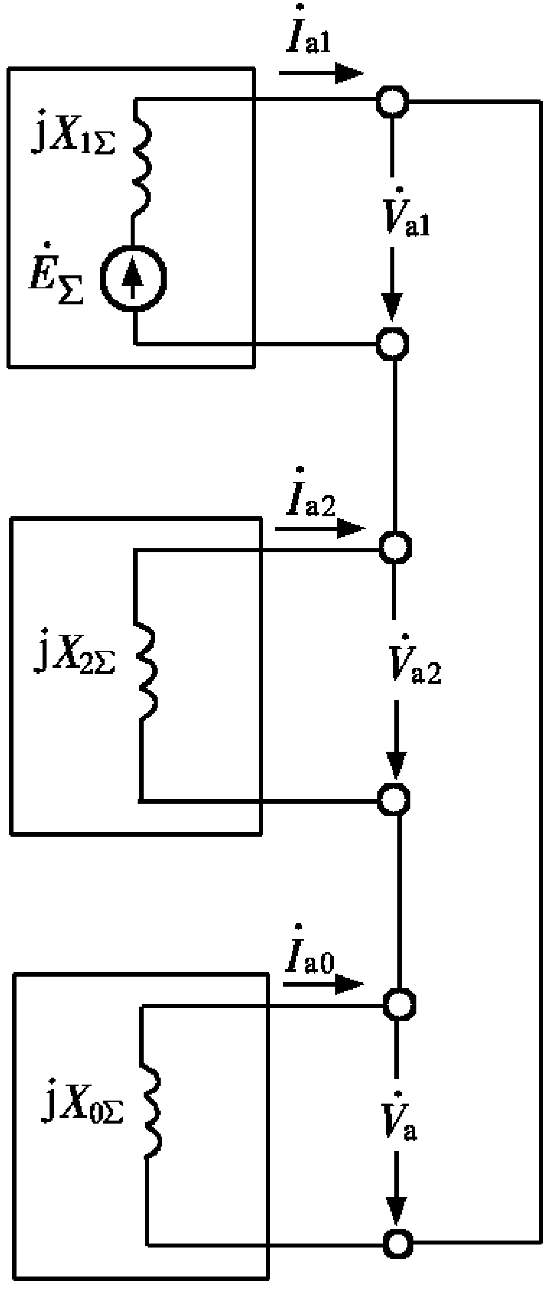
$$\dot{I}_{a1}=\frac{\dot{E}_{\Sigma}}{j(X_{1\Sigma}+X_{2\Sigma}+X_{0\Sigma})}$$

$$\left.\begin{aligned}&\dot{I}_{a2}=\dot{I}_{a0}=\dot{I}_{a1}\\ &\dot{V}_{a1}=\dot{E}_{\Sigma}-jX_{1\Sigma}\dot{I}_{a1}=j(X_{2\Sigma}+X_{0\Sigma})\dot{I}_{a1}\\ &\dot{V}_{a2}=-jX_{2\Sigma}\dot{I}_{a1}\\ &\dot{V}_{a0}=-jX_{0\Sigma}\dot{I}_{a1}\end{aligned}\right\}$$

2、单相接地故障的复合序网

$$\left.\begin{aligned}&\dot{V}_{a1}+\dot{V}_{a2}+\dot{V}_{a0}=0\\&\dot{I}_{a1}=\dot{I}_{a2}=\dot{I}_{a0}\end{aligned}\right\}$$

$$\left.\begin{aligned}&\dot{E}_{\Sigma}-jX_{1\Sigma}\dot{I}_{a1}=\dot{V}_{a1}\\&-jX_{2\Sigma}\dot{I}_{a2}=\dot{V}_{a2}\\&-jX_{0\Sigma}\dot{I}_{a0}=\dot{V}_{a0}\end{aligned}\right\}$$



$$\dot{I}_{a1}=\frac{\dot{E}_{\Sigma}}{j(X_{1\Sigma}+X_{2\Sigma}+X_{0\Sigma})}$$

$$\left.\begin{aligned}&\dot{I}_{a2}=\dot{I}_{a0}=\dot{I}_{a1}\\&\dot{V}_{a1}=\dot{E}_{\Sigma}-jX_{1\Sigma}\dot{I}_{a1}=j(X_{2\Sigma}+X_{0\Sigma})\dot{I}_{a1}\\&\dot{V}_{a2}=-jX_{2\Sigma}\dot{I}_{a1}\\&\dot{V}_{a0}=-jX_{0\Sigma}\dot{I}_{a1}\end{aligned}\right\}$$

3、单相接地的短路电流和短路点非故障相电压

$$\dot{I}_f^{(1)} = \dot{I}_a = \dot{I}_{a1} + \dot{I}_{a2} + \dot{I}_{a0} = 3\dot{I}_{a1}$$

不计电阻影响，设三相短路电流为 $\dot{I}^{(3)}$ ， $X_{1\Sigma} \approx X_{2\Sigma}$ $k_0 = X_{0\Sigma}/X_{1\Sigma}$

$$\dot{I}_f^{(1)} = 3\dot{I}_{a1} = \frac{3\dot{E}_\Sigma}{Z_{1\Sigma} + Z_{2\Sigma} + Z_{0\Sigma}} \approx \frac{3\dot{I}^{(3)}}{2 + k_0}$$

结论： $k_0 \le 1, I_f^{(1)} \ge I^{(3)};\quad k_0 > 1, I_f^{(1)} < I^{(3)}$

$$\begin{aligned}\dot{V}_{fb} &= a^2\dot{V}_{a1} + a\dot{V}_{a2} + \dot{V}_{a0} \\ &= a^2\left(\dot{U}_{fa|0|} - j\dot{I}_{a1}X_{1\Sigma}\right) \\ &\quad + a\left(-j\dot{I}_{a2}X_{2\Sigma}\right) - j\dot{I}_{a0}X_{0\Sigma} \\ &= \dot{U}_{fb|0|} - j\dot{I}_{a1}\left(X_{0\Sigma} - X_{1\Sigma}\right)\end{aligned}$$

$$\dot{V}_{fb} = \dot{V}_{fb|0|} - \dot{V}_{fa|0|}\frac{k_0 - 1}{2 + k_0}$$

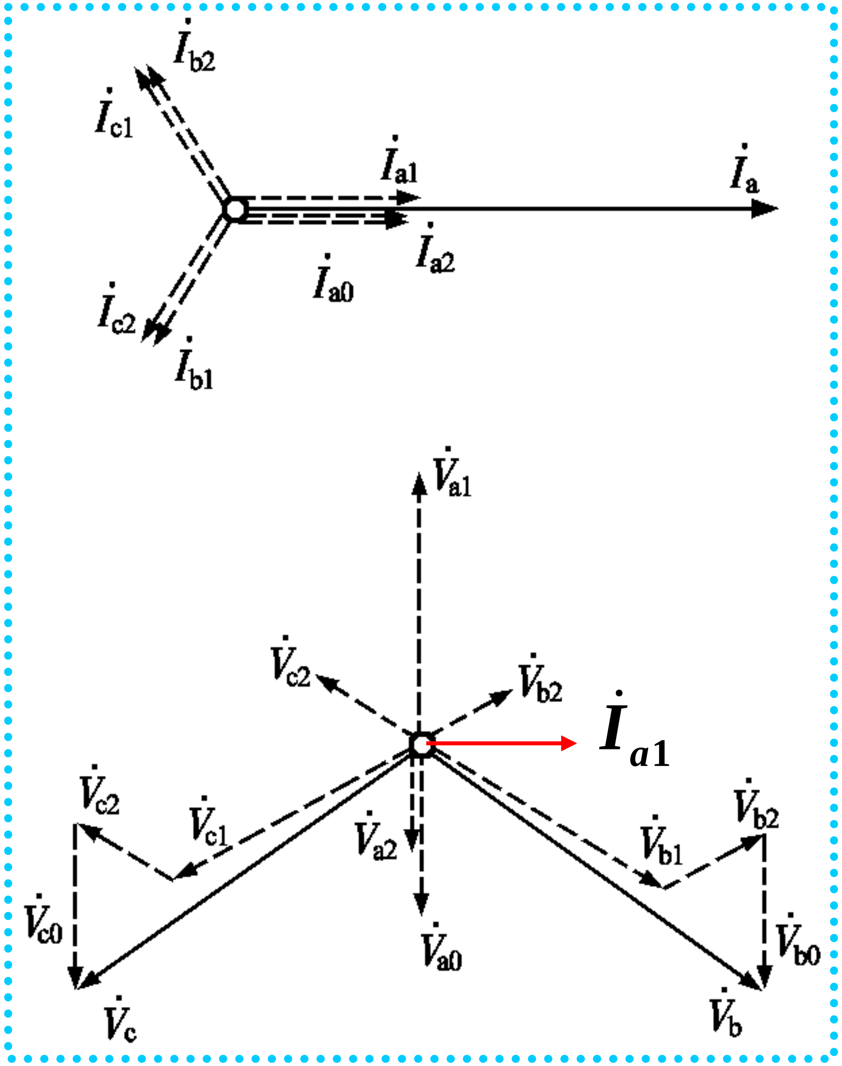
$$k_0 = 0, \dot{V}_{fb} = \frac{\sqrt{3}}{2}\dot{V}_{fb|0|}\angle 30^\circ$$

$$k_0 = 1, \dot{V}_{fb} = \dot{V}_{fb|0|}$$

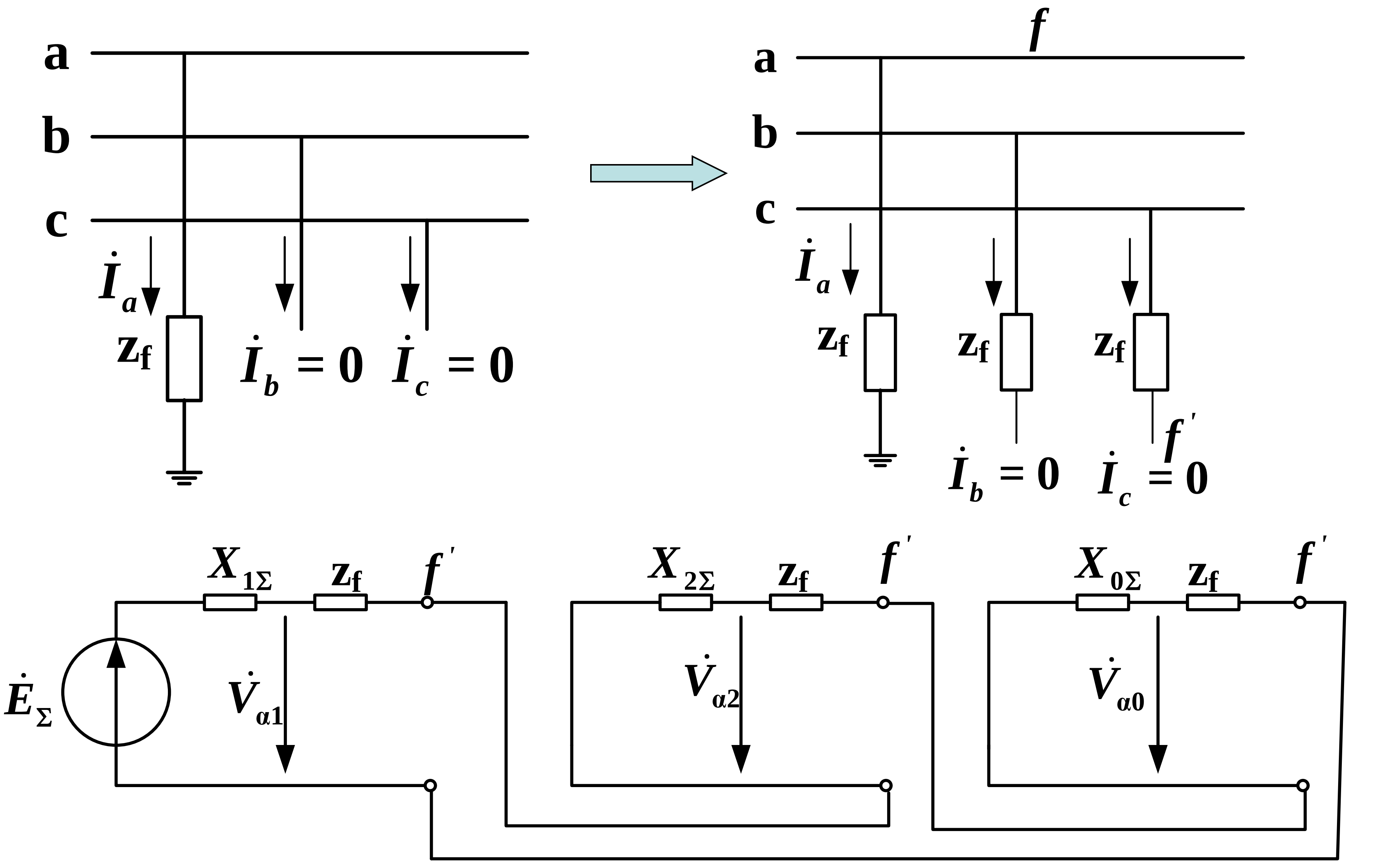
$$k_0 = \infty, \dot{V}_{fb} = \sqrt{3}\dot{V}_{fb|0|}\angle -30^\circ$$

4、相量图

$$\left.\begin{aligned}
&\dot{I}_{a2}=\dot{I}_{a0}=\dot{I}_{a1}=\frac{\dot{E}_{\Sigma}}{j(X_{1\Sigma}+X_{2\Sigma}+X_{0\Sigma})}\\
&\dot{V}_{a1}=\dot{E}_{\Sigma}-jX_{1\Sigma}\dot{I}_{a1}=j(X_{2\Sigma}+X_{0\Sigma})\dot{I}_{a1}\\
&\dot{V}_{a2}=-jX_{2\Sigma}\dot{I}_{a1}\\
&\dot{V}_{a0}=-jX_{0\Sigma}\dot{I}_{a1}
\end{aligned}\right\}$$

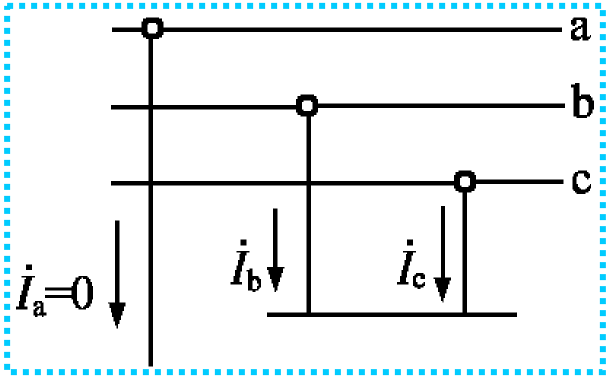


5、单相短路经阻抗接地



a
b
c
$\dot{I}_a$
z_f
$\dot{I}_b = 0$
$\dot{I}_c = 0$
f
f'
$\dot{E}_\Sigma$
$X_{1\Sigma}$
$X_{2\Sigma}$
$X_{0\Sigma}$
$\dot{V}_{\alpha 1}$
$\dot{V}_{\alpha 2}$
$\dot{V}_{\alpha 0}$

二、两相短路



1、边界条件

$$\left.\begin{aligned} &\dot{I}_a = 0 \\ &\dot{I}_b + \dot{I}_c = 0 \\ &\dot{V}_b = \dot{V}_c \end{aligned}\right\}$$

$$\left.\begin{aligned} &\dot{I}_{a1} + \dot{I}_{a2} + \dot{I}_{a0} = 0 \\ &a^2\dot{I}_{a1} + a\dot{I}_{a2} + \dot{I}_{a0} + a\dot{I}_{a1} + a^2\dot{I}_{a2} + \dot{I}_{a0} = 0 \\ &a^2\dot{V}_{a1} + a\dot{V}_{a2} + \dot{V}_{a0} = a\dot{V}_{a1} + a^2\dot{V}_{a2} + \dot{V}_{a0} \end{aligned}\right\}$$

$$\left.\begin{aligned} &\dot{I}_{a0} = 0 \\ &\dot{I}_{a1} + \dot{I}_{a2} = 0 \\ &\dot{V}_{a1} = \dot{V}_{a2} \end{aligned}\right\}$$

$$\left.\begin{aligned} \dot{E}_{\Sigma} - jX_{1\Sigma}\dot{I}_{a1} &= \dot{V}_{a1} \\ - jX_{2\Sigma}\dot{I}_{a2} &= \dot{V}_{a2} \\ - jX_{0\Sigma}\dot{I}_{a0} &= \dot{V}_{a0} \end{aligned}\right\}$$

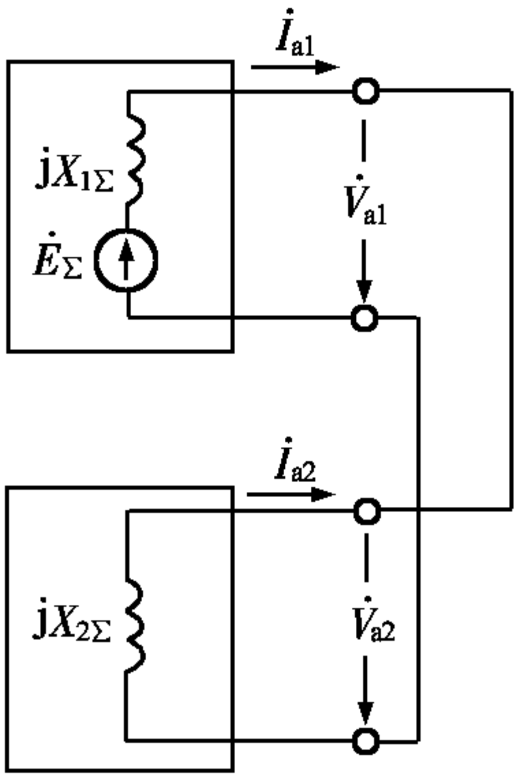
$$\dot{I}_{a1} = \frac{\dot{E}_{\Sigma}}{j(X_{1\Sigma} + X_{2\Sigma})}$$

$$\left.\begin{aligned} &\boldsymbol{\dot{I}_{a2} = -\dot{I}_{a1}} \\ &\boldsymbol{\dot{I}_{a0} = 0} \\ &\boldsymbol{\dot{V}_{a1} = \dot{V}_{a2} = -jX_{2\Sigma}\dot{I}_{a2} = jX_{2\Sigma}\dot{I}_{a1}} \\ &\boldsymbol{\dot{V}_{a0} = 0} \end{aligned}\right\}$$

2、两相短路的复合序网

$$\left.\begin{aligned}&\dot{I}_{a0}=0\\&\dot{I}_{a1}+\dot{I}_{a2}=0\\&\dot{V}_{a1}=\dot{V}_{a2}\end{aligned}\right\}$$

$$\left.\begin{aligned}\dot{E}_{\Sigma}-jX_{1\Sigma}\dot{I}_{a1}&=\dot{V}_{a1}\\-jX_{2\Sigma}\dot{I}_{a2}&=\dot{V}_{a2}\\-jX_{0\Sigma}\dot{I}_{a0}&=\dot{V}_{a0}\end{aligned}\right\}$$



$$\dot{I}_{a1}=\frac{\dot{E}_{\Sigma}}{j(X_{1\Sigma}+X_{2\Sigma})}$$

$$\left.\begin{aligned}&\dot{I}_{a2}=-\dot{I}_{a1}\\&\dot{I}_{a0}=0\\&\dot{V}_{a1}=\dot{V}_{a2}=-jX_{2\Sigma}\dot{I}_{a2}=jX_{2\Sigma}\dot{I}_{a1}\\&\dot{V}_{a0}=0\end{aligned}\right\}$$

3、两相短路的短路电流和电压

$$\left.\begin{aligned}\dot{I}_b &= a^2\dot{I}_{a1} + a\dot{I}_{a2} + \dot{I}_{a0} = (a^2 - a)\dot{I}_{a1} = -j\sqrt{3}\dot{I}_{a1} \\ \dot{I}_c &= -\dot{I}_b = j\sqrt{3}\dot{I}_{a1}\end{aligned}\right\}$$

$$I_f^{(2)} = I_b = I_c = \sqrt{3}I_{a1} = \frac{\sqrt{3}E_\Sigma}{X_{1\Sigma} + X_{2\Sigma}} \approx \frac{\sqrt{3}}{2}I^{(3)}$$

结论：两相短路电流小于三相短路电流

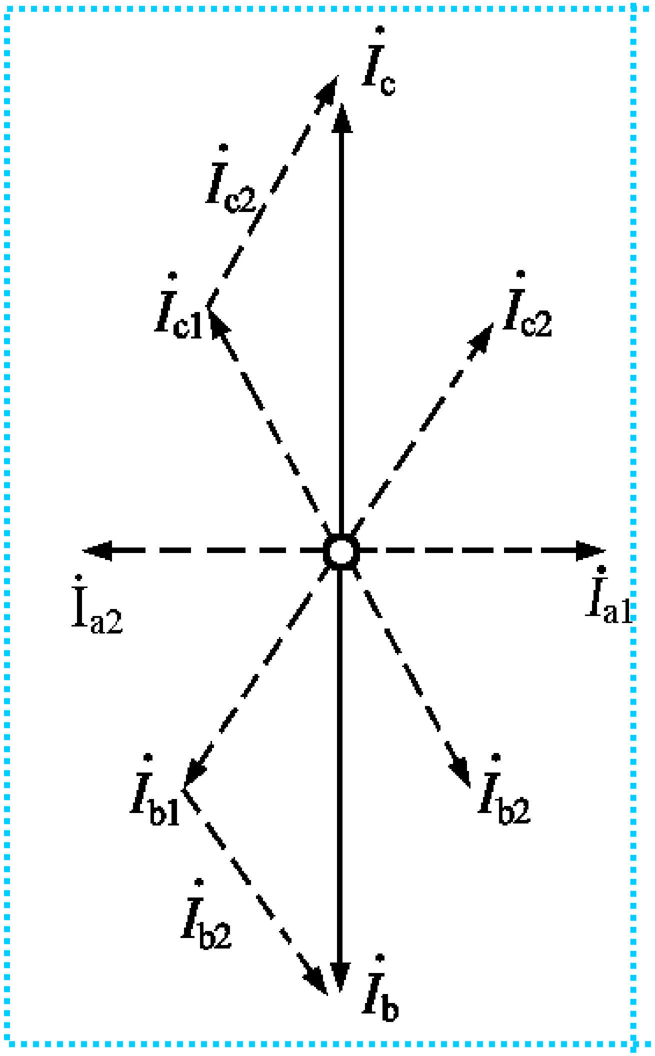
$$\left.\begin{aligned}\dot{V}_a &= \dot{V}_{a1} + \dot{V}_{a2} + \dot{V}_{a0} = 2\dot{V}_{a1} = j2X_{2\Sigma}\dot{I}_{a1} \\ \dot{V}_b &= a^2\dot{V}_{a1} + a\dot{V}_{a2} + \dot{V}_{a0} = -\dot{V}_{a1} = -\frac{1}{2}\dot{V}_a \\ \dot{V}_c &= \dot{V}_b = -\dot{V}_{a1} = -\frac{1}{2}\dot{V}_a\end{aligned}\right\} \quad \begin{aligned}\dot{V}_a &= \dot{V}_{fa|0|} \\ \dot{V}_b &= \dot{V}_c = -\dot{V}_{fa|0|}/2\end{aligned}$$

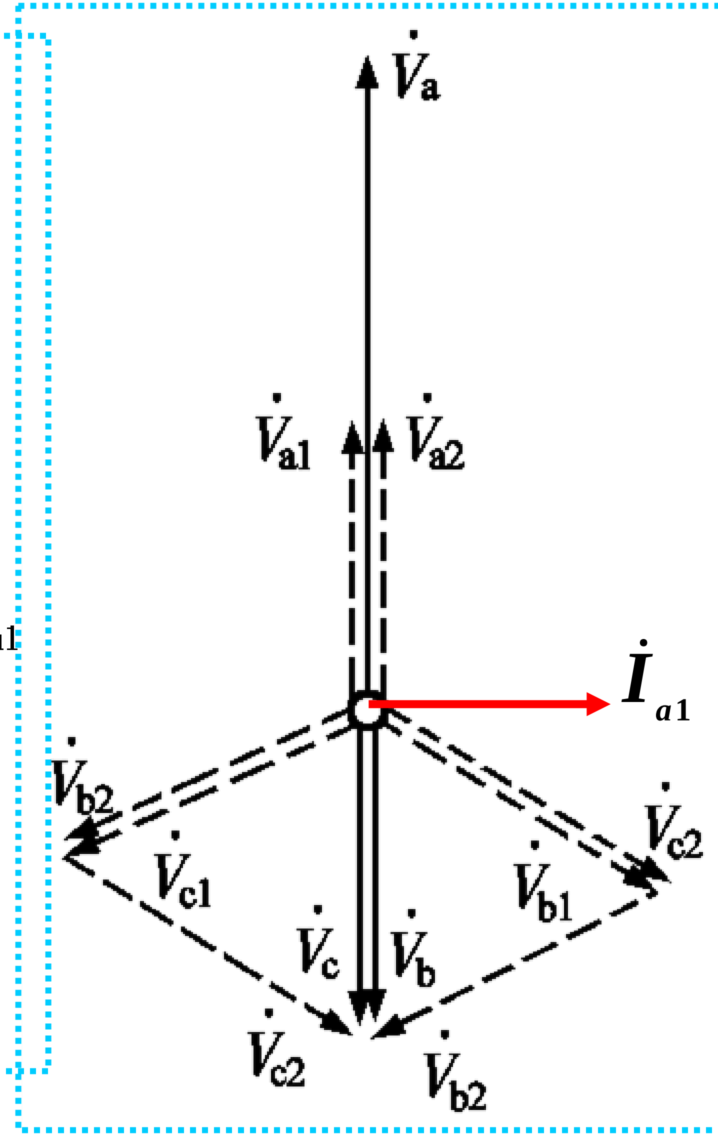
结论：故障相电压是正常电压的二分之一。

4、相量图

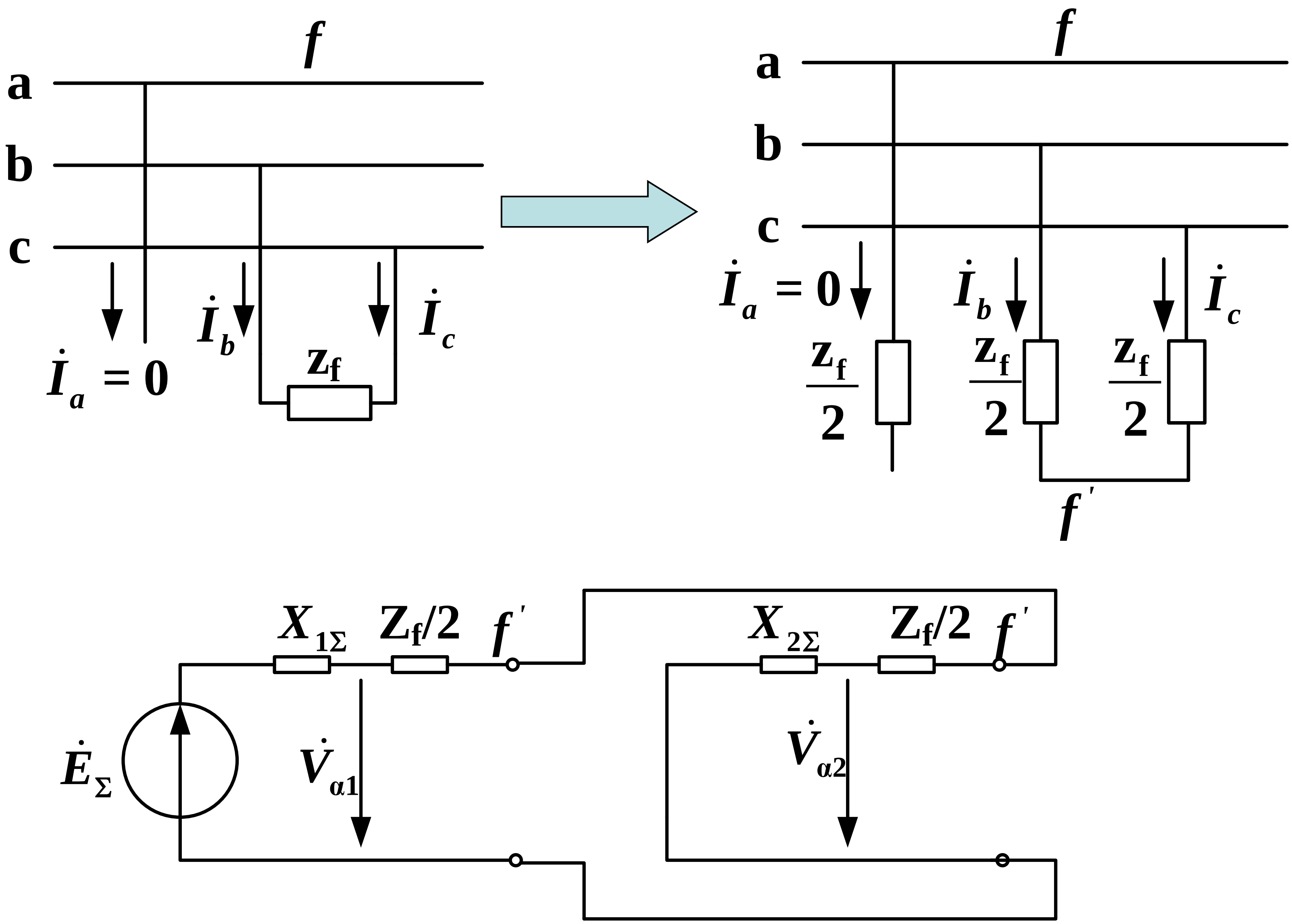
$$\dot{I}_{a1}=\frac{\dot{E}_{\Sigma}}{j(X_{1\Sigma}+X_{2\Sigma})}$$

$$\left.\begin{aligned}\dot{I}_{a2}&=-\dot{I}_{a1}\\ \dot{V}_{a1}&=\dot{V}_{a2}\\ &=-jX_{2\Sigma}\dot{I}_{a2}\\ &=jX_{2\Sigma}\dot{I}_{a1}\end{aligned}\right\}$$



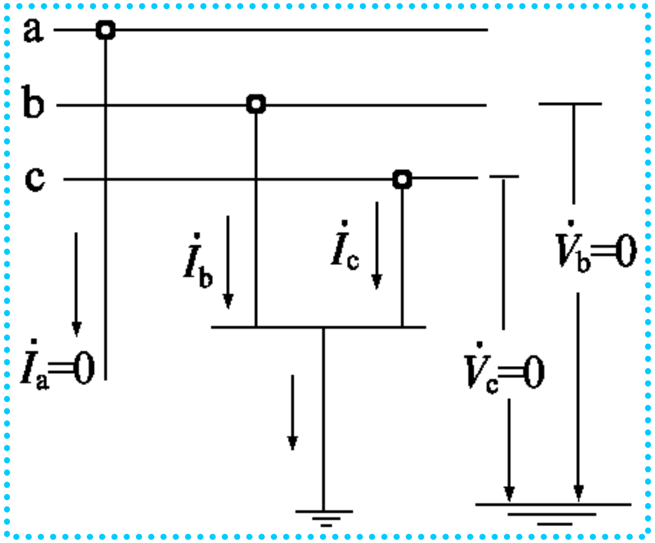


5、两相经过阻抗短路



a
b
c
f
$\dot{I}_a = 0$
$\dot{I}_b$
$\dot{I}_c$
z_f
$\frac{z_f}{2}$
f'
$\dot{E}_\Sigma$
$X_{1\Sigma}$
$Z_f/2$
$\dot{V}_{\alpha 1}$
$X_{2\Sigma}$
$\dot{V}_{\alpha 2}$

三、两相短路接地



1、边界条件

$$\left.\begin{aligned}\dot{V}_a &= 0\\ \dot{I}_b &= 0\\ \dot{I}_c &= 0\end{aligned}\right\} \Rightarrow \left.\begin{aligned}&\dot{V}_{a1} + \dot{V}_{a2} + \dot{V}_{a0} = 0\\ &\dot{I}_{a1} = \dot{I}_{a2} = \dot{I}_{a0}\end{aligned}\right\}$$

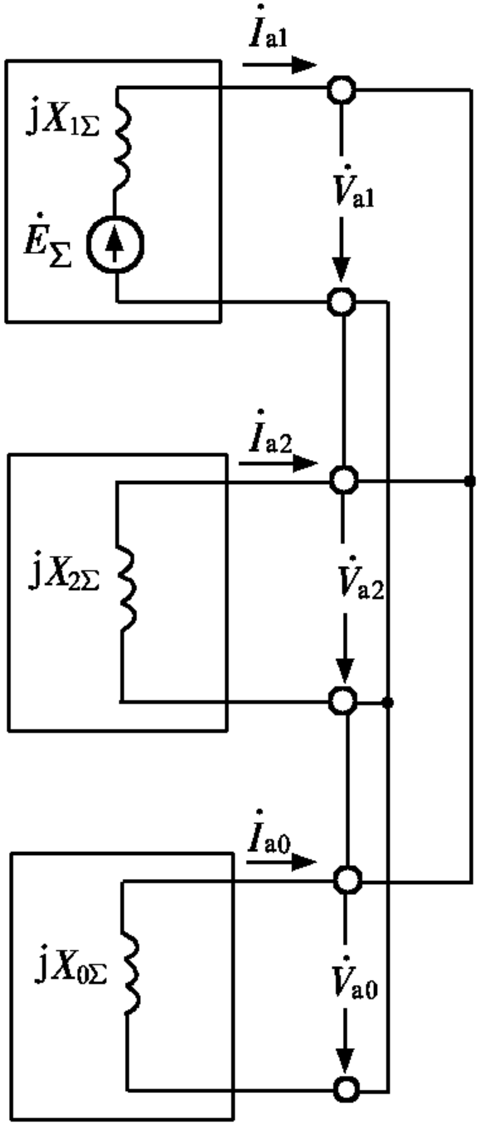
$$\left.\begin{aligned}\dot{I}_a &= 0\\ \dot{V}_b &= 0\\ \dot{V}_c &= 0\end{aligned}\right\} \Rightarrow \left.\begin{aligned}&\dot{I}_{a1} + \dot{I}_{a2} + \dot{I}_{a0} = 0\\ &\dot{V}_{a1} = \dot{V}_{a2} = \dot{V}_{a0}\end{aligned}\right\}$$

$$\left.\begin{aligned}\dot{E}_\Sigma - jX_{1\Sigma}\dot{I}_{a1} &= \dot{V}_{a1}\\ - jX_{2\Sigma}\dot{I}_{a2} &= \dot{V}_{a2}\\ - jX_{0\Sigma}\dot{I}_{a0} &= \dot{V}_{a0}\end{aligned}\right\}$$

2、两相短路接地复合序网图

$$\left.\begin{aligned}\dot I_{a1}+\dot I_{a2}+\dot I_{a0}=0\\ \dot V_{a1}=\dot V_{a2}=\dot V_{a0}\end{aligned}\right\}$$

$$\left.\begin{aligned}\dot E_{\Sigma}-jX_{1\Sigma}\dot I_{a1}&=\dot V_{a1}\\ -jX_{2\Sigma}\dot I_{a2}&=\dot V_{a2}\\ -jX_{0\Sigma}\dot I_{a0}&=\dot V_{a0}\end{aligned}\right\}$$



$$\dot I_{a1}=\frac{\dot E_{\Sigma}}{j(X_{1\Sigma}+X_{2\Sigma}//X_{0\Sigma})}$$

$$\left.\begin{aligned}\dot I_{a2}&=-\frac{X_{0\Sigma}}{X_{2\Sigma}+X_{0\Sigma}}\dot I_{a1}\\ I_{a0}&=-\frac{X_{2\Sigma}}{X_{2\Sigma}+X_{0\Sigma}}\dot I_{a1}\\ \dot V_{a1}=\dot V_{a2}=\dot V_{a0}&=j\frac{X_{2\Sigma}X_{0\Sigma}}{X_{2\Sigma}+X_{0\Sigma}}\dot I_{a1}\end{aligned}\right\}$$

3、两相短路接地故障相电流

$$\left.\begin{aligned}
\dot{I}_b &= a^2\dot{I}_{a1} + a\dot{I}_{a2} + \dot{I}_{a0} = \left(a^2 - \frac{X_{2\Sigma} + aX_{0\Sigma}}{X_{2\Sigma} + X_{0\Sigma}}\right)\dot{I}_{a1} \\
&= \frac{-3X_{2\Sigma} - j\sqrt{3}(X_{2\Sigma} + 2X_{0\Sigma})}{2(X_{2\Sigma} + X_{0\Sigma})}\dot{I}_{a1} \\
\dot{I}_c &= a\dot{I}_{a1} + a^2\dot{I}_{a2} + \dot{I}_{a0} = \left(a - \frac{X_{2\Sigma} + a^2X_{0\Sigma}}{X_{2\Sigma} + X_{0\Sigma}}\right)\dot{I}_{a1} \\
&= \frac{-3X_{2\Sigma} + j\sqrt{3}(X_{2\Sigma} + 2X_{0\Sigma})}{2(X_{2\Sigma} + X_{0\Sigma})}\dot{I}_{a1}
\end{aligned}\right\}$$

$$I_f^{(1,1)} = I_b = I_c = \sqrt{3}\sqrt{1 - \frac{X_{2\Sigma}X_{0\Sigma}}{(X_{2\Sigma} + X_{0\Sigma})^2}}I_{a1}$$

$$= \sqrt{3}\sqrt{1 - \frac{k_0}{(1+k_0)^2}}\frac{1+k_0}{1+2k_0}I^{(3)}$$

$$k_0 = 0, I_b = I_c = \sqrt{3}I^{(3)}$$

$$k_0 = 1, I_b = I_c = I^{(3)}$$

$$k_0 = \infty, I_b = I_c = \frac{\sqrt{3}}{2}I^{(3)}$$

注意:两相短路接地故障相电流的变化规律同单相接地非故障相电压变化规律有相似之处

4、非故障相电压

$$\dot{V}_a = 3\dot{V}_{a1} = j\frac{3X_{2\Sigma}X_{0\Sigma}}{X_{2\Sigma}+X_{0\Sigma}}\dot{I}_{a1} = \dot{V}_{a|0|}\frac{3k_0}{1+2k_0}$$

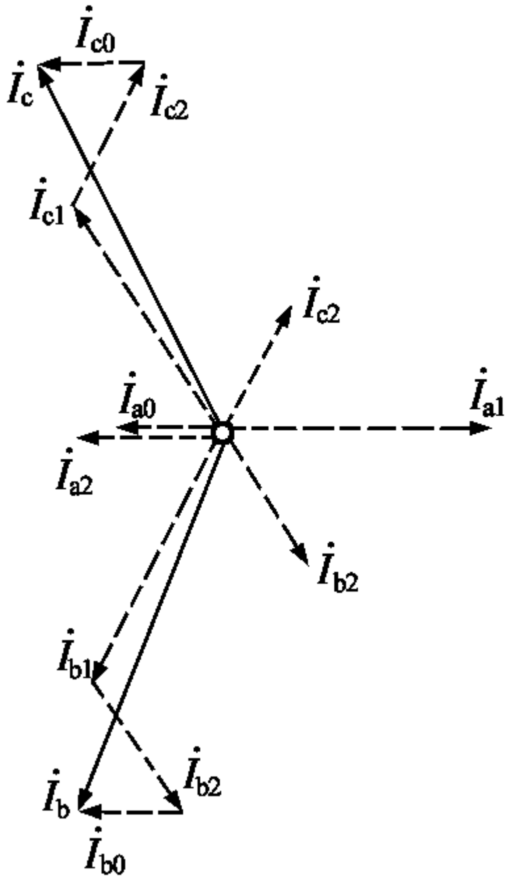
$$k_0 = 0, \dot{V}_a = 0$$

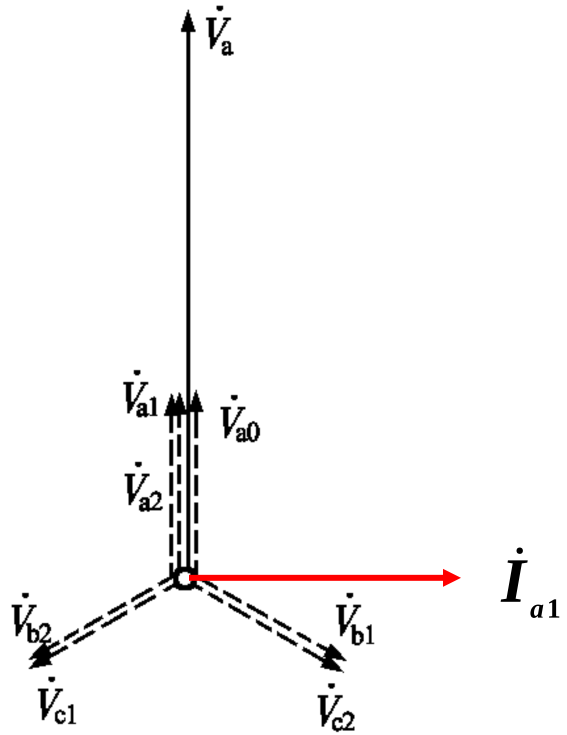
$$k_0 = 1, \dot{V}_a = \dot{V}_{a|0|}$$

$$k_0 = \infty, \dot{V}_a = \frac{3}{2}\dot{V}_{a|0|}$$

注意:两相短路接地非故障相电压变化规律同单相接地故障相电流的变化规律有相似之处

5、两相短路接地相量图

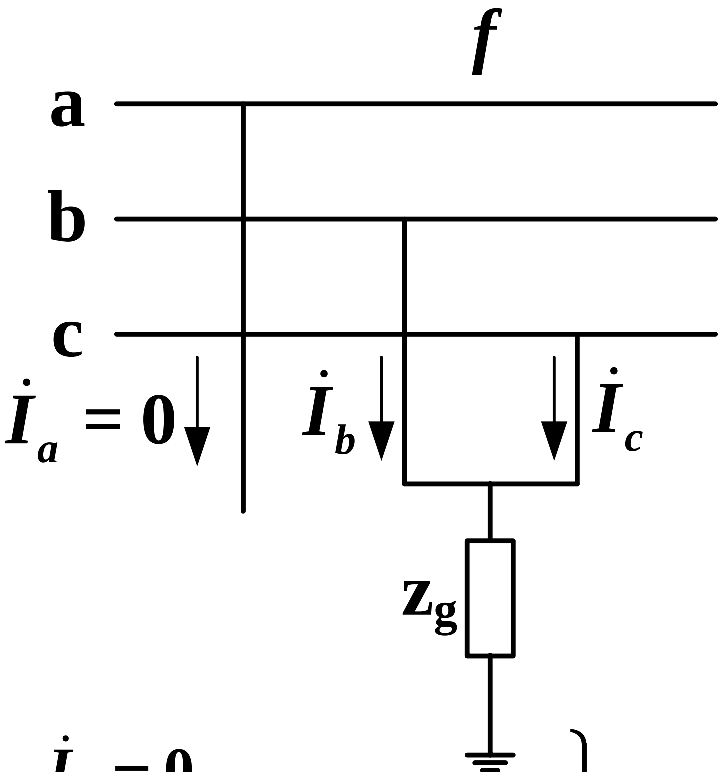




$$\dot{I}_{a1} = \frac{\dot{E}_{\Sigma}}{j(X_{1\Sigma} + X_{2\Sigma} // X_{0\Sigma})}$$

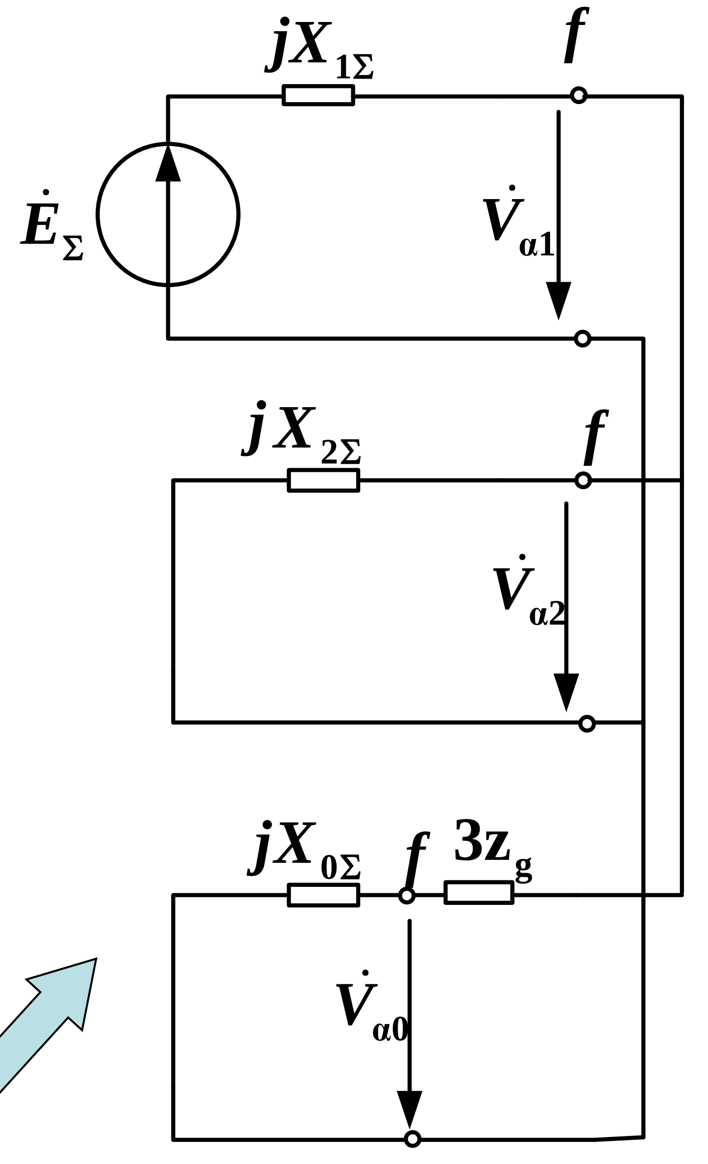
$$\left.\begin{aligned}
\dot{I}_{a2} &= -\frac{X_{0\Sigma}}{X_{2\Sigma} + X_{0\Sigma}}\dot{I}_{a1} \\
I_{a0} &= -\frac{X_{2\Sigma}}{X_{2\Sigma} + X_{0\Sigma}}\dot{I}_{a1} \\
\dot{V}_{a1} = \dot{V}_{a2} = \dot{V}_{a0} &= j\frac{X_{2\Sigma}X_{0\Sigma}}{X_{2\Sigma} + X_{0\Sigma}}\dot{I}_{a1}
\end{aligned}\right\}$$

6、两相短路经过阻抗接地



$$\left.\begin{aligned}&\dot{I}_a = 0\\&\dot{V}_b = \dot{V}_c = (\dot{I}_b + \dot{I}_c)Zg\end{aligned}\right\}$$

$$\left.\begin{aligned}&\dot{I}_{a(1)} + \dot{I}_{a(2)} + \dot{I}_{a(0)} = 0\\&\dot{V}_{a(1)} = \dot{V}_{a(2)} = \dot{V}_{a(0)} - 3\dot{I}_{a(0)}Zg\end{aligned}\right\}$$



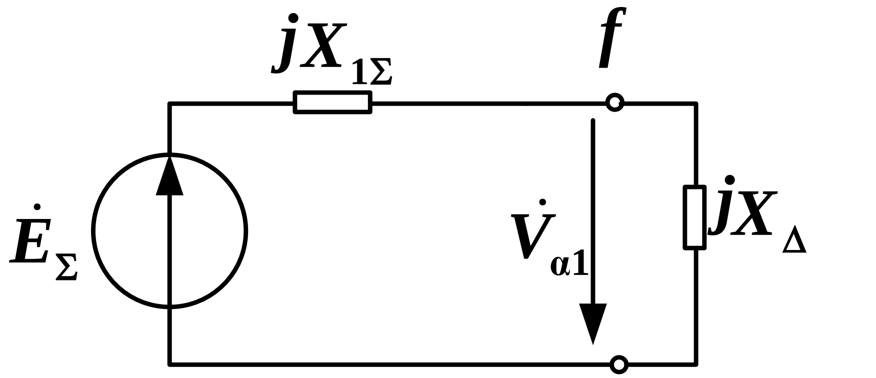
四、正序等效定则

1.正序分量的计算

$$\dot{I}_{a1} = \frac{\dot{E}_{\Sigma}}{j(X_{1\Sigma} + X_{2\Sigma} + X_{0\Sigma})}$$

$$\dot{I}_{a1} = \frac{\dot{E}_{\Sigma}}{j(X_{1\Sigma} + X_{2\Sigma})}$$

$$\dot{I}_{a1} = \frac{\dot{E}_{\Sigma}}{j(X_{1\Sigma} + X_{2\Sigma} // X_{0\Sigma})}$$



$$\dot{I}_{a1} = \frac{\dot{E}_{\Sigma}}{j(X_{1\Sigma} + X_{\Delta}^{(n)})}$$

四、正序等效定则

2.短路电流的计算

$$I_f^{(1)} = 3I_{a1}$$

$$I_f^{(2)} = I_b = I_c = \sqrt{3} I_{a1}$$

$$I_f^{(1,1)} = I_b = I_c$$

$$= \sqrt{3}\sqrt{1 - \frac{X_{2\Sigma} X_{0\Sigma}}{(X_{2\Sigma} + X_{0\Sigma})^2}} I_{a1}$$

$$\Rightarrow \quad I_f^{(n)} = m^{(n)} I_{a1}^{(n)}$$

附加电抗和比例系数

短路类型$f^{(n)}$	$X_{\Delta}^{(n)}$	$m^{(n)}$
三相短路$f^{(3)}$	0	1
两相短路接地$f^{(1,1)}$	$\dfrac{X_{2\Sigma}X_{0\Sigma}}{X_{2\Sigma}+X_{0\Sigma}}$	$\sqrt{3}\sqrt{1-\dfrac{X_{2\Sigma}X_{0\Sigma}}{(X_{2\Sigma}+X_{0\Sigma})^{2}}}$
两相短路$f^{(2)}$	$X_{2\Sigma}$	$\sqrt{3}$
单相接地短路$f^{(1)}$	$X_{2\Sigma}+X_{0\Sigma}$	3

- 选特殊相作基准相——故障处与另两相情况不同的一相

四、正序等效定则

3. 任意时刻短路电流的计算

$I_f^{(1)}, I_f^{(1,1)}, I_f^{(2)}$ 为t=0时的短路电流

计算任意时刻的短路电流可用运算曲线法

- 在正序网络中的故障点f上接一阻抗 x_{Δ} 。
- 用运算曲线求经过 x_{Δ} 后发生三相短路时任意时刻的电流，即为f点不对称短路的正序电流。
- 其它计算步骤与前述相同。

8.4 非故障处电流电压的计算

- 电力系统设计运行中，除需要知道故障点的短路电流和电压外，有时还需要知道网络中某些支路电流和节点电压。

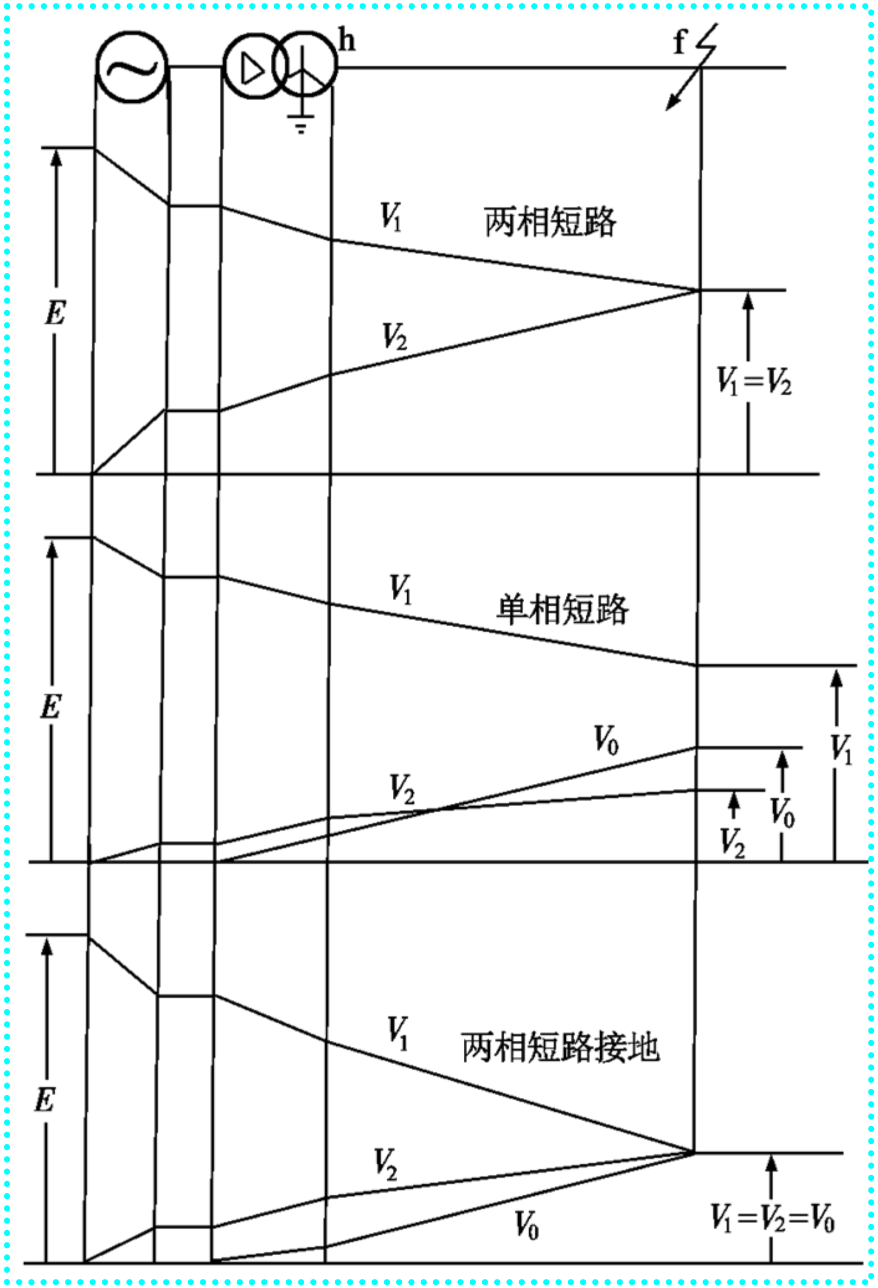
- 基本思路：先求出电流电压的各序分量在网络中的分布，然后将相应的各序分量进行合成求得各相电流和相电压。

一、网络中电流电压的分布计算

1. 电流分布计算

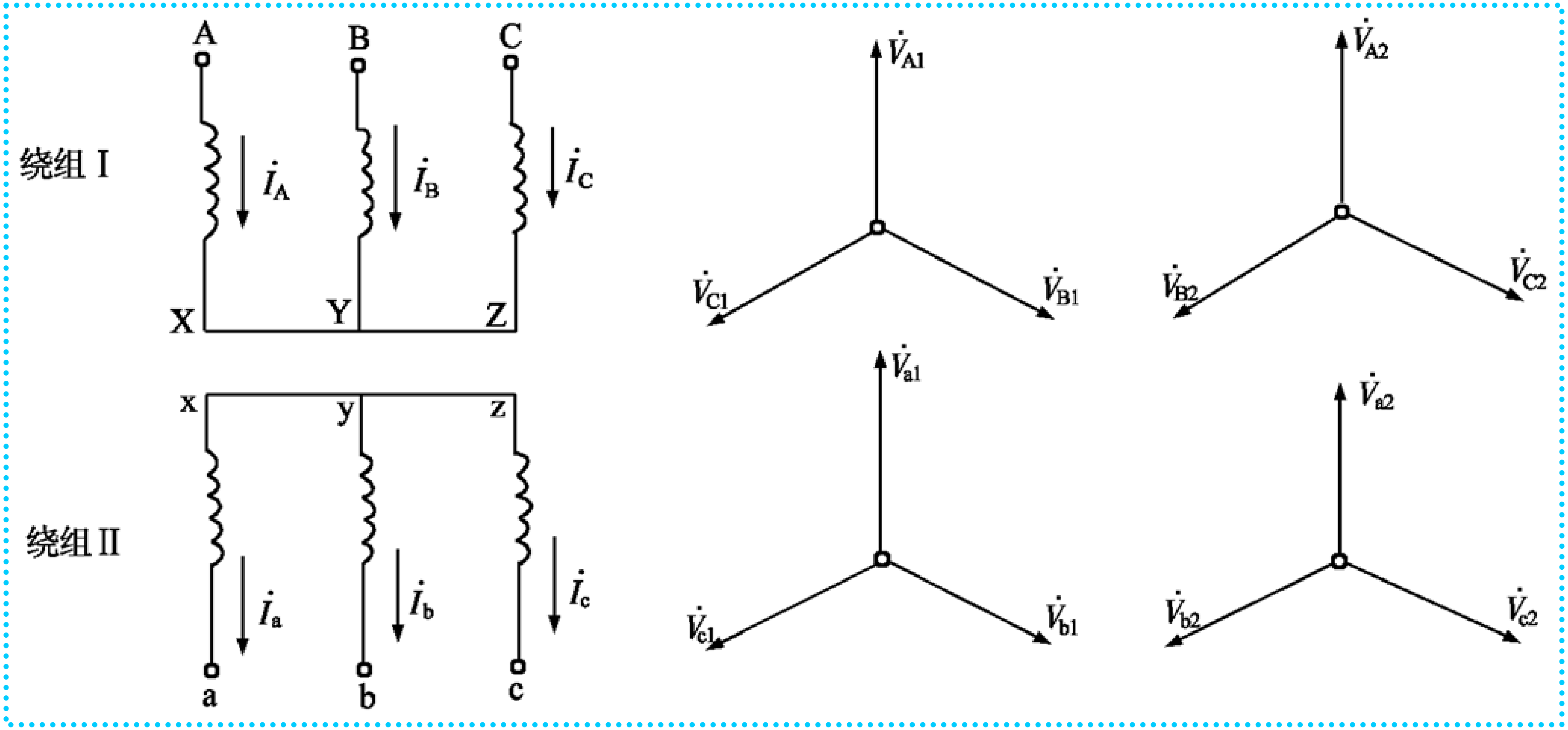
2. 电压分布的计算：

$$\left.\begin{aligned}\dot{V}_{h1} &= \dot{V}_{f1} + jX_1\dot{I}_1 \\ \dot{V}_{h2} &= \dot{V}_{f2} + jX_2\dot{I}_2 \\ \dot{V}_{h0} &= \dot{V}_{f0} + jX_0\dot{I}_0\end{aligned}\right\}$$

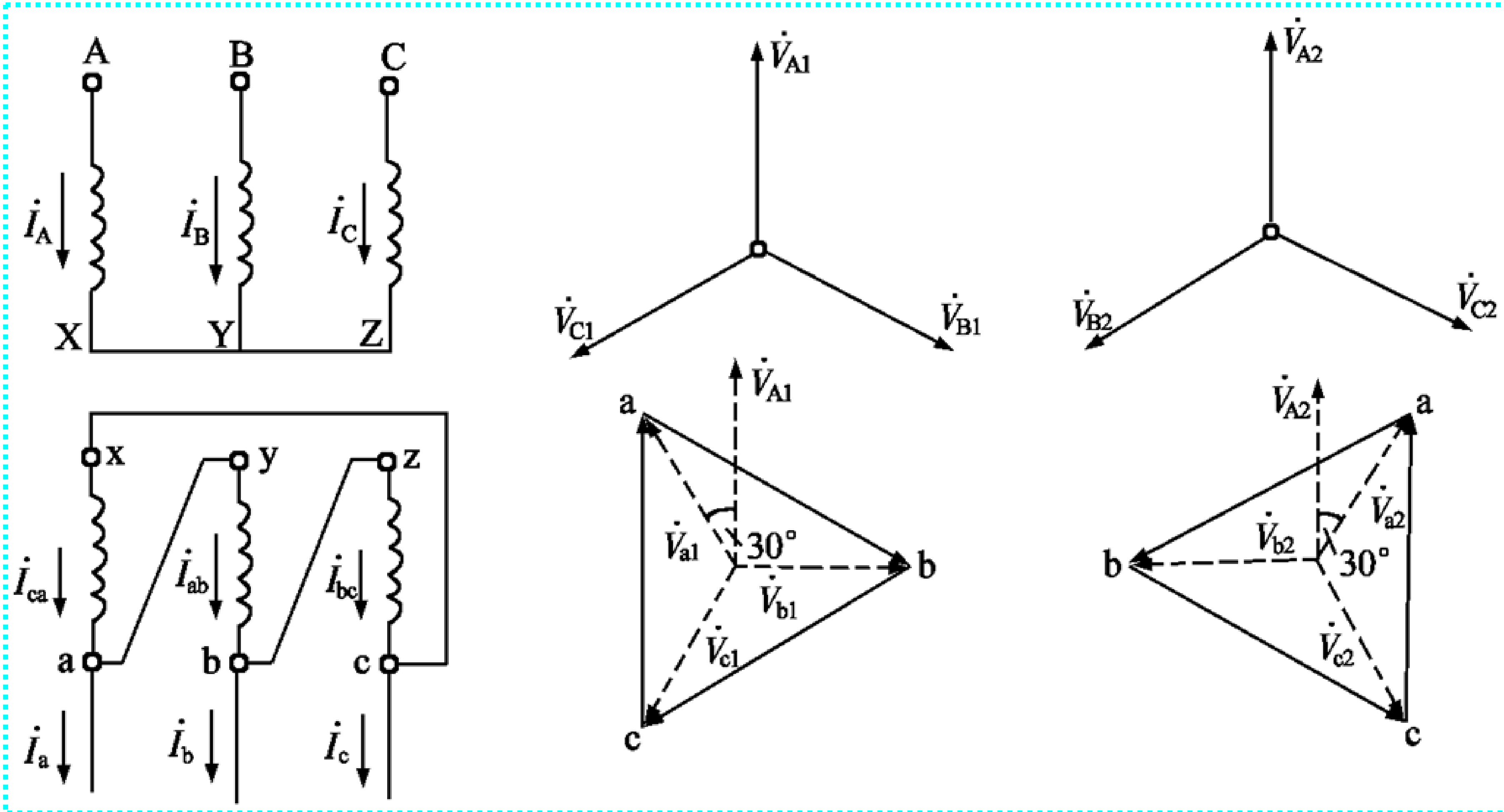


二、对称分量经变压器后的相位变化

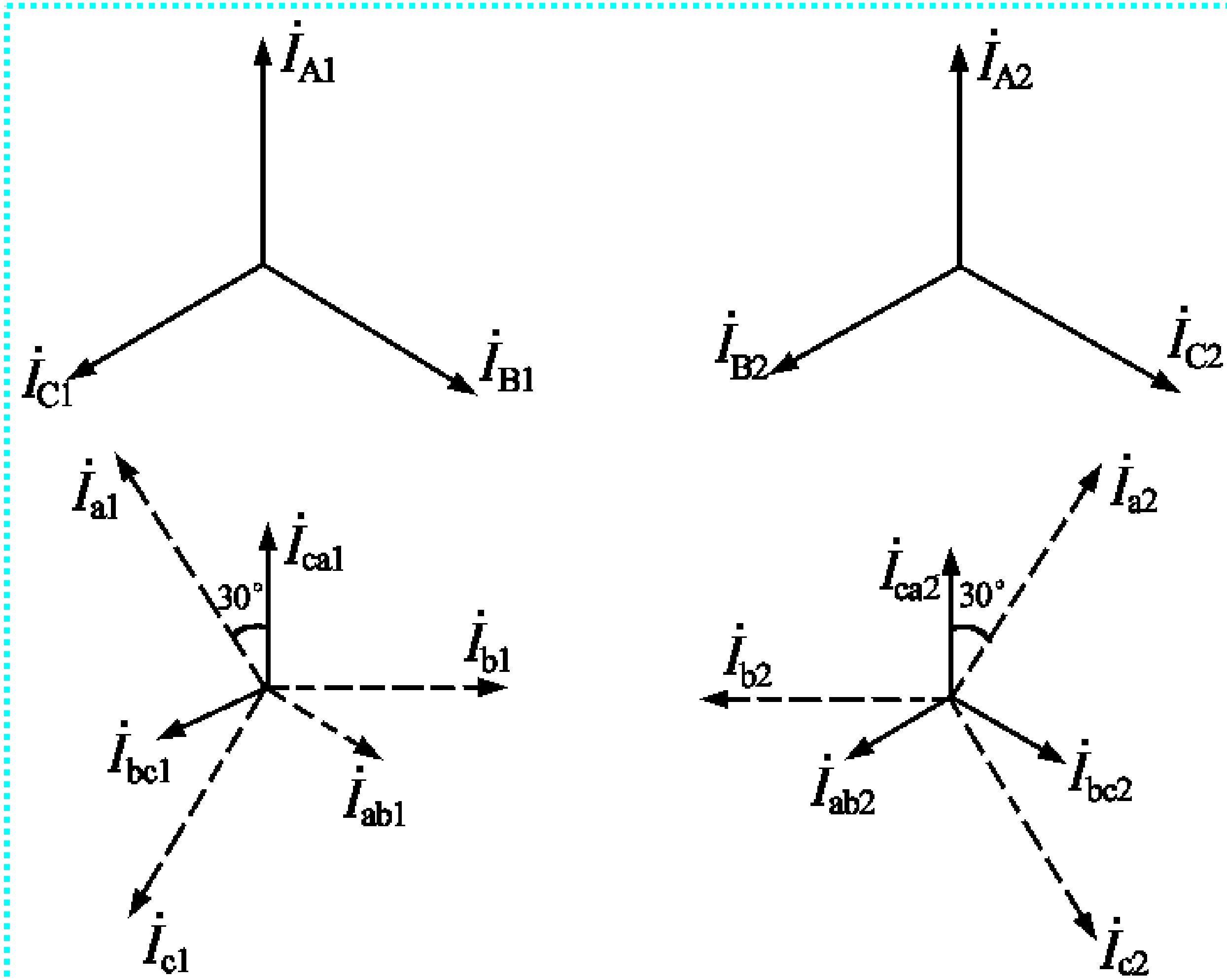
1. Y／Y—12连接的变压器：不发生相位移动。



2. Y / △－11连接的变压器:移相30^0



2. Y / △－11连接的变压器



$\dot{I}_{A1}$
$\dot{I}_{C1}$
$\dot{I}_{B1}$
$\dot{I}_{A2}$
$\dot{I}_{B2}$
$\dot{I}_{C2}$
$\dot{I}_{a1}$
$\dot{I}_{ca1}$
30°
$\dot{I}_{b1}$
$\dot{I}_{bc1}$
$\dot{I}_{ab1}$
$\dot{I}_{c1}$
$\dot{I}_{a2}$
$\dot{I}_{ca2}$
30°
$\dot{I}_{b2}$
$\dot{I}_{ab2}$
$\dot{I}_{bc2}$
$\dot{I}_{c2}$

2. Y / △—11连接的变压器

$$\left.\begin{aligned}\dot{V}_{a1} &= \dot{V}_{A1}e^{j30^0} = \dot{V}_{A1}e^{-j330^0}\\ \dot{V}_{a2} &= \dot{V}_{A2}e^{-j30^0} = \dot{V}_{A2}e^{j330^0}\end{aligned}\right\}\qquad \left.\begin{aligned}\dot{I}_{a1} &= \dot{I}_{A1}e^{j30^0} = \dot{I}_{A1}e^{-j330^0}\\ \dot{I}_{a2} &= \dot{I}_{A2}e^{-j30^0} = \dot{I}_{A2}e^{j330^0}\end{aligned}\right\}$$

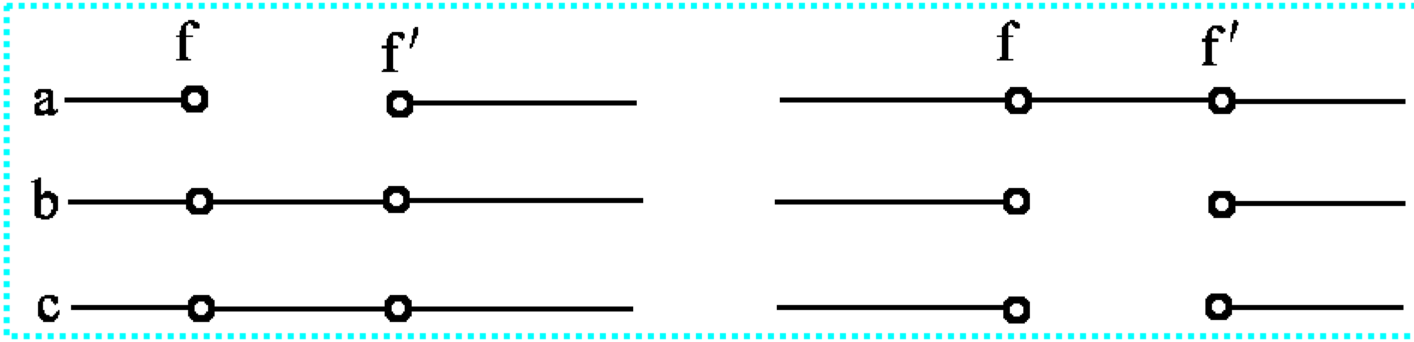
$$\left.\begin{aligned}\dot{I}_a &= \dot{I}_{a1} + \dot{I}_{a2} = \dot{I}_{A1}e^{j30^0} + \dot{I}_{A2}e^{-j30^0} = -j[a\dot{I}_{A1} + a^2(-\dot{I}_{A2})]\\ \dot{I}_b &= a^2\dot{I}_{a1} + a\dot{I}_{a2} = a^2\dot{I}_{A1}e^{j30^0} + a\dot{I}_{A2}e^{-j30^0} = -j[\dot{I}_{A1} + (-\dot{I}_{A2})]\\ \dot{I}_c &= a\dot{I}_{a1} + a^2\dot{I}_{a2} = a\dot{I}_{A1}e^{j30^0} + a^2\dot{I}_{A2}e^{-j30^0} = -j[a^2\dot{I}_{A1} + a(-\dot{I}_{A2})]\end{aligned}\right\}$$

3. Y / △—k连接的变压器

$$\left.\begin{aligned}\dot{V}_{a1} &= \dot{V}_{A1}e^{-jk\times30^0}\\ \dot{V}_{a2} &= \dot{V}_{A2}e^{jk\times30^0}\end{aligned}\right\}\qquad \left.\begin{aligned}\dot{I}_{a1} &= \dot{I}_{A1}e^{-jk\times30^0}\\ \dot{I}_{a2} &= \dot{I}_{A2}e^{jk\times30^0}\end{aligned}\right\}$$

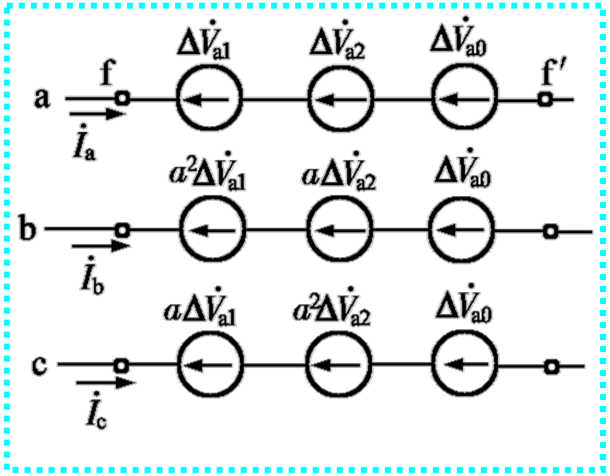
8.5 非全相断线的分析计算

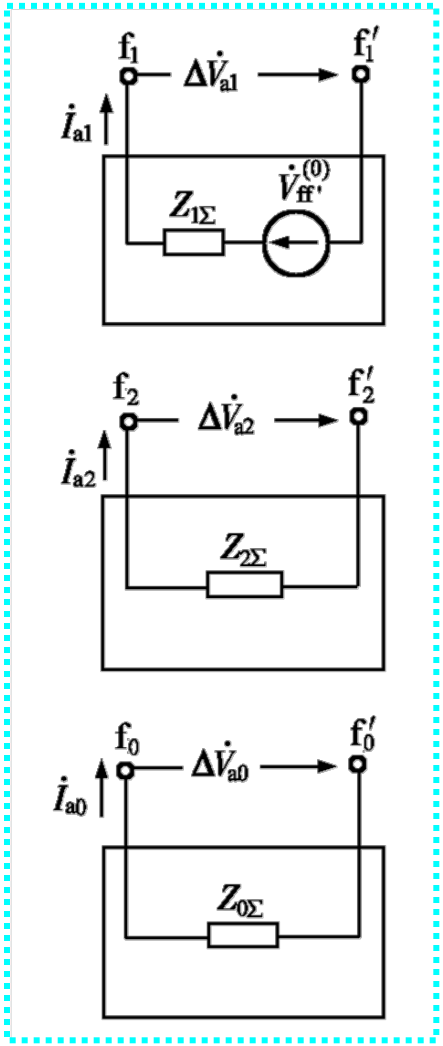
- 非全相断线



- 横向故障和纵向故障

8.5 非全相断线的分析计算





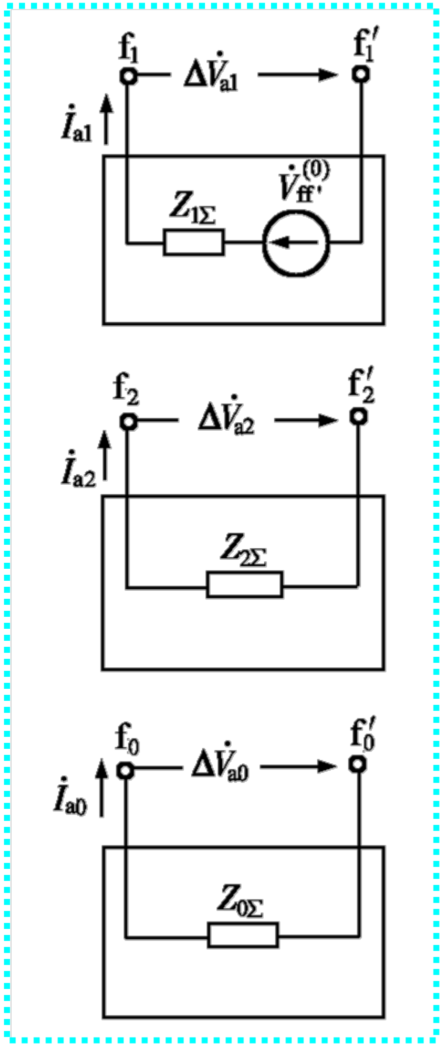
$$\left.\begin{aligned}
\dot{V}_{ff'}^{(0)} - Z_{1\Sigma}\dot{I}_{a1} &= \Delta\dot{V}_{a1} \\
- Z_{2\Sigma}\dot{I}_{a2} &= \Delta\dot{V}_{a2} \\
- Z_{0\Sigma}\dot{I}_{a0} &= \Delta\dot{V}_{a0}
\end{aligned}\right\}$$

8.5 非全相断线的分析计算

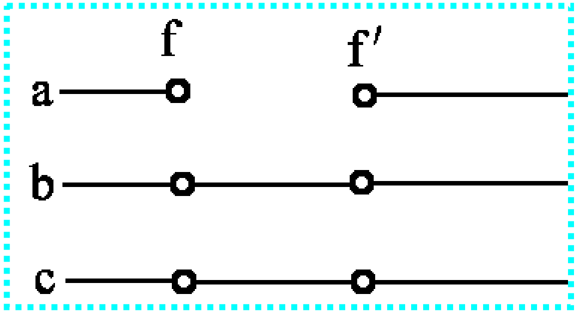
- 断线故障时电流不像短路电流那么大，电压也不致很低，所以要**计及负荷等值阻抗**。

- 正、负、零序阻抗分别为正、负、零序网络**从断口看入**的等值阻抗。

- $\dot{U}_{ff'|0|}$ 的计算：先由正常潮流求各电源的 $\dot{E}''$ ，然后应用正序网络计算f、f'断开后的 $\dot{U}_{ff'|0|}$ 。



一、单相断开



$$\left.\begin{aligned}&\dot{I}_a = 0\\&\Delta\dot{V}_b = \Delta\dot{V}_c = 0\end{aligned}\right\}$$

$$\left.\begin{aligned}&\dot{I}_{a1} + \dot{I}_{a2} + \dot{I}_{a0} = 0\\&\Delta\dot{V}_{a1} = \Delta\dot{V}_{a2} = \Delta\dot{V}_{a0}\end{aligned}\right\}$$

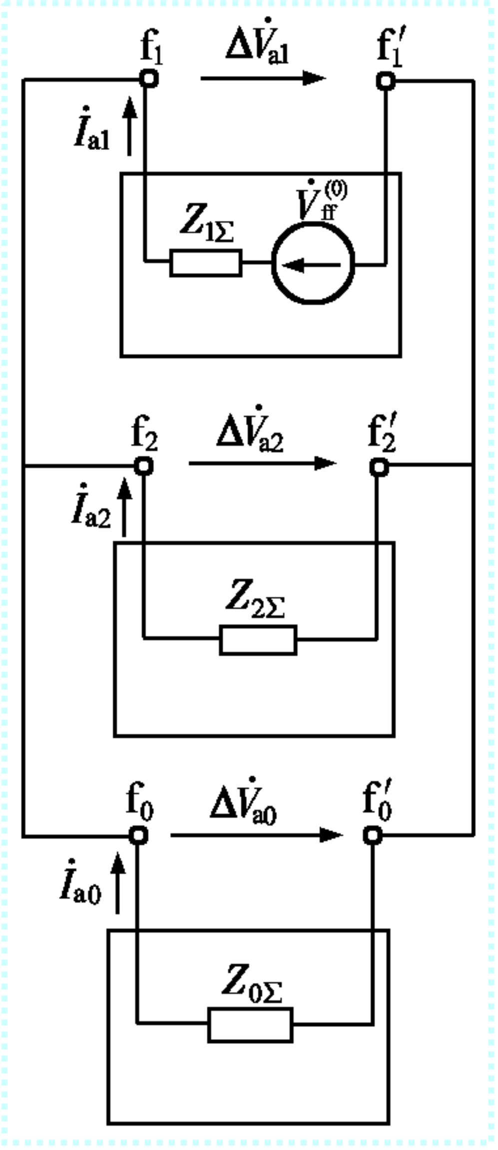
$$\left.\begin{aligned}\dot{V}_{ff'}^{(0)} - jX_{1\Sigma}\dot{I}_{a1} &= \Delta\dot{V}_{a1}\\- jX_{2\Sigma}\dot{I}_{a2} &= \Delta\dot{V}_{a2}\\- jX_{0\Sigma}\dot{I}_{a0} &= \Delta\dot{V}_{a0}\end{aligned}\right\}$$

$$\left.\begin{aligned}\dot{I}_{a1} &= \frac{\dot{V}_{ff'}^{(0)}}{j(X_{1\Sigma} + X_{2\Sigma} // X_{0\Sigma})}\\\dot{I}_{a2} &= -\frac{X_{0\Sigma}}{X_{2\Sigma} + X_{0\Sigma}}\dot{I}_{a1}\\\dot{I}_{a0} &= -\frac{X_{2\Sigma}}{X_{2\Sigma} + X_{0\Sigma}}\dot{I}_{a1}\end{aligned}\right\}$$

单相断开的复合序网

$$\left.\begin{aligned} \dot{V}_{ff'}^{(0)} - Z_{1\Sigma}\dot{I}_{a1} &= \Delta\dot{V}_{a1} \\ -Z_{2\Sigma}\dot{I}_{a2} &= \Delta\dot{V}_{a2} \\ -Z_{0\Sigma}\dot{I}_{a0} &= \Delta\dot{V}_{a0} \end{aligned}\right\}$$

$$\left.\begin{aligned} \dot{I}_{a1} + \dot{I}_{a2} + \dot{I}_{a0} = 0 \\ \Delta\dot{V}_{a1} = \Delta\dot{V}_{a2} = \Delta\dot{V}_{a0} \end{aligned}\right\}$$



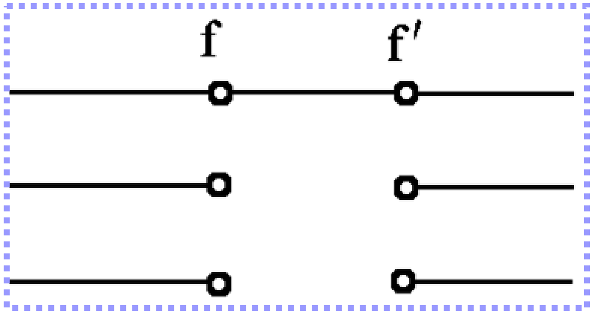
非故障相电流和断口电压

$$\left.\begin{aligned}\dot{I}_b &= \left(a^2 - \frac{X_{2\Sigma} + aX_{0\Sigma}}{X_{2\Sigma} + X_{0\Sigma}}\right)\dot{I}_{a1} = \frac{-3X_{2\Sigma} - j\sqrt{3}(X_{2\Sigma} + 2X_{0\Sigma})}{2(X_{2\Sigma} + X_{0\Sigma})}\dot{I}_{a1} \\ \dot{I}_c &= \left(a - \frac{X_{2\Sigma} + a^2 X_{0\Sigma}}{X_{2\Sigma} + X_{0\Sigma}}\right)\dot{I}_{a1} = \frac{-3X_{2\Sigma} + j\sqrt{3}(X_{2\Sigma} + 2X_{0\Sigma})}{2(X_{2\Sigma} + X_{0\Sigma})}\dot{I}_{a1}\end{aligned}\right\}$$

$$I_b = I_c = \sqrt{3}\sqrt{1 - \frac{X_{2\Sigma}X_{0\Sigma}}{(X_{2\Sigma} + X_{0\Sigma})^2}}I_{a1}$$

$$\Delta\dot{V}_a = 3\Delta\dot{V}_{a1} = j\frac{3X_{2\Sigma}X_{0\Sigma}}{X_{2\Sigma} + X_{0\Sigma}}\dot{I}_{a1}$$

二、两相断开



$$\dot{I}_b = \dot{I}_c = 0$$

$$\Delta\dot{V}_a = 0$$

$$\left.\begin{aligned} &\dot{I}_{a1} = \dot{I}_{a2} = \dot{I}_{a0} \\ &\Delta\dot{V}_{a1} + \Delta\dot{V}_{a2} + \Delta\dot{V}_{a0} = 0 \end{aligned}\right\}$$

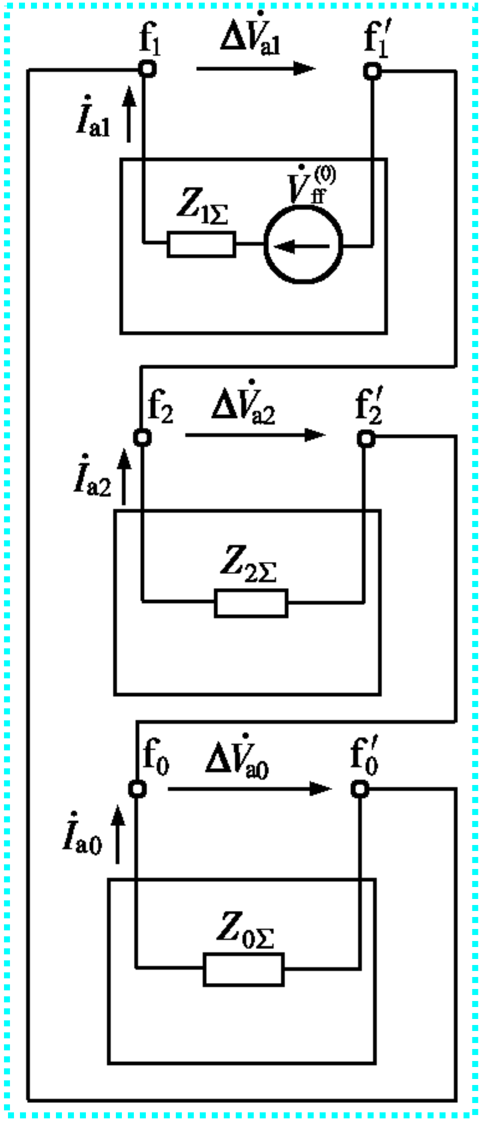
$$\left.\begin{aligned} \dot{V}_{ff'}^{(0)} - jX_{1\Sigma}\dot{I}_{a1} &= \Delta\dot{V}_{a1} \\ - jX_{2\Sigma}\dot{I}_{a2} &= \Delta\dot{V}_{a2} \\ - jX_{0\Sigma}\dot{I}_{a0} &= \Delta\dot{V}_{a0} \end{aligned}\right\}$$

$$\dot{I}_{a1} = \dot{I}_{a2} = \dot{I}_{a0} = \frac{\dot{V}_{ff'}^{(0)}}{j(X_{1\Sigma} + X_{2\Sigma} + X_{0\Sigma})}$$

两相断开时的复合序网

$$\left.\begin{aligned}\dot{I}_{a1} = \dot{I}_{a2} = \dot{I}_{a0} \\ \Delta V_{a1} + \Delta\dot{V}_{a2} + \Delta\dot{V}_{a0} = 0\end{aligned}\right\}$$

$$\left.\begin{aligned}\dot{V}_{ff'}^{(0)} - jX_{1\Sigma}\dot{I}_{a1} &= \Delta\dot{V}_{a1} \\ - jX_{2\Sigma}\dot{I}_{a2} &= \Delta\dot{V}_{a2} \\ - jX_{0\Sigma}\dot{I}_{a0} &= \Delta\dot{V}_{a0}\end{aligned}\right\}$$

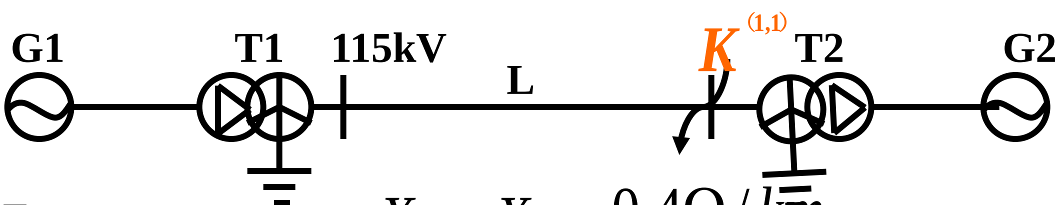


非故障相电流和断口电压

$$\dot{I}_a = 3\dot{I}_{a1}$$

$$\left.\begin{aligned}
\Delta\dot{V}_b &= j[(a^2 - a)X_{2\Sigma} + (a^2 - 1)X_{0\Sigma}]\dot{I}_{a1} \\
&= \frac{\sqrt{3}}{2}[(2X_{2\Sigma} + X_{0\Sigma})] - j\sqrt{3}X_{0\Sigma}]\dot{I}_{a1} \\
\Delta\dot{V}_c &= j[(a - a^2)X_{2\Sigma} + (a - 1)X_{0\Sigma}]\dot{I}_{a1} \\
&= \frac{\sqrt{3}}{2}[-(2X_{2\Sigma} + X_{0\Sigma})] - j\sqrt{3}X_{0\Sigma}]\dot{I}_{a1}
\end{aligned}\right\}$$

例题：已知某系统接线如图所示，各元件电抗均已知，当k点发生BC两相接地短路时，求短路点的各序电压、电流及各相电压和电流，并绘出短路点电压、电流的向量图。



G1: $10.5KV$, $50MVA$, $\cos\phi = 0.85$, $X_d'' = 0.125$, $X_2 = 0.16$

T1: $60MVA$, $10.5/121KV$, $U_k\% = 10.5$

L: $X_1 = X_2 = 0.4\Omega/km$, $l = 50\,km$, $X_0 = 2X_1$

T2: $31.5MVA$, $10.5/121KV$, $U_k\% = 10.5$

G2: $10.5KV$, $25MVA$, $\cos\phi = 0.85$, $X_d'' = 0.125$, $X_2 = 0.16$

1）计算各元件电抗（取S_B=100MVA，U_B=U_{av}）

发电机G1：
$$X_{G1(1)} = X''_d \frac{S_B}{S_N} = 0.125 \times \frac{100}{50} = 0.25$$

$$X_{G1(2)} = X_2 \frac{S_B}{S_N} = 0.16 \times \frac{100}{50} = 0.32$$

发电机G2：
$$X_{G2(1)} = X''_d \frac{S_B}{S_N} = 0.125 \times \frac{100}{25} = 0.5$$

$$X_{G2(2)} = X_2 \frac{S_B}{S_N} = 0.16 \times \frac{100}{25} = 0.64$$

变压器T1:

$$X_{T1(1)}=X_{T1(2)}=X_{T1(0)}=\frac{U_k\%}{100}\frac{S_B}{S_{T2N}}=\frac{10.5}{100}\times\frac{100}{60}=0.175$$

变压器T2:

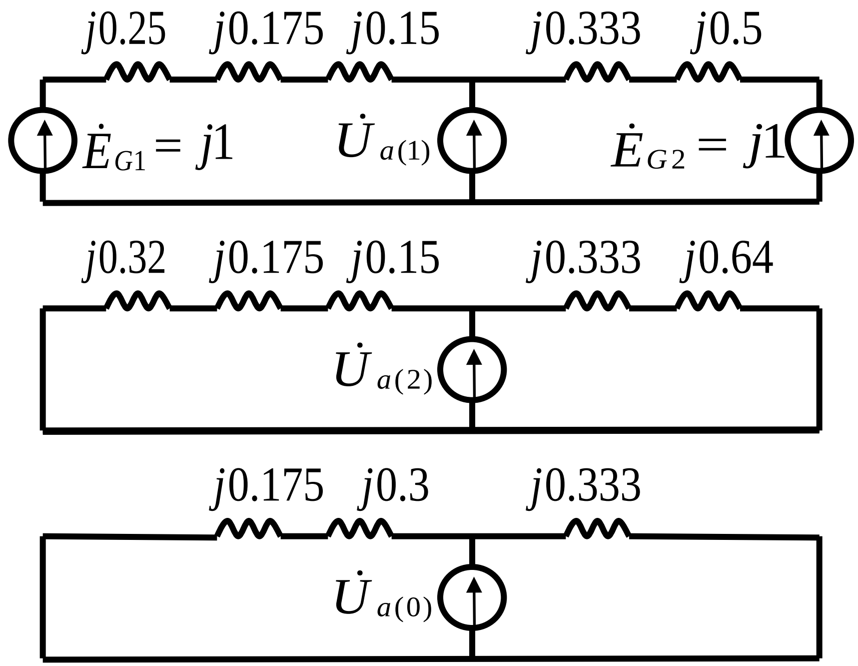
$$X_{T2(1)}=X_{T2(2)}=X_{T2(0)}=\frac{U_k\%}{100}\frac{S_B}{S_{T1N}}=\frac{10.5}{100}\times\frac{100}{31.5}=0.333$$

线路L:

$$X_{l(1)}=X_{l(2)}=x_1 l\frac{S_B}{U_B^2}=0.4\times50\times\frac{100}{115^2}=0.15$$

$$X_{l(0)}=2X_{l(1)}=0.3$$

2）以A相作基准相作出各序网图，求等值电抗



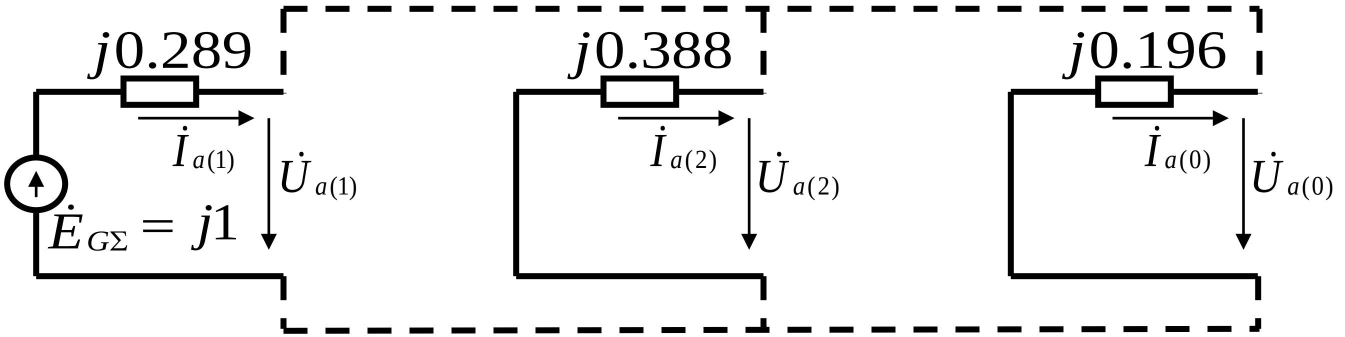
$$\dot{E}_{G\Sigma} = j1$$

$$X_{\Sigma(1)} = j(0.25 + 0.175 + 0.15) // (0.333 + 0.5) = j0.289$$

$$X_{\Sigma(2)} = j(0.32 + 0.175 + 0.15) // (0.333 + 0.64) = j0.388$$

$$X_{\Sigma(0)} = j(0.175 + 0.3) // 0.333 = j0.196$$

3）复合序网图



4）由复合序网求得各序的电流和电压为：

$$\dot{I}_{a(1)}=\frac{\dot{E}_{G\Sigma}}{j(X_{\Sigma(1)}+X_{\Sigma(2)}//X_{\Sigma(0)})}=2.358$$

$$\dot{I}_{a(2)}=-\dot{I}_{a(1)}\frac{X_{\Sigma(0)}}{(X_{\Sigma(0)}+X_{\Sigma(2)})}=-0.80$$

$$\dot{I}_{a(0)}=-\dot{I}_{a(1)}-\dot{I}_{a(2)}=-1.585$$

$$\dot{U}_{a(1)}=\dot{U}_{a(2)}=\dot{U}_{a(0)}=\dot{I}_{a(1)}(X_{\Sigma(2)}//X_{\Sigma(0)})=j0.311$$

5）求各相电流和电压

$$\dot{I}_a = \dot{I}_{a(1)} + \dot{I}_{a(2)} + \dot{I}_{a(0)} = 2.385 - 0.800 - 1.585 = 0$$

$$\dot{I}_b = a^2\dot{I}_{a(1)} + a\dot{I}_{a(2)} + \dot{I}_{a(0)} = 2.385\angle 240^\circ + (-0.8)\angle 120^\circ + (-1.585)$$

$$= 3.642\angle -130.77^\circ$$

$$\dot{I}_c = a\dot{I}_{a(1)} + a^2\dot{I}_{a(2)} + \dot{I}_{a(0)} = 2.385\angle 120^\circ + (-0.8)\angle 240^\circ + (-1.585)$$

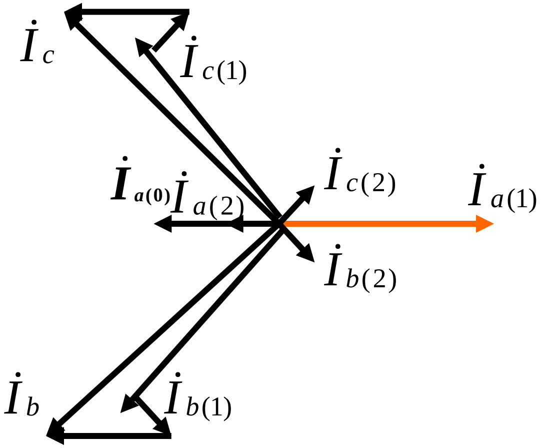
$$= 3.642\angle 130.77^\circ$$

$$\dot{U}_a = \dot{U}_{a(1)} + \ddot{U}_{a(2)} + \dot{U}_{a(0)} = 3\dot{U}_{a(1)} = j0.933$$

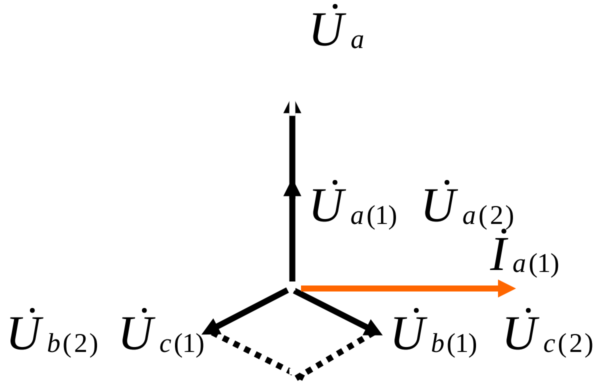
$$\dot{U}_b = \alpha^2\dot{U}_{a(1)} + \alpha\ddot{U}_{a(2)} + \dot{U}_{a(0)} = 0$$

$$\dot{U}_c = \alpha\dot{U}_{a(1)} + \alpha^2\ddot{U}_{a(2)} + \dot{U}_{a(0)} = 0$$

6）画电流和电压相量图



$\dot{I}_c$
$\dot{I}_{c(1)}$
$\dot{I}_{a(0)}$
$\dot{I}_{a(2)}$
$\dot{I}_{c(2)}$
$\dot{I}_{a(1)}$
$\dot{I}_{b(2)}$
$\dot{I}_b$
$\dot{I}_{b(1)}$



$\dot{U}_a$
$\dot{U}_{a(1)}$
$\dot{U}_{a(2)}$
$\dot{I}_{a(1)}$
$\dot{U}_{b(2)}$
$\dot{U}_{c(1)}$
$\dot{U}_{b(1)}$
$\dot{U}_{c(2)}$